Topics in Functional Analysis

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Preface

This is the lecture note for the graduate course *Functional Analysis* taught by Prof. Yutian Li in 2023 Fall at the Chinese University of Hong Kong, Shenzhen. The note is edited and revised with the support of ChatGPT. The materials presume a basic knowledge of functional analysis, so contents that are close to the boundary of these prerequisites may be omitted. Topics of the note include: review of linear functional analysis, weak convergence, spectral theory, semigroups, Sobolev spaces, and calculus of variations.

Contents

Chapter 1 Review of the Great Theorems

1.1 Hahn–Banach theory

Theorem 1.1.1 (Hahn–Banach theorem over real vector spaces)**.** Let X be a real vector space, let $Y \subseteq X$ be a linear subspace, and let $p\colon X \to \mathbb{R}$ be sublinear:

$$p(\alpha x) = \alpha p(x) \quad (\alpha \geq 0), \qquad p(x+y) \leq p(x) + p(y).$$

If $f\colon Y \to \mathbb{R}$ is linear and $f(y) \leq p(y)$ for every $y \in Y$, then there is a linear functional $\widetilde{f}\colon X \to \mathbb{R}$ such that

$$\widetilde{f}|_Y = f, \qquad \widetilde{f}(x) \leq p(x) \quad (x \in X).$$

Proof. We follow the maximal-extension argument indicated by the Zorn-lemma discussion in the source. Let $\mathscr{P}$ be the set of pairs (Z, g) such that

$$Y \subseteq Z \subseteq X, \qquad Z \text{ is a linear subspace}, \qquad g\colon Z \to \mathbb{R} \text{ is linear},$$

and

$$g|_Y = f, \qquad g(z) \leq p(z) \quad (z \in Z).$$

Order $\mathscr{P}$ by extension:

$$(Z_1, g_1) \preccurlyeq (Z_2, g_2) \quad\iff\quad Z_1 \subseteq Z_2 \ \text{ and } \ g_2|_{Z_1} = g_1.$$

The pair (Y, f) belongs to $\mathscr{P}$, so $\mathscr{P}$ is nonempty.

Let $\mathscr{C} \subseteq \mathscr{P}$ be a chain and set

$$Z_{\mathscr{C}} = \bigcup_{(Z,g)\in\mathscr{C}} Z.$$

Because the members of the chain are totally ordered by inclusion, $Z_{\mathscr{C}}$ is a linear subspace. For $z \in Z_{\mathscr{C}}$ choose $(Z, g) \in \mathscr{C}$ with $z \in Z$ and define

$$g_{\mathscr{C}}(z) = g(z).$$

This is well defined: if z lies in two members of the chain, one extends the other and the two functionals agree on the smaller domain. The same chain argument shows that $g_{\mathscr{C}}$ is linear, extends f, and satisfies $g_{\mathscr{C}} \leq p$. Thus every chain has an upper bound, and Zorn's lemma gives a maximal pair (M, g).

Suppose that $M \neq X$ and choose $x_0 \in X \setminus M$. Every linear extension of g to $M + \mathbb{R}x_0$ has the form

$$g_\alpha(m + tx_0) = g(m) + t\alpha \qquad (m \in M,\ t \in \mathbb{R})$$

for some $\alpha \in \mathbb{R}$. Define

$$L = \sup_{m \in M} \big(g(m) - p(m - x_0)\big), \qquad U = \inf_{m \in M} \big(p(m + x_0) - g(m)\big).$$

For arbitrary $m, n \in M$,

$$\begin{aligned} g(m) + g(n) = g(m+n) &\le p(m+n) \\ &\le p(m - x_0) + p(n + x_0), \end{aligned}$$

so

$$g(m) - p(m - x_0) \le p(n + x_0) - g(n).$$

Taking the supremum in m and the infimum in n gives $L \le U$. Choose $\alpha \in [L, U]$.

If $t > 0$, then, with $m/t \in M$,

$$\begin{aligned} g_\alpha(m + tx_0) &= t\,\big(g(m/t) + \alpha\big) \\ &\le t\,\big(g(m/t) + p(m/t + x_0) - g(m/t)\big) \\ &= p(m + tx_0). \end{aligned}$$

If $t < 0$, write $t = -s$ with $s > 0$. Since $\alpha \ge L$,

$$\begin{aligned} g_\alpha(m - sx_0) &= s\,\big(g(m/s) - \alpha\big) \\ &\le s\,p(m/s - x_0) = p(m - sx_0). \end{aligned}$$

The case $t = 0$ is the original domination on M. Thus $(M + \mathbb{R}x_0, g_\alpha)$ is a proper extension of (M, g), contradicting maximality. Therefore $M = X$ and g is the required extension. $\square$

Corollary 1.1.2 (Norm-preserving Hahn–Banach extension)**.** Let X be a normed space, let $Y \subseteq X$ be a linear subspace, and let $f \in Y^*$. There is $\widetilde{f} \in X^*$ satisfying

$$\widetilde{f}|_Y = f, \qquad \|\widetilde{f}\| = \|f\|.$$

Proof. If the scalar field is real, apply the preceding theorem with $p(x) = \|f\|\,\|x\|$. Since this sublinear function is symmetric, domination of both x and $-x$ gives

$$|\widetilde{f}(x)| \le \|f\|\,\|x\|,$$

and therefore $\|\widetilde{f}\| \le \|f\|$. Restriction to Y gives the reverse inequality.

For completeness, suppose that X is complex. Regard X and Y as real spaces and put $h = \operatorname{Re} f$. The real theorem gives a real-linear extension $\widetilde{h}$ satisfying

$$|\widetilde{h}(x)| \le \|f\|\,\|x\|.$$

Define

$$\widetilde{f}(x) = \widetilde{h}(x) - i\widetilde{h}(ix).$$

Then $\widetilde{f}(ix) = i\widetilde{f}(x)$, so $\widetilde{f}$ is complex linear, and on Y one has

$$\widetilde{f}(y) = \operatorname{Re} f(y) - i \operatorname{Re} f(iy) = f(y).$$

For $|\zeta| = 1$,

$$\operatorname{Re}(\zeta \widetilde{f}(x)) = \widetilde{h}(\zeta x) \le \|f\| \, \|x\|.$$

Taking the supremum over unimodular ζ yields $|\widetilde{f}(x)| \le \|f\| \, \|x\|$. Again restriction gives equality of the norms. □

Remark 1.1.3 (A useful uniqueness criterion)**.** If X^* is strictly convex, then every norm-preserving Hahn–Banach extension is unique. Indeed, if F_1 and F_2 are two distinct extensions of f with $\|F_1\| = \|F_2\| = \|f\| > 0$, then $(F_1 + F_2)/2$ is another extension and must have norm at least $\|f\|$; this contradicts strict convexity after normalization. The converse is not automatic from the one-line argument in the source and is therefore not asserted here without an additional global unique-extension hypothesis.

Proposition 1.1.4 (Norming functionals)**.** Let X be a normed space and let $x \in X \setminus \{0\}$. There is $f_x \in X^*$ such that

$$f_x(x) = \|x\|, \qquad \|f_x\| = 1.$$

Consequently, continuous linear functionals separate points: if $x \neq y$, then some $f \in X^*$ satisfies $f(x) \neq f(y)$.

Proof. Define $g(tx) = t\|x\|$ on $\operatorname{span}\{x\}$. Then $\|g\| = 1$ and a norm-preserving Hahn–Banach extension gives f_x. Apply this to $x - y$ for the separation statement. □

1.2 Geometric separation

The order inequalities in this section are stated for real normed spaces. In a complex normed space one applies the result to the underlying real space and then writes the resulting bounded real-linear functional h as

$$h = \operatorname{Re} f, \qquad f(x) = h(x) - ih(ix),$$

with f complex linear. Thus the corresponding complex statements use $\operatorname{Re} f$ in place of f.

Lemma 1.2.1 (Minkowski functional)**.** Let X be a normed space and let $C \subseteq X$ be open and convex with $0 \in C$. Define

$$p_C(x) = \inf\{t > 0 : x \in tC\}.$$

Then p_C is finite and sublinear. Moreover, for some $M > 0$,

$$p_C(x) \le M\|x\| \quad (x \in X), \qquad C = \{x \in X : p_C(x) < 1\}.$$

Proof. Because C is open and contains 0, there is $r > 0$ such that $B(0,r) \subseteq C$. For $x \neq 0$ and every $t > \|x\|/r$, we have $\|x/t\| < r$, hence $x/t \in C$ and therefore $x \in tC$. Consequently,

$$0 \le p_C(x) \le \frac{\|x\|}{r},$$

and the same inequality is trivial for $x = 0$.

For $\lambda > 0$,

$$\begin{aligned} p_C(\lambda x) &= \inf\{t > 0 : \lambda x \in tC\} \\ &= \inf\{\lambda s : s > 0,\ x \in sC\} = \lambda p_C(x). \end{aligned}$$

Also $p_C(0) = 0$, so positive homogeneity holds for every $\lambda \geq 0$.

To prove subadditivity, fix $\varepsilon > 0$. By the definition of the infimum choose $a, b > 0$ such that

$$x \in aC, \quad y \in bC, \quad a < p_C(x) + \varepsilon, \quad b < p_C(y) + \varepsilon.$$

Then $x/a, y/b \in C$, and convexity gives

$$\frac{x+y}{a+b} = \frac{a}{a+b}\frac{x}{a} + \frac{b}{a+b}\frac{y}{b} \in C.$$

Thus

$$p_C(x+y) \leq a + b < p_C(x) + p_C(y) + 2\varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves subadditivity.

Finally, if $x \in C$, continuity of $s \mapsto sx$ and openness of C give $s > 1$ sufficiently close to 1 with $sx \in C$. Hence $x \in s^{-1}C$ and $p_C(x) \leq s^{-1} < 1$. Conversely, if $p_C(x) < 1$, choose $t \in (p_C(x), 1)$ with $x \in tC$. Writing $x = tc$ with $c \in C$, convexity of C and $0 \in C$ imply $tc \in C$. Therefore

$$C = \{x \in X : p_C(x) < 1\}.$$

□

Lemma 1.2.2 (Separation of an open convex set and a point)**.** Let X be a real normed space, let $C \subseteq X$ be open and convex, and let $x_0 \notin C$. Then there are $f \in X^* \setminus \{0\}$ and $\alpha \in \mathbb{R}$ such that

$$f(x) < \alpha = f(x_0) \qquad (x \in C).$$

Proof. Choose $c_0 \in C$ and replace C by $C - c_0$ and x_0 by $x_0 - c_0$. After this translation, $0 \in C$ and still $x_0 \notin C$. Let p_C be the Minkowski functional of the translated set. Since $C = \{p_C < 1\}$, $p_C(x_0) \geq 1$.

Define

$$g(tx_0) = t \qquad (t \in \mathbb{R})$$

on $\operatorname{span}\{x_0\}$. For $t \geq 0$,

$$p_C(tx_0) = tp_C(x_0) \geq t = g(tx_0),$$

whereas for $t < 0$ we have $g(tx_0) < 0 \leq p_C(tx_0)$. Thus $g \leq p_C$ on its domain. Hahn–Banach gives a linear extension f with $f \leq p_C$ on X and $f(x_0) = 1$.

The norm estimate for p_C also proves continuity: for every $x \in X$,

$$|f(x)| \leq \max\{p_C(x), p_C(-x)\} \leq M'\|x\|.$$

Hence $f \in X^*$. If $x \in C$, then

$$f(x) \leq p_C(x) < 1 = f(x_0).$$

Undoing the translation gives the required strict separation, with $\alpha = f(x_0)$. □

Theorem 1.2.3 (Geometric Hahn–Banach separation). Let A, B be nonempty disjoint convex subsets of a real normed space X.

(a). If A is open, then there are $f \in X^* \setminus \{0\}$ and $\alpha \in \mathbb{R}$ such that

$$f(a) < \alpha \le f(b) \qquad (a \in A,\ b \in B).$$

(b). If A is closed and B is compact, then there are $f \in X^* \setminus \{0\}$, $\alpha \in \mathbb{R}$, and $\varepsilon > 0$ such that

$$f(a) + \varepsilon \le \alpha \le f(b) \qquad (a \in A,\ b \in B).$$

Proof. For the first assertion, put

$$C = A - B = \{a - b : a \in A,\ b \in B\}.$$

The set C is convex and open, and $0 \notin C$. Apply the preceding lemma to C and the point 0. After changing the sign of the functional if necessary, we obtain $f \in X^* \setminus \{0\}$ such that

$$f(a - b) < 0 \qquad (a \in A,\ b \in B).$$

Hence $f(a) < f(b)$ for all $a \in A$ and $b \in B$, and

$$-\infty < \sup_{a \in A} f(a) \le \inf_{b \in B} f(b) < \infty.$$

Let $\alpha = \sup_A f$. Since A is open, no point of A can maximize the nonzero functional f: for $a \in A$, choose h with $f(h) > 0$; then $a + th \in A$ for sufficiently small $t > 0$ and $f(a + th) > f(a)$. Therefore

$$f(a) < \alpha \le f(b) \qquad (a \in A,\ b \in B).$$

For the second assertion, $A - B$ is closed. Indeed, if $a_n - b_n \to z$, compactness of B gives a subsequence $b_{n_k} \to b \in B$; then $a_{n_k} \to z + b \in A$, because A is closed, and so $z \in A - B$. Since $0 \notin A - B$, choose $r > 0$ with

$$B(0, r) \cap (A - B) = \varnothing.$$

Apply the first part to the open convex set $B(0, r)$ and the convex set $A - B$. There are $g \in X^* \setminus \{0\}$ and $\beta \in \mathbb{R}$ such that

$$g(u) < \beta \le g(a - b) \qquad (\|u\| < r,\ a \in A,\ b \in B).$$

Taking the supremum over the open ball gives

$$r\|g\| \le g(a - b) \qquad (a \in A,\ b \in B).$$

Set $f = -g$ and $\delta = r\|g\| > 0$. Then

$$f(a) + \delta \le f(b) \qquad (a \in A,\ b \in B),$$

so $\sup_A f + \delta \le \inf_B f$. With

$$\varepsilon = \frac{\delta}{2}, \qquad \alpha = \frac{1}{2}\left(\sup_A f + \inf_B f\right),$$

we obtain

$$f(a) + \varepsilon \le \alpha \le f(b) \qquad (a \in A,\ b \in B).$$

□

1.3 Baire category and uniform boundedness

Definition 1.3.1. A subset N of a topological space is *nowhere dense* if $\operatorname{int}(\overline{N}) = \varnothing$. A space is of the *first category* if it is a countable union of nowhere dense sets, and of the *second category* otherwise.

Lemma 1.3.2. For a subset A of a topological space X, the following are equivalent:

(a). A is dense in X;

(b). every nonempty open subset of X meets A;

(c). $\operatorname{int}(X \setminus A) = \varnothing$.

Proof. Assume first that $\overline{A} = X$, and let $O \subseteq X$ be nonempty and open. Choose $x \in O$. Since $x \in \overline{A}$, every neighbourhood of x, in particular O, meets A. Thus (1) implies (2).

If (2) holds and $\operatorname{int}(X \setminus A)$ were nonempty, that interior would be a nonempty open set disjoint from A, contradicting (2). Hence (2) implies (3).

Finally, the elementary identity

$$X \setminus \overline{A} = \operatorname{int}(X \setminus A)$$

shows that (3) is equivalent to $X \setminus \overline{A} = \varnothing$, that is, $\overline{A} = X$. This proves (3) implies (1) and completes the equivalence. □

Theorem 1.3.3 (Baire category theorem)**.** Every nonempty complete metric space is of the second category. Equivalently:

(a). a countable union of closed sets with empty interior has empty interior;

(b). a countable intersection of open dense sets is dense.

Proof. We prove the open-dense formulation, following the nested-ball construction in the source but using closed balls so that the limiting point is retained. Let $(U_n)_{n\geq 1}$ be open and dense, and let $O \subseteq X$ be nonempty and open. Since $O \cap U_1$ is nonempty and open, choose $x_1 \in X$ and $0 < r_1 < 1$ such that

$$\overline{B}(x_1, r_1) \subseteq O \cap U_1.$$

Inductively, once $\overline{B}(x_n, r_n)$ has been chosen, the set $B(x_n, r_n) \cap U_{n+1}$ is nonempty and open. Choose $x_{n+1} \in X$ and

$$0 < r_{n+1} < \min\left\{\frac{r_n}{2}, \frac{1}{n+1}\right\}$$

with

$$\overline{B}(x_{n+1}, r_{n+1}) \subseteq B(x_n, r_n) \cap U_{n+1}.$$

The closed balls are nested and $r_n \to 0$. If $m > n$, then $x_m \in \overline{B}(x_n, r_n)$, so $d(x_m, x_n) \leq r_n$. Thus (x_n) is Cauchy. Completeness gives $x_n \to x \in X$.

For every fixed n, all x_m with $m \geq n$ lie in the closed ball $\overline{B}(x_n, r_n)$, hence

$$x \in \overline{B}(x_n, r_n) \subseteq U_n.$$

Also $x \in \overline{B}(x_1, r_1) \subseteq O$. Consequently,

$$O \cap \bigcap_{n=1}^{\infty} U_n \neq \varnothing.$$

Since every nonempty open set meets the intersection, $\bigcap_n U_n$ is dense.

If F_n is closed with empty interior, then $U_n = X \setminus F_n$ is open and dense. The preceding result says that $\bigcap_n U_n$ is dense, so its complement $\bigcup_n F_n$ has empty interior. This is the closed-set formulation and proves the category theorem. □

Example 1.3.4. A normed space with a countably infinite Hamel basis cannot be complete: it is the union of the finite-dimensional spans of the first n basis vectors, each of which is closed and has empty interior. Baire's theorem also underlies the existence of continuous nowhere differentiable functions on $[0, 1]$.

Theorem 1.3.5 (Uniform boundedness principle)**.** Let X be a Banach space, let Y be a normed space, and let $\{T_i : i \in I\} \subseteq \mathcal{B}(X, Y)$. If

$$\sup_{i \in I} \|T_i x\| < \infty \qquad (x \in X),$$

then

$$\sup_{i \in I} \|T_i\| < \infty.$$

Proof. For $n \in \mathbb{N}$, define

$$F_n = \{x \in X : \|T_i x\| \leq n \text{ for every } i \in I\} = \bigcap_{i \in I} T_i^{-1}\big(\overline{B}_Y(0, n)\big).$$

Each F_n is closed. Pointwise boundedness says that every $x \in X$ lies in some F_n, so $X = \bigcup_n F_n$. Baire's theorem gives $N \in \mathbb{N}$, $x_0 \in X$, and $r > 0$ such that

$$B(x_0, r) \subseteq F_N.$$

To move the estimate to the origin, take $h \in X$ with $\|h\| < r$. Both $x_0 + h$ and $x_0 - h$ lie in F_N, and therefore, for every $i \in I$,

$$\begin{aligned} 2\|T_i h\| &= \|T_i(x_0 + h) - T_i(x_0 - h)\| \\ &\leq \|T_i(x_0 + h)\| + \|T_i(x_0 - h)\| \leq 2N. \end{aligned}$$

Hence $\|T_i h\| \leq N$ for all $\|h\| < r$, uniformly in i. If $\|x\| \leq 1$, put $h = (r/2)x$. Then

$$\|T_i x\| = \frac{2}{r}\|T_i h\| \leq \frac{2N}{r}.$$

Taking the supremum over the unit ball gives

$$\sup_{i \in I} \|T_i\| \leq \frac{2N}{r} < \infty.$$

□

Theorem 1.3.6 (Banach–Steinhaus pointwise-limit theorem)**.** Let X be Banach, let Y be normed, and let

$T_n \in \mathcal{B}(X,Y)$. Suppose $T_n x$ converges for every $x \in X$, and define $Tx = \lim_n T_n x$. Then $T \in \mathcal{B}(X,Y)$ and

$$\|T\| \le \liminf_{n\to\infty} \|T_n\|.$$

Proof. Linearity is inherited from the pointwise limit. For every x, the sequence $(T_n x)$ is bounded, so uniform boundedness yields $M := \sup_n \|T_n\| < \infty$. Hence $\|Tx\| \le M\|x\|$. More precisely, for $\|x\| = 1$,

$$\|Tx\| = \lim_n \|T_n x\| \le \liminf_n \|T_n\|,$$

and taking the supremum over the unit sphere gives the claim. □

Theorem 1.3.7 (Weak boundedness implies norm boundedness)**.** Let $B \subseteq X$, where X is a normed space. If $f(B)$ is bounded in $\mathbb{K}$ for every $f \in X^*$, then B is norm bounded.

Proof. For $b \in B$, define $J_b \in (X^*)^*$ by $J_b(f) = f(b)$. The family $\{J_b : b \in B\}$ is pointwise bounded on the Banach space X^*, so uniform boundedness gives $\sup_{b\in B} \|J_b\| < \infty$. Hahn–Banach gives $\|J_b\| = \|b\|$, and the result follows. □

1.4 Open mapping, bounded inverse, and equivalent norms

Definition 1.4.1. A map between topological spaces is *open* if it sends open sets to open sets.

Theorem 1.4.2 (Banach open mapping theorem)**.** Let X and Y be Banach spaces and let $T \in \mathcal{B}(X,Y)$ be surjective. Then T is an open map.

Proof. We retain the four steps of the handwritten proof.

Step 1: obtain interior in the closure of an image. Surjectivity gives

$$Y = \bigcup_{n=1}^{\infty} T(nB_X) \subseteq \bigcup_{n=1}^{\infty} \overline{T(nB_X)}.$$

Thus the closed sets $\overline{T(nB_X)}$ cover the Banach space Y. Baire's theorem yields $n_0 \in \mathbb{N}$, $y_0 \in Y$, and $r > 0$ such that

$$B_Y(y_0, r) \subseteq \overline{T(n_0 B_X)}.$$

Step 2: center the ball at the origin. Let $\|z\| < r$. Since y_0 and $y_0 + z$ both lie in the displayed ball, choose $x_k, u_k \in n_0 B_X$ with

$$Tx_k \to y_0 + z, \qquad Tu_k \to y_0.$$

Then $T(x_k - u_k) \to z$ and $x_k - u_k \in 2n_0 B_X$. Hence

$$B_Y(0, r) \subseteq \overline{T(2n_0 B_X)}.$$

After scaling, there is $\delta > 0$ such that

$$B_Y(0, \delta) \subseteq \overline{T(B_X)}.$$

For the correction argument we normalize this to

$$B_Y(0, 1) \subseteq \overline{T(B_X)}.$$

Step 3: remove the closure by successive corrections. Let $\|y\| < 1/2$ and set $r_0 = y$. We construct $x_k \in X$ such that

$$\|x_k\| < 2^{-k}, \qquad r_k = y - T\sum_{j=1}^{k} x_j, \qquad \|r_k\| < 2^{-k-1}.$$

Suppose r_{k-1} is known and $\|r_{k-1}\| < 2^{-k}$. Then $2^k r_{k-1} \in B_Y(0,1)$. By the closure inclusion, choose $u_k \in B_X$ such that

$$\|2^k r_{k-1} - Tu_k\| < \frac{1}{2}.$$

Set $x_k = 2^{-k}u_k$. Then $\|x_k\| < 2^{-k}$ and

$$\|r_{k-1} - Tx_k\| < 2^{-k-1},$$

which is exactly the desired estimate for r_k.

Because $\sum_k \|x_k\| < 1$, the series $\sum_k x_k$ converges in X to some $x \in B_X$. Moreover, $r_k \to 0$, so continuity of T gives

$$Tx = T\left(\sum_{k=1}^{\infty} x_k\right) = \lim_{k\to\infty} T\sum_{j=1}^{k} x_j = y.$$

Therefore $B_Y(0,1/2) \subseteq T(B_X)$. Undoing the normalization, there is $c > 0$ such that

$$B_Y(0,c) \subseteq T(B_X).$$

Step 4: pass to an arbitrary open set. Let $O \subseteq X$ be open and let $y = Tx \in T(O)$. Choose $\varepsilon > 0$ with $x + \varepsilon B_X \subseteq O$. Then

$$T(x + \varepsilon B_X) = y + \varepsilon T(B_X) \supseteq B_Y(y, \varepsilon c).$$

Thus every point of $T(O)$ is an interior point, and $T(O)$ is open. □

Corollary 1.4.3 (Bounded inverse theorem)**.** A bounded bijective linear map between Banach spaces has a bounded inverse.

Proof. The open mapping theorem says that T^{-1} is continuous at 0, hence bounded. □

Example 1.4.4 (A priori estimate for a boundary-value problem)**.** Let

$$X = \{u \in C^2[0,1] : u(0) = u(1) = 0\}, \quad \|u\|_X = \|u\|_\infty + \|u'\|_\infty + \|u''\|_\infty,$$

and let $Y = C[0,1]$ with the supremum norm. For continuous coefficients a, b, c, the operator

$$Lu = au'' + bu' + cu$$

is bounded from X to Y. If the boundary-value problem $Lu = f$ has a unique solution $u \in X$ for every $f \in Y$—equivalently, if L is bijective—then the bounded inverse theorem gives a constant C such that

$$\|u\|_X \le C\|Lu\|_\infty.$$

The bijectivity hypothesis is essential; it does not follow merely from the formula for L.

Theorem 1.4.5 (Equivalence of complete norms). Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be norms making the same vector space X Banach. If

$$\|x\|_2 \le C\|x\|_1 \qquad (x \in X),$$

then there is $C' > 0$ such that

$$\|x\|_1 \le C'\|x\|_2 \qquad (x \in X).$$

Thus the two norms are equivalent and induce the same topology.

Proof. The identity map $(X, \|\cdot\|_1) \to (X, \|\cdot\|_2)$ is a bounded bijection between Banach spaces. Apply the bounded inverse theorem. □

1.5 Closed operators and the closed graph theorem

Definition 1.5.1. For a linear operator $T\colon \mathrm{D}(T) \subseteq X \to Y$, its graph is

$$\mathrm{Graph}(T) = \{(x, Tx) : x \in \mathrm{D}(T)\} \subseteq X \times Y.$$

The operator is *closed* if its graph is closed. Equivalently, whenever $x_n \in \mathrm{D}(T)$, $x_n \to x$ in X, and $Tx_n \to y$ in Y, one has $x \in \mathrm{D}(T)$ and $Tx = y$.

Remark 1.5.2. Every bounded linear operator is closed. The converse fails when the domain is not complete. For example,

$$D\colon C^1[0,1] \subset C[0,1] \to C[0,1], \qquad Du = u',$$

is closed when both spaces use the supremum norm, but it is unbounded.

Theorem 1.5.3 (Banach closed graph theorem). Let X and Y be Banach spaces. An everywhere-defined linear map $T\colon X \to Y$ is bounded if and only if it is closed.

Proof. If T is bounded and $x_n \to x$ while $Tx_n \to y$, then continuity gives $Tx_n \to Tx$ and hence $y = Tx$. Thus bounded operators are closed.

Conversely, suppose T is closed and equip X with the graph norm

$$\|x\|_T = \|x\|_X + \|Tx\|_Y.$$

We first check the completeness assertion used in the source. If (x_n) is Cauchy in $\|\cdot\|_T$, then (x_n) is Cauchy in X and (Tx_n) is Cauchy in Y. Since both spaces are Banach, there are $x \in X$ and $y \in Y$ with

$$x_n \to x, \qquad Tx_n \to y.$$

Closedness gives $y = Tx$, and therefore

$$\|x_n - x\|_T = \|x_n - x\|_X + \|Tx_n - Tx\|_Y \longrightarrow 0.$$

Hence $(X, \|\cdot\|_T)$ is Banach.

The identity map

$$I\colon (X, \|\cdot\|_T) \longrightarrow (X, \|\cdot\|_X)$$

is a bounded bijection because $\|x\|_X \le \|x\|_T$. By the bounded inverse theorem, its inverse is bounded: there is $C > 0$ such that

$$\|x\|_T \le C\|x\|_X \qquad (x \in X).$$

In particular,

$$\|Tx\|_Y \le \|x\|_T \le C\|x\|_X,$$

so T is bounded. $\square$

Theorem 1.5.4 (Everywhere-defined symmetric maps)**.** Let H be a real Hilbert space and let $T\colon H \to H$ be a map satisfying

$$\langle Tx, y\rangle = \langle x, Ty\rangle \qquad (x, y \in H).$$

Then T is linear and bounded. The same conclusion holds over a complex Hilbert space when the inner product is taken linear in the first variable and the symmetric identity $\langle Tx, y\rangle = \langle x, Ty\rangle$ is imposed.

Proof. We first prove linearity without assuming continuity. For $x_1, x_2, y \in H$,

$$\begin{aligned}\langle T(x_1 + x_2) - Tx_1 - Tx_2, y\rangle &= \langle x_1 + x_2, Ty\rangle - \langle x_1, Ty\rangle - \langle x_2, Ty\rangle \\ &= 0.\end{aligned}$$

Since a vector orthogonal to every y is zero, $T(x_1 + x_2) = Tx_1 + Tx_2$. If α is a scalar and the inner product is linear in the first variable, then

$$\langle T(\alpha x) - \alpha Tx, y\rangle = \langle \alpha x, Ty\rangle - \alpha \langle x, Ty\rangle = 0 \qquad (y \in H).$$

Thus $T(\alpha x) = \alpha Tx$, and T is linear. With the opposite inner-product convention the same computation is made in the other variable.

Now suppose $x_n \to x$ and $Tx_n \to z$. For each $y \in H$,

$$\begin{aligned}\langle z, y\rangle &= \lim_{n\to\infty} \langle Tx_n, y\rangle \\ &= \lim_{n\to\infty} \langle x_n, Ty\rangle \\ &= \langle x, Ty\rangle = \langle Tx, y\rangle .\end{aligned}$$

Hence $z = Tx$, so T is closed. The closed graph theorem now applies to the everywhere-defined linear operator T and shows that T is bounded. $\square$

1.6 Adjoints, compact maps, and closed ranges

Definition 1.6.1. For normed spaces X, Y and $T \in \mathcal{B}(X, Y)$, the adjoint $T^*\colon Y^* \to X^*$ is defined by

$$(T^*y^*)(x) = y^*(Tx).$$

A bounded linear map $T\colon X \to Y$ is *compact* if it maps bounded sets to relatively compact sets, equivalently, if every bounded sequence (x_n) has a subsequence for which (Tx_{n_k}) converges in Y.

Theorem 1.6.2 (Schauder theorem)**.** A bounded linear operator $T\colon X \to Y$ is compact if and only if its adjoint $T^*\colon Y^* \to X^*$ is compact.

Remark 1.6.3. A proof is given in Chapter 6, where compact operators are studied systematically.

Theorem 1.6.4 (Closed range theorem)**.** Let X, Y be Banach spaces and $T \in \mathcal{B}(X, Y)$. Then the following are equivalent:

(a). $\mathrm{Ran}(T)$ is norm closed in Y;

(b). $\mathrm{Ran}(T^*)$ is norm closed in X^*;

(c). $\mathrm{Ran}(T^*)$ is weak-$*$ closed in X^*.

Whenever these conditions hold,

$$\mathrm{Ran}(T) = {}^{\perp}\mathrm{Ker}(T^*), \qquad \mathrm{Ran}(T^*) = \mathrm{Ker}(T)^{\perp}.$$

Without a closed-range hypothesis one always has

$$\overline{\mathrm{Ran}(T)} = {}^{\perp}\mathrm{Ker}(T^*), \qquad \overline{\mathrm{Ran}(T^*)}^{w^*} = \mathrm{Ker}(T)^{\perp}.$$

Corollary 1.6.5. For $T \in \mathcal{B}(X, Y)$ between Banach spaces, the following are equivalent:

(a). T is surjective;

(b). there is $c > 0$ such that $\|T^*y^*\| \geq c\|y^*\|$ for every $y^* \in Y^*$;

(c). T^* is injective and has closed range.

Proof. If T is surjective, its induced map $X/\operatorname{Ker} T \to Y$ is a bounded bijection, so the bounded inverse theorem yields the lower bound on T^*. Conversely, the lower bound makes T^* injective with closed range. The closed range theorem then gives $\overline{\mathrm{Ran}(T)} = {}^{\perp}\mathrm{Ker}(T^*) = Y$, while $\mathrm{Ran}(T)$ is closed, so $\mathrm{Ran}(T) = Y$. $\square$

Chapter 2 Weak Convergence and Weak Topology

2.1 Topologies generated by maps

Definition 2.1.1. A *topology* τ on a set X is a family of subsets of X such that $\varnothing, X \in \tau$, arbitrary unions of members of τ belong to τ, and finite intersections of members of τ belong to τ. The pair (X, τ) is a topological space.

A family $\mathcal{B} \subseteq \tau$ is a *base* if every open set is a union of members of $\mathcal{B}$. A family $\mathcal{S}$ is a *subbase* if the finite intersections of members of $\mathcal{S}$ form a base.

Definition 2.1.2. A map $F\colon (X, \tau_X) \to (Y, \tau_Y)$ is continuous if $F^{-1}(O) \in \tau_X$ for every $O \in \tau_Y$. A sequence (x_n) converges to x if, for every neighbourhood U of x, one has $x_n \in U$ for all sufficiently large n.

Definition 2.1.3 (Initial topology)**.** Let X be a set and let $\phi_i\colon X \to (Y_i, \tau_i)$ be maps indexed by $i \in I$. The *initial topology* generated by the family (ϕ_i) is the topology with subbase

$$\{\phi_i^{-1}(O) : i \in I,\ O \in \tau_i\}.$$

It is the weakest topology on X for which every ϕ_i is continuous.

Proposition 2.1.4. For the initial topology generated by (ϕ_i):

(a). $x_n \to x$ if and only if $\phi_i(x_n) \to \phi_i(x)$ for every i;

(b). a map $\psi\colon Z \to X$ is continuous if and only if every $\phi_i \circ \psi$ is continuous.

Proof. The forward implications follow from continuity of the coordinate maps. For the converse in the first statement, a basic neighbourhood of x is a finite intersection

$$\bigcap_{j=1}^{m} \phi_{i_j}^{-1}(O_j), \qquad \phi_{i_j}(x) \in O_j,$$

and coordinatewise convergence eventually places x_n in every factor. The second statement follows because inverse images commute with finite intersections and arbitrary unions. $\square$

2.2 The weak topology

Definition 2.2.1. Let X be a normed space. The *weak topology* $\sigma(X, X^*)$ is the initial topology generated by all $f \in X^*$. A sequence converges weakly, written $x_n \rightharpoonup x$, if

$$f(x_n) \longrightarrow f(x) \qquad (f \in X^*).$$

Theorem 2.2.2. The weak topology on a normed space is Hausdorff.

Proof. If $x \neq y$, Hahn–Banach gives $f \in X^*$ with $f(x) \neq f(y)$. Disjoint open intervals about these two scalar values have disjoint inverse images containing x and y. □

Proposition 2.2.3 (Basic weak neighbourhoods)**.** For $x_0 \in X$, the sets

$$V(x_0; f_1, \ldots, f_m; \varepsilon) = \{x \in X : |f_j(x - x_0)| < \varepsilon, \ 1 \leq j \leq m\}$$

form a neighbourhood base at x_0. Allowing different positive radii in the coordinates gives the same topology.

Proof. The weak topology is generated by the subbasic sets $f^{-1}(U)$ with $f \in X^*$ and $U \subseteq \mathbb{K}$ open. Hence a basic neighbourhood of x_0 is a finite intersection

$$\bigcap_{j=1}^{m} f_j^{-1}(U_j), \qquad f_j(x_0) \in U_j.$$

For each j, choose $\varepsilon_j > 0$ such that the open disc of radius ε_j about $f_j(x_0)$ is contained in U_j. Then

$$\{x : |f_j(x - x_0)| < \varepsilon_j, \quad 1 \leq j \leq m\}$$

is contained in the given basic neighbourhood. Conversely, every such finite-coordinate set is a finite intersection of inverse images of open discs, so it is weakly open. Replacing the separate radii by $\varepsilon = \min_j \varepsilon_j$ proves that one common radius gives the same neighbourhood base. □

Theorem 2.2.4 (Finite-dimensional case)**.** If X is finite dimensional, weak convergence and norm convergence coincide. Equivalently, the weak and norm topologies on X are the same.

Proof. Choose a basis $e_1, \ldots, e_d$ and its coordinate functionals $e_1^*, \ldots, e_d^*$. If $x_n \rightharpoonup x$, then every coordinate of $x_n - x$ converges to zero. All norms on the finite-dimensional coordinate space are equivalent, so $\|x_n - x\| \to 0$. □

Example 2.2.5. If (e_n) is an orthonormal sequence in a Hilbert space, then $e_n \rightharpoonup 0$: Bessel's inequality implies $\langle x, e_n \rangle \to 0$ for every x. In contrast, in ℓ^1 weak convergence of sequences is equivalent to norm convergence (Schur's theorem).

Proposition 2.2.6 (Uniqueness, boundedness, and lower semicontinuity). Let X be a normed space.

(a). Weak limits are unique.

(b). If $x_n \rightharpoonup x$, then (x_n) is norm bounded.

(c). If $x_n \rightharpoonup x$, then

$$\|x\| \le \liminf_{n\to\infty} \|x_n\|.$$

Proof. Weak limits are unique because the weak topology is Hausdorff.

For boundedness, regard each x_n as the evaluation functional

$$J_{x_n} \in (X^*)^*, \qquad J_{x_n}(f) = f(x_n).$$

For every fixed $f \in X^*$, weak convergence gives $f(x_n) \to f(x)$, so the scalar sequence $(J_{x_n}(f))$ is bounded. The uniform boundedness principle, applied on the Banach space X^*, yields

$$\sup_n \|J_{x_n}\| < \infty.$$

Hahn–Banach gives $\|J_{x_n}\| = \|x_n\|$, and hence (x_n) is norm bounded.

For weak lower semicontinuity, if $x = 0$ there is nothing to prove. Otherwise choose $f \in X^*$ with $\|f\| = 1$ and, after multiplying by a unimodular scalar in the complex case, $f(x) = \|x\|$. Then

$$\begin{aligned}\|x\| &= \operatorname{Re} f(x) = \lim_{n\to\infty} \operatorname{Re} f(x_n)\\ &\le \liminf_{n\to\infty} |f(x_n)| \le \liminf_{n\to\infty} \|x_n\|.\end{aligned}$$

□

Theorem 2.2.7 (Weak closure of the unit sphere). Let X be infinite dimensional and let $S_X = \{x : \|x\| = 1\}$. Then

$$\overline{S_X}^{w} = B_X = \{x : \|x\| \le 1\}.$$

Consequently, the weak topology is strictly weaker than the norm topology.

Proof. Weak lower semicontinuity of the norm shows that

$$B_X = \{x : \|x\| \le 1\}$$

is weakly closed. Since $S_X \subseteq B_X$, this gives $\overline{S_X}^{w} \subseteq B_X$.

For the reverse inclusion, points of S_X already belong to the weak closure, so let $\|x_0\| < 1$. Consider a basic weak neighbourhood

$$V = \{x \in X : |f_j(x - x_0)| < \varepsilon,\ 1 \le j \le m\}.$$

The linear map

$$F\colon X \to \mathbb{K}^m, \qquad F(x) = (f_1(x), \dots, f_m(x)),$$

has finite-dimensional range. If its kernel were $\{0\}$, then X would be isomorphic to a subspace of $\mathbb{K}^m$ and would be finite dimensional. Hence

$$M := \bigcap_{j=1}^{m} \operatorname{Ker} f_j \neq \{0\}.$$

Choose $0 \neq y \in M$. The function

$$h(t) = \|x_0 + ty\|, \qquad t \geq 0,$$

is continuous, satisfies $h(0) < 1$, and tends to infinity as $t \to \infty$. The intermediate value theorem therefore gives $t_0 > 0$ with $\|x_0 + t_0 y\| = 1$. Since $f_j(y) = 0$ for every j,

$$f_j(x_0 + t_0 y - x_0) = 0,$$

so $x_0 + t_0 y \in V \cap S_X$. Every basic weak neighbourhood of x_0 thus meets the sphere, proving $x_0 \in \overline{S_X}^{w}$. □

2.3 Weak continuity and compactness

Proposition 2.3.1. Let X, Y be normed spaces.

(a). If $A \in \mathcal{B}(X, Y)$ and $x_n \rightharpoonup x$, then $Ax_n \rightharpoonup Ax$.

(b). If $A \in \mathcal{K}(X, Y)$ and $x_n \rightharpoonup x$, then $\|Ax_n - Ax\| \to 0$.

(c). If $B\colon X \times Y \to Z$ is a bounded bilinear map, then $x_n \to x$ and $y_n \to y$ in norm imply $B(x_n, y_n) \to B(x, y)$ in norm.

Proof. For the first assertion, $g(Ax_n) = (A^*g)(x_n) \to (A^*g)(x)$ for every $g \in Y^*$. For the second, weak convergence makes (x_n) bounded. If $Ax_n \not\to Ax$, a subsequence stays a fixed distance away. Compactness gives a further norm-convergent subsequence, while the first assertion forces its weak limit to be Ax; hence its norm limit is Ax, a contradiction. The bilinear estimate

$$\|B(x_n, y_n) - B(x, y)\| \leq \|B\| \big(\|x_n - x\| \, \|y_n\| + \|x\| \, \|y_n - y\| \big)$$

proves the third assertion. □

Theorem 2.3.2 (Mazur's theorem). If $x_n \rightharpoonup x$ in a normed space, then for every n there is a finite convex combination of the tail,

$$y_n \in \operatorname{conv}\{x_k : k \geq n\},$$

such that $\|y_n - x\| \to 0$.

Proof. Let $C_n = \overline{\operatorname{conv}\{x_k : k \geq n\}}^{\|\cdot\|}$. If $x \notin C_n$, the geometric Hahn–Banach theorem gives $f \in X^*$ and $\alpha \in \mathbb{R}$ with

$$\operatorname{Re} f(x) > \alpha \geq \operatorname{Re} f(z) \qquad (z \in C_n).$$

(Over a real space the real parts may of course be omitted.) But $f(x_k) \to f(x)$, contradicting this inequality for all $k \geq n$. Thus $x \in C_n$, and one may choose $y_n \in \operatorname{conv}\{x_k : k \geq n\}$ with $\|y_n - x\| < 1/n$. □

Corollary 2.3.3 (Banach–Saks–Mazur block form)**.** If $x_n \rightharpoonup x$, one can choose integers $1 \le m_1 < m_2 < \cdots$ and convex combinations

$$z_j \in \operatorname{conv}\{x_k : m_j \le k < m_{j+1}\}$$

such that $z_j \to x$ in norm. In particular, weak convergence always admits strongly convergent convex blocks.

Proof. Set $m_1 = 1$. Suppose m_j has been chosen. Mazur's theorem applied to the tail $(x_k)_{k \ge m_j}$ gives a finite convex combination

$$z_j = \sum_{k=m_j}^{N_j} \alpha_k^{(j)} x_k, \qquad \alpha_k^{(j)} \ge 0, \quad \sum_{k=m_j}^{N_j} \alpha_k^{(j)} = 1,$$

with

$$\|z_j - x\| < \frac{1}{j}.$$

Choose $m_{j+1} > N_j$. Then the blocks are disjoint and $z_j \in \operatorname{conv}\{x_k : m_j \le k < m_{j+1}\}$. The displayed estimate gives $z_j \to x$ in norm. ☐

Theorem 2.3.4 (Convex closures)**.** For a convex subset C of a normed space,

$$\overline{C}^{\|\cdot\|} = \overline{C}^{w}.$$

Hence a convex set is norm closed if and only if it is weakly closed.

Proof. The weak topology is coarser, so norm closure is contained in weak closure. If $x \notin \overline{C}^{\|\cdot\|}$, separate the closed convex set $\overline{C}^{\|\cdot\|}$ from x by a continuous linear functional: there are $f \in X^*$ and $\alpha \in \mathbb{R}$ such that

$$\operatorname{Re} f(x) > \alpha \ge \operatorname{Re} f(z) \qquad (z \in \overline{C}^{\|\cdot\|}).$$

Then

$$\{y \in X : \operatorname{Re} f(y) > \alpha\}$$

is a weakly open neighbourhood of x disjoint from C. Hence x is not in the weak closure, proving the reverse inclusion. ☐

Corollary 2.3.5. A convex norm-lower-semicontinuous function $\varphi\colon X \to (-\infty, \infty]$ is weakly lower semicontinuous.

Proof. Every sublevel set $\{x : \varphi(x) \le a\}$ is convex and norm closed, hence weakly closed. ☐

Theorem 2.3.6 (Weak continuity of linear maps)**.** Let E and F be Banach spaces and let $T\colon E \to F$ be linear. Then T is norm continuous if and only if it is continuous from $\sigma(E, E^*)$ to $\sigma(F, F^*)$.

Proof. Norm continuity implies weak continuity because $f \circ T \in E^*$ for every $f \in F^*$. Conversely, weak continuity makes the graph weakly closed in $E \times F$. A weakly closed set is norm closed, so the closed graph theorem implies that T is bounded. ☐

2.4 The Weak-Star Topology

Definition 2.4.1. The *weak-$*$ topology* $\sigma(E^*, E)$ on E^* is the initial topology generated by the evaluation maps

$$\phi_x(f) = f(x), \qquad x \in E.$$

Thus $f_\alpha \xrightarrow{*} f$ if and only if $f_\alpha(x) \to f(x)$ for every $x \in E$.

Proposition 2.4.2. The weak-$*$ topology is Hausdorff. A neighbourhood base at $f_0 \in E^*$ is given by

$$V(f_0; x_1, \dots, x_m; \varepsilon) = \{f \in E^* : |f(x_j) - f_0(x_j)| < \varepsilon,\ 1 \le j \le m\}.$$

Moreover,

$$\text{norm convergence} \Longrightarrow \text{weak convergence} \Longrightarrow \text{weak-}* \text{ convergence}$$

for nets in E^*, after identifying the weak topology as $\sigma(E^*, E^{**})$.

Proof. If $f \neq g$, then $f(x) \neq g(x)$ for some $x \in E$. Disjoint scalar neighbourhoods of these two values have disjoint inverse images under the evaluation map $h \mapsto h(x)$, so the weak-$*$ topology is Hausdorff. Its neighbourhood base is obtained by taking finite intersections of the evaluation subbasic sets, exactly as for the weak topology.

If $f_\alpha \to f$ in norm, then for every $F \in E^{**}$,

$$|F(f_\alpha - f)| \le \|F\|\,\|f_\alpha - f\| \longrightarrow 0,$$

so norm convergence implies weak convergence. If $f_\alpha \rightharpoonup f$ in E^*, apply the convergence to the canonical functional $J_E x \in E^{**}$:

$$f_\alpha(x) = J_E x(f_\alpha) \longrightarrow J_E x(f) = f(x) \qquad (x \in E).$$

Thus weak convergence implies weak-$*$ convergence. $\square$

Lemma 2.4.3 (Finite functional dependence)**.** Let $\ell, \ell_1, \dots, \ell_m$ be linear functionals on a vector space V. If

$$\bigcap_{j=1}^{m} \operatorname{Ker} \ell_j \subseteq \operatorname{Ker} \ell,$$

then ℓ belongs to $\operatorname{span}\{\ell_1, \dots, \ell_m\}$.

Proof. Define

$$L \colon V \to \mathbb{K}^m, \qquad L(v) = (\ell_1(v), \dots, \ell_m(v)).$$

If $L(v) = L(w)$, then $v - w \in \bigcap_j \operatorname{Ker} \ell_j \subseteq \operatorname{Ker} \ell$, so $\ell(v) = \ell(w)$. Consequently the formula

$$\widetilde{\ell}(L(v)) = \ell(v)$$

defines a well-defined linear functional on the subspace $L(V)$ of $\mathbb{K}^m$. Extend $\widetilde{\ell}$ to a linear functional on all of $\mathbb{K}^m$. There are scalars $a_1, \ldots, a_m$ such that

$$\widetilde{\ell}(z_1, \ldots, z_m) = \sum_{j=1}^{m} a_j z_j.$$

Therefore, for every $v \in V$,

$$\ell(v) = \widetilde{\ell}(L(v)) = \sum_{j=1}^{m} a_j \ell_j(v),$$

which proves $\ell \in \operatorname{span}\{\ell_1, \ldots, \ell_m\}$. □

Theorem 2.4.4 (Dual of the weak-$*$ dual)**.** A linear functional $\Lambda \colon E^* \to \mathbb{K}$ is weak-$*$ continuous if and only if there is an $x \in E$ such that

$$\Lambda(f) = f(x) \qquad (f \in E^*).$$

In other words, the continuous dual of $(E^*, \sigma(E^*, E))$ is the canonical copy of E.

Proof. For each $x \in E$, the evaluation map $\phi_x(f) = f(x)$ is weak-$*$ continuous by definition, so every finite linear combination of evaluations is weak-$*$ continuous.

Conversely, suppose Λ is weak-$*$ continuous. Continuity at the origin gives $x_1, \ldots, x_m \in E$ and $\varepsilon > 0$ such that

$$|f(x_j)| < \varepsilon \quad (1 \le j \le m) \qquad \Longrightarrow \qquad |\Lambda(f)| < 1.$$

Let

$$M = \bigcap_{j=1}^{m} \operatorname{Ker} \phi_{x_j} = \{f \in E^* : f(x_j) = 0 \text{ for every } j\}.$$

If $f \in M$, then tf belongs to the displayed neighbourhood for every scalar t. Hence

$$|t|\,|\Lambda(f)| = |\Lambda(tf)| < 1 \qquad (t \in \mathbb{K}),$$

which forces $\Lambda(f) = 0$. Thus $M \subseteq \operatorname{Ker} \Lambda$. Applying the finite functional-dependence lemma on the vector space E^* gives scalars $a_1, \ldots, a_m$ such that

$$\Lambda = \sum_{j=1}^{m} a_j \phi_{x_j}.$$

Therefore

$$\Lambda(f) = \sum_{j=1}^{m} a_j f(x_j) = f\left(\sum_{j=1}^{m} a_j x_j\right),$$

so Λ is evaluation at $x = \sum_j a_j x_j$. □

Corollary 2.4.5. A hyperplane in E^* is weak-$*$ closed if and only if it has the form

$$\{f \in E^* : f(x) = a\}$$

for some $x \in E$ and $a \in \mathbb{K}$.

Proof. Every set of the displayed form is the inverse image of the closed singleton $\{a\}$ under the weak-$*$ continuous evaluation $f \mapsto f(x)$, and hence is weak-$*$ closed.

Conversely, write the affine hyperplane as

$$H = \{f \in E^* : \Lambda(f) = a\},$$

where Λ is a nonzero algebraic linear functional on E^*. Choose $f_0 \notin H$. Since H is weak-$*$ closed, there is a balanced weak-$*$ neighbourhood W of 0 such that

$$(f_0 + W) \cap H = \varnothing.$$

Put $d = \Lambda(f_0) - a \neq 0$. We claim that $|\Lambda(h)| < |d|$ for every $h \in W$. Otherwise, for some $h \in W$ the scalar $s = -d/\Lambda(h)$ would satisfy $|s| \leq 1$. Balancedness gives $sh \in W$, while

$$\Lambda(f_0 + sh) = \Lambda(f_0) - d = a,$$

contradicting $(f_0 + W) \cap H = \varnothing$. Thus Λ is bounded on a neighbourhood of 0 and is therefore weak-$*$ continuous. By the theorem on the dual of $(E^*, \sigma(E^*, E))$, there is $x \in E$ such that $\Lambda(f) = f(x)$. Hence

$$H = \{f \in E^* : f(x) = a\},$$

as required. $\square$

2.5 Banach–Alaoglu, Goldstine, and reflexivity

Theorem 2.5.1 (Banach–Alaoglu theorem). The closed dual unit ball B_{E^*} is compact in the weak-$*$ topology.

Proof. For every $x \in E$, let

$$K_x = \{z \in \mathbb{K} : |z| \leq \|x\|\},$$

and form the product

$$K = \prod_{x \in E} K_x.$$

Each K_x is compact, so K is compact by Tychonoff's theorem. Define

$$\Phi\colon B_{E^*} \to K, \qquad \Phi(f) = (f(x))_{x \in E}.$$

The map is injective, and the subspace topology induced from the product is exactly the weak-$*$ topology, because the product coordinate projections are the evaluation maps $f \mapsto f(x)$.

We show that $\Phi(B_{E^*})$ is closed in K. An element $\omega = (\omega_x)_{x \in E} \in K$ belongs to the image precisely when it satisfies

$$\omega_{x+y} = \omega_x + \omega_y, \qquad \omega_{\lambda x} = \lambda \omega_x$$

for all $x, y \in E$ and scalars λ. Each of these is a closed coordinate condition in the product topology. Their intersection is therefore closed. If ω satisfies them, the formula

$$f_\omega(x) = \omega_x$$

defines a linear functional, and the fact that $\omega \in K$ gives

$$|f_\omega(x)| = |\omega_x| \le \|x\| \qquad (x \in E).$$

Thus $f_\omega \in B_{E^*}$ and $\Phi(f_\omega) = \omega$. Hence the closed intersection is exactly $\Phi(B_{E^*})$.

A closed subset of the compact space K is compact, so $\Phi(B_{E^*})$ is compact. Since Φ is a homeomorphism onto its image, B_{E^*} is weak-$*$ compact. $\square$

Definition 2.5.2. The canonical embedding $J\colon E \to E^{**}$ is

$$(Jx)(f) = f(x).$$

The space E is *reflexive* if $J(E) = E^{**}$.

Theorem 2.5.3 (Goldstine theorem)**.** The canonical image $J(B_E)$ is weak-$*$ dense in $B_{E^{**}}$.

Proof. Fix $x^{**} \in B_{E^{**}}$. A basic weak-$*$ neighbourhood of x^{**} is determined by $f_1, \dots, f_m \in E^*$ and $\varepsilon > 0$; we must find $x \in B_E$ such that

$$|f_j(x) - x^{**}(f_j)| < \varepsilon \qquad (1 \le j \le m).$$

Define

$$Q\colon E \to \mathbb{K}^m, \qquad Qx = (f_1(x), \dots, f_m(x)),$$

and put

$$y = (x^{**}(f_1), \dots, x^{**}(f_m)).$$

We claim that $y \in \overline{Q(B_E)}$.

Suppose not. The closed convex set $\overline{Q(B_E)}$ and the point y can be strictly separated in the finite-dimensional real vector space underlying $\mathbb{K}^m$. Thus there is a real-linear functional of the form

$$z \longmapsto \operatorname{Re} \sum_{j=1}^{m} a_j z_j$$

for which

$$\operatorname{Re} \sum_{j=1}^{m} a_j x^{**}(f_j) > \sup_{x \in B_E} \operatorname{Re} \sum_{j=1}^{m} a_j f_j(x).$$

Let $f = \sum_j a_j f_j$. The left side is $\operatorname{Re} x^{**}(f)$, while the right side is $\|f\|$: after multiplying a nearly norming vector by a unimodular scalar, the real part can be made arbitrarily close to $\|f\|$. We would therefore have

$$\operatorname{Re} x^{**}(f) > \|f\|,$$

contradicting $|x^{**}(f)| \le \|x^{**}\| \, \|f\| \le \|f\|$. Hence $y \in \overline{Q(B_E)}$.

Choose $x \in B_E$ with $\|Qx - y\|_\infty < \varepsilon$. Then the displayed coordinate inequalities hold, so the given weak-$*$ neighbourhood of x^{**} meets $J(B_E)$. This proves weak-$*$ density. $\square$

Theorem 2.5.4 (Kakutani's characterization). A Banach space E is reflexive if and only if its closed unit ball is weakly compact.

Proof. If E is reflexive, J identifies B_E with $B_{E^{**}}$, which is weak-$*$ compact by Banach–Alaoglu; under J this is the weak topology of E. Conversely, if B_E is weakly compact, then $J(B_E)$ is weak-$*$ compact and hence weak-$*$ closed in E^{**}. Goldstine says it is weak-$*$ dense in $B_{E^{**}}$, so $J(B_E) = B_{E^{**}}$. Scaling gives $J(E) = E^{**}$. $\square$

Theorem 2.5.5 (Eberlein–Šmulian theorem). A subset of a Banach space is relatively weakly compact if and only if every sequence in it has a weakly convergent subsequence whose limit belongs to its weak closure. In particular, a Banach space E is reflexive if and only if B_E is weakly sequentially compact.

The direction proved by the separable-reduction idea in the source. Assume that E is reflexive and let (x_n) be a sequence in B_E. Set

$$M_0 = \operatorname{span}\{x_n : n \geq 1\}, \qquad M = \overline{M_0}^{\|\cdot\|}.$$

Then M is separable. It is also reflexive, because a closed subspace of a reflexive Banach space is reflexive. Hence B_M is weakly compact by Kakutani's theorem.

To recover the metrizability step used in the handwritten proof, note first that separability of M makes B_{M^*} weak-$*$ compact and metrizable. Because M is reflexive, the weak and weak-$*$ topologies on B_{M^*} coincide. Thus B_{M^*} is a compact metric space and has a countable weakly dense subset (g_j). The norm-closed convex hull of (g_j) is all of B_{M^*}: otherwise Hahn–Banach would separate a point of B_{M^*} from that convex hull by an element of $M^{**} = M$, which is a weakly continuous functional and would contradict weak density. Therefore M^* is norm separable.

Choose a norm-dense sequence (f_j) in B_{M^*}. On B_M, the metric

$$d(u, v) = \sum_{j=1}^{\infty} 2^{-j} \frac{|f_j(u-v)|}{1 + |f_j(u-v)|}$$

induces the weak topology. Hence the weakly compact set B_M is compact and metrizable, and therefore sequentially compact. The sequence (x_n) thus has a subsequence converging weakly in M, and hence weakly in E. This proves the implication

$$E \text{ reflexive} \quad \Longrightarrow \quad B_E \text{ weakly sequentially compact}$$

by exactly the separable-subspace strategy of the notes.

The converse implication and the general subset statement constitute the deep half of the Eberlein–Šmulian theorem; the source does not contain the additional diagonal and closure arguments needed for a complete proof of that half. $\square$

Corollary 2.5.6. If E is reflexive, every closed bounded convex subset of E is weakly compact.

Proof. Such a set lies in a scalar multiple of B_E, which is weakly compact. It is weakly closed because convex norm-closed sets are weakly closed. $\square$

2.6 Separability and metrizability

Theorem 2.6.1. If E^* is separable, then E is separable.

Proof. Choose a norm-dense sequence (f_n) in S_{E^*}. For each n, choose $x_n \in S_E$ with $|f_n(x_n)| > 1/2$. Let $M = \overline{\operatorname{span}\{x_n : n \geq 1\}}$. If $M \neq E$, Hahn–Banach gives $f \in S_{E^*}$ with $f|_M = 0$. Choose f_n with $\|f - f_n\| < 1/2$. Then

$$|f_n(x_n)| = |(f_n - f)(x_n)| < 1/2,$$

a contradiction. Thus $M = E$, and rational linear combinations of the x_n form a countable dense set. □

Theorem 2.6.2. For a Banach space E, the following are equivalent:

(a). E is separable;

(b). B_{E^*} is metrizable in the weak-$*$ topology.

Proof. Assume first that E is separable and choose a norm-dense sequence (x_n) in B_E. Define, for $f, g \in B_{E^*}$,

$$d(f,g) = \sum_{n=1}^{\infty} 2^{-n} \frac{|(f-g)(x_n)|}{1 + |(f-g)(x_n)|}.$$

This is a metric. Indeed, if $d(f,g) = 0$, then f and g agree on every x_n, hence by continuity on B_E and therefore on all of E.

Weak-$*$ convergence implies convergence for d, because each summand tends to zero and the summands are dominated by 2^{-n}; a finite-head plus small-tail argument proves convergence of the series. Conversely, suppose $d(f_k, f) \to 0$ with $f_k, f \in B_{E^*}$. Then $f_k(x_n) \to f(x_n)$ for every n. Fix $x \in E$ and $\varepsilon > 0$. For $x \neq 0$, choose x_n such that

$$\left\| \frac{x}{\|x\|} - x_n \right\| < \frac{\varepsilon}{4\|x\|};$$

for $x = 0$ there is nothing to prove. Since $\|f_k - f\| \leq 2$,

$$\begin{aligned} |(f_k - f)(x)| &\leq |(f_k - f)(x - \|x\| x_n)| + \|x\| \, |(f_k - f)(x_n)| \\ &\leq 2\|x - \|x\| x_n\| + \|x\| \, |(f_k - f)(x_n)|. \end{aligned}$$

The first term is $< \varepsilon/2$, and the second is eventually $< \varepsilon/2$. Hence $f_k(x) \to f(x)$ for every $x \in E$. Thus d induces the weak-$*$ topology on B_{E^*}.

Conversely, suppose B_{E^*} is weak-$*$ metrizable. Banach–Alaoglu makes it a compact metric space. Therefore $C(B_{E^*})$, with the supremum norm, is separable. Define

$$J \colon E \to C(B_{E^*}), \qquad (Jx)(f) = f(x).$$

Weak-$*$ continuity of each evaluation shows $Jx \in C(B_{E^*})$, and Hahn–Banach gives

$$\|Jx\|_\infty = \sup_{\|f\| \leq 1} |f(x)| = \|x\|.$$

Thus J is an isometric embedding. Every subspace of a separable metric space is separable, so E is separable. □

Theorem 2.6.3. The weak topology on B_E is metrizable if and only if E^* is separable.

Proof. Suppose first that E^* is separable and choose a norm-dense sequence (f_n) in B_{E^*}. Define

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n} \frac{|f_n(x-y)|}{1+|f_n(x-y)|}, \qquad x, y \in B_E.$$

The function d is a metric because the dense family (f_n) separates points. Weak convergence in B_E implies d-convergence by dominated convergence for the series. Conversely, if $d(x_k, x) \to 0$, then $f_n(x_k - x) \to 0$ for every n. For arbitrary $f \in B_{E^*}$ and $\varepsilon > 0$, choose f_n with $\|f - f_n\| < \varepsilon/4$. Since $x_k, x \in B_E$,

$$\begin{aligned} |f(x_k - x)| &\le |(f - f_n)(x_k - x)| + |f_n(x_k - x)| \\ &\le 2\|f - f_n\| + |f_n(x_k - x)|, \end{aligned}$$

which is eventually $< \varepsilon$. By homogeneity the same holds for every $f \in E^*$, so $x_k \rightharpoonup x$. Thus d induces the weak topology on B_E.

Conversely, suppose the relative weak topology on B_E is metrizable. It is then first countable at 0, so choose a countable neighbourhood base (U_n) at 0 in B_E. For each n, choose a basic weak neighbourhood

$$V_n = \{x \in E : |f(x)| < \varepsilon_n \text{ for every } f \in F_n\}$$

with $F_n \subseteq E^*$ finite and

$$0 \in B_E \cap V_n \subseteq U_n.$$

Let

$$\mathcal{F} = \bigcup_{n=1}^{\infty} F_n, \qquad M = \overline{\operatorname{span} \mathcal{F}}^{\|\cdot\|} \subseteq E^*.$$

We prove $M = E^*$. Otherwise choose $g \in E^* \setminus M$. Hahn–Banach, applied in the Banach space E^*, gives $x^{**} \in E^{**}$ such that

$$x^{**}|_M = 0, \qquad x^{**}(g) \neq 0.$$

After normalization, assume $\|x^{**}\| \le 1$. Goldstine's theorem gives a net $(x_\alpha) \subseteq B_E$ such that

$$Jx_\alpha \overset{*}{\rightharpoonup} x^{**} \quad \text{in } E^{**}.$$

For every $f \in \mathcal{F}$, we have $f(x_\alpha) \to x^{**}(f) = 0$. Hence, for each n, the net is eventually in $B_E \cap V_n \subseteq U_n$. Since (U_n) is a neighbourhood base, this means $x_\alpha \rightharpoonup 0$ in E. In particular, $g(x_\alpha) \to 0$. Goldstine convergence, however, also gives

$$g(x_\alpha) = (Jx_\alpha)(g) \longrightarrow x^{**}(g) \neq 0,$$

a contradiction. Thus $M = E^*$, and the countable set $\mathcal{F}$ has norm- dense linear span. Therefore E^* is separable. □

Corollary 2.6.4. If E is separable, then every bounded sequence in E^* has a weak-$*$ convergent subsequence.

Proof. After scaling, the sequence lies in B_{E^*}, which is weak-$*$ compact and metrizable. Compact metric spaces are sequentially compact. $\square$

2.7 Uniform convexity and reflexivity

Definition 2.7.1. A Banach space E is *uniformly convex* if, for every $\varepsilon > 0$, there is $\delta > 0$ such that

$$\|x\|, \|y\| \le 1, \quad \|x - y\| \ge \varepsilon \quad \Longrightarrow \quad \left\| \frac{x+y}{2} \right\| \le 1 - \delta.$$

Proposition 2.7.2 (Radon–Riesz property)**.** Let E be uniformly convex. If $x_n \rightharpoonup x$ and $\|x_n\| \to \|x\|$, then $x_n \to x$ in norm.

Proof. The case $x = 0$ is immediate. Otherwise normalize $y_n = x_n/\|x_n\|$ and $y = x/\|x\|$. Then $y_n \rightharpoonup y$ and $\|y_n\| = \|y\| = 1$. If a subsequence satisfied $\|y_n - y\| \ge \varepsilon$, uniform convexity would give $\|(y_n + y)/2\| \le 1 - \delta$. But weak lower semicontinuity applied to $(y_n + y)/2 \rightharpoonup y$ gives $1 \le \liminf \|(y_n + y)/2\|$, a contradiction. $\square$

Theorem 2.7.3 (Milman–Pettis theorem)**.** Every uniformly convex Banach space is reflexive.

Proof. Let $x^{**} \in S_{E^{**}}$. By Goldstine there is a net $(x_\alpha) \subseteq B_E$ with $Jx_\alpha \overset{*}{\rightharpoonup} x^{**}$. Fix $\varepsilon > 0$ and choose the corresponding uniform-convexity constant $\delta > 0$. After multiplying a norming functional by a unimodular scalar, choose $f \in S_{E^*}$ with $\operatorname{Re} x^{**}(f) > 1 - \delta/2$. Eventually $\operatorname{Re} f(x_\alpha) > 1 - \delta$, and hence for large α, β,

$$\left\| \frac{x_\alpha + x_\beta}{2} \right\| \ge \operatorname{Re} f\left(\frac{x_\alpha + x_\beta}{2} \right) > 1 - \delta.$$

Uniform convexity gives $\|x_\alpha - x_\beta\| < \varepsilon$. Thus the net is norm Cauchy and converges to some $x \in B_E$. Since norm convergence implies weak-$*$ convergence under J, uniqueness of the weak-$*$ limit gives $Jx = x^{**}$. Scaling proves that J is surjective. $\square$

Chapter 3 Lebesgue Spaces

Throughout this chapter, $\Omega \subseteq \mathbb{R}^d$ is measurable and all L^p spaces are taken with respect to Lebesgue measure unless stated otherwise. For $1 < p < \infty$, the conjugate exponent is $p' = p/(p-1)$.

3.1 Uniform convexity, reflexivity, and duality

Theorem 3.1.1. For $1 < p < \infty$, the space $L^p(\Omega)$ is uniformly convex and therefore reflexive.

Proof. Let $f, g \in L^p(\Omega)$ with $\|f\|_p, \|g\|_p \leq 1$ and $\|f - g\|_p \geq \varepsilon$. When $2 \leq p < \infty$, Clarkson's inequality is

$$\left\|\frac{f+g}{2}\right\|_p^p + \left\|\frac{f-g}{2}\right\|_p^p \leq \frac{\|f\|_p^p + \|g\|_p^p}{2} \leq 1.$$

Consequently,

$$\left\|\frac{f+g}{2}\right\|_p^p \leq 1 - \left(\frac{\varepsilon}{2}\right)^p,$$

and hence

$$\left\|\frac{f+g}{2}\right\|_p \leq 1 - \delta_p(\varepsilon), \qquad \delta_p(\varepsilon) := 1 - (1 - (\varepsilon/2)^p)^{1/p} > 0.$$

For $1 < p \leq 2$, put $q = p' = p/(p-1)$. The second Clarkson inequality reads

$$\|f+g\|_p^q + \|f-g\|_p^q \leq 2\big(\|f\|_p^p + \|g\|_p^p\big)^{q/p}.$$

After division by 2^q, the assumptions $\|f\|_p, \|g\|_p \leq 1$ give

$$\left\|\frac{f+g}{2}\right\|_p^q + \left\|\frac{f-g}{2}\right\|_p^q \leq 1.$$

Thus the same argument, now with exponent q, gives

$$\left\|\frac{f+g}{2}\right\|_p \leq (1 - (\varepsilon/2)^q)^{1/q} < 1.$$

This is uniform convexity. The Milman–Pettis theorem therefore implies that $L^p(\Omega)$ is reflexive. □

Theorem 3.1.2 (Riesz representation for L^p)**.** Let $1 < p < \infty$. For each $g \in L^{p'}(\Omega)$, the formula

$$T_g(f) = \int_\Omega fg \, \mathrm{d}x$$

defines an element of $(L^p(\Omega))^*$, and

$$\|T_g\| = \|g\|_{p'}.$$

Every element of $(L^p(\Omega))^*$ is of this form. Hence

$$(L^p(\Omega))^* \cong L^{p'}(\Omega)$$

isometrically.

Proof. For $g \in L^{p'}(\Omega)$, Hölder's inequality gives

$$|T_g(f)| \le \|f\|_p \|g\|_{p'}, \qquad f \in L^p(\Omega),$$

so $\|T_g\| \le \|g\|_{p'}$. If $g \neq 0$, define

$$f_g(x) = \frac{|g(x)|^{p'-2}\overline{g(x)}}{\|g\|_{p'}^{p'-1}},$$

with the value 0 where $g = 0$. Since $(p'-1)p = p'$, one has $\|f_g\|_p = 1$, and

$$T_g(f_g) = \frac{1}{\|g\|_{p'}^{p'-1}} \int_\Omega |g|^{p'} \, \mathrm{d}x = \|g\|_{p'}.$$

Thus $\|T_g\| = \|g\|_{p'}$.

Let

$$J \colon L^{p'}(\Omega) \longrightarrow (L^p(\Omega))^*, \qquad Jg = T_g.$$

The map J is an isometry, so $J(L^{p'})$ is norm closed. Suppose that $\Phi \in (L^p)^{**}$ annihilates this range. Since L^p is reflexive, there is $f \in L^p$ such that $\Phi = J_{L^p} f$. Therefore

$$0 = \Phi(Jg) = Jg(f) = \int_\Omega fg \, \mathrm{d}x \qquad (g \in L^{p'}).$$

Taking

$$g = |f|^{p-2}\overline{f}$$

(with value 0 at the zeros of f) is legitimate because $|g|^{p'} = |f|^p$. It gives

$$0 = \int_\Omega |f|^p \, \mathrm{d}x,$$

so $f = 0$. Hence the annihilator of $J(L^{p'})$ in $(L^p)^{**}$ is trivial. A proper closed linear subspace of a normed space has a nonzero annihilator by Hahn–Banach; therefore $J(L^{p'}) = (L^p)^*$. □

Theorem 3.1.3 (The endpoint duality theorem)**.** For a σ-finite measure space, and in particular for Lebesgue measure on $\Omega \subseteq \mathbb{R}^d$,

$$(L^1(\Omega))^* \cong L^\infty(\Omega)$$

isometrically through the same pairing $T_g(f) = \int_\Omega fg$.

Proof. For $g \in L^\infty(\Omega)$, Hölder's inequality gives

$$\left| \int_\Omega fg \, \mathrm{d}x \right| \le \|f\|_1 \|g\|_\infty,$$

so $g \mapsto T_g$ is a linear contraction from L^∞ into $(L^1)^*$. We first prove surjectivity and then verify that the norm is preserved.

Let $T \in (L^1(\Omega))^*$. Since the measure space is σ-finite, choose measurable sets

$$E_1 \subseteq E_2 \subseteq \cdots, \qquad \mu(E_n) < \infty, \qquad \bigcup_{n=1}^{\infty} E_n = \Omega.$$

For a measurable set $A \subseteq E_n$, define

$$\nu_n(A) = T(\mathbf{1}_A).$$

If (A_k) are pairwise disjoint subsets of E_n, then

$$\mathbf{1}_{\cup_{k=1}^{m} A_k} \longrightarrow \mathbf{1}_{\cup_{k=1}^{\infty} A_k} \quad \text{in } L^1$$

because the omitted tail has measure tending to zero. Continuity of T therefore shows that ν_n is countably additive. Moreover,

$$|\nu_n(A)| \leq \|T\| \, \|\mathbf{1}_A\|_1 = \|T\| \, \mu(A),$$

so $\nu_n \ll \mu|_{E_n}$. The Radon–Nikodym theorem gives a measurable function g_n on E_n such that

$$\nu_n(A) = \int_A g_n \, \mathrm{d}\mu \qquad (A \subseteq E_n \text{ measurable}).$$

The displayed domination implies $|g_n| \leq \|T\|$ almost everywhere. Indeed, if $|g_n| > \|T\| + \varepsilon$ on a set of positive measure, restricting to a subset on which the complex phase of g_n lies in a sufficiently short arc (and using the sign in the real case) would contradict $|\nu_n(A)| \leq \|T\|\mu(A)$.

For $m > n$, the measures represented by g_m and g_n agree on every subset of E_n. Uniqueness in the Radon–Nikodym theorem gives $g_m = g_n$ almost everywhere on E_n. The functions therefore patch to a measurable g on Ω satisfying

$$\|g\|_\infty \leq \|T\|.$$

If s is a simple function supported in some E_n, linearity gives

$$T(s) = \int_\Omega sg \, \mathrm{d}\mu.$$

Such simple functions are dense in $L^1(\Omega)$. If $s_k \to f$ in L^1, then

$$T(s_k) \to T(f) \quad \text{and} \quad \int s_k g \to \int fg,$$

so $T(f) = \int fg$ for every $f \in L^1$.

It remains to prove equality of norms. We already know $\|T_g\| \leq \|g\|_\infty$. Put $M = \|g\|_\infty$ and let $\varepsilon > 0$. If $M > 0$, the set $\{|g| > M - \varepsilon\}$ has positive measure. Its intersection with some E_n contains a measurable subset A with $0 < \mu(A) < \infty$. Define

$$f = \frac{\mathbf{1}_A \, \overline{\operatorname{sgn} g}}{\mu(A)}, \qquad \operatorname{sgn} g = \begin{cases} g/|g|, & g \neq 0, \\ 0, & g = 0. \end{cases}$$

Then $\|f\|_1 = 1$ and

$$|T_g(f)| = \frac{1}{\mu(A)} \int_A |g| \, \mathrm{d}\mu \geq M - \varepsilon.$$

Letting $\varepsilon \downarrow 0$ yields $\|T_g\| \geq M$. Hence the identification is isometric. □

Remark 3.1.4. Except in the finite-dimensional case (equivalently, when the measure algebra has only finitely many atoms), $L^1(\Omega)$ is not reflexive. Indeed, if it were reflexive then its dual $L^\infty(\Omega)$ would also be reflexive, which fails on every genuinely infinite measure algebra. Likewise $L^\infty(\Omega)$ is nonreflexive except in the same finite-dimensional situation.

3.2 Separability and dense classes

Theorem 3.2.1. For $1 \le p < \infty$, the space $L^p(\mathbb{R}^d)$ is separable. The space $L^\infty(\mathbb{R}^d)$ is not separable.

Proof. Finite linear combinations, with rational coefficients, of indicators of rectangles with rational endpoints form a countable set. Simple-function approximation and regularity of Lebesgue measure show that this set is dense in L^p for $p < \infty$. For L^∞, choose infinitely many pairwise measurably distinguishable sets. More explicitly, for $t \in \mathbb{R}$ let

$$A_t = (-\infty, t) \times \mathbb{R}^{d-1}.$$

If $s \neq t$, then $A_s \triangle A_t$ has positive measure, and therefore

$$\|\mathbf{1}_{A_s} - \mathbf{1}_{A_t}\|_\infty = 1.$$

Thus $L^\infty(\mathbb{R}^d)$ contains an uncountable 1-separated set and cannot be separable. □

Theorem 3.2.2. For $1 \le p < \infty$, $C_c(\mathbb{R}^d)$ and $C_c^\infty(\mathbb{R}^d)$ are dense in $L^p(\mathbb{R}^d)$.

Proof. Fix $f \in L^p(\mathbb{R}^d)$ and $\varepsilon > 0$. Truncating first in space and then in size gives

$$f_{R,M} = \mathbf{1}_{B(0,R)} \max\{-M, \min\{f, M\}\}$$

in the real case, with the analogous radial truncation in the complex case, such that $\|f - f_{R,M}\|_p < \varepsilon/3$ for large R and M. The bounded, finite-support function $f_{R,M}$ can be approximated in L^p by a simple function

$$s = \sum_{j=1}^{m} a_j \mathbf{1}_{E_j},$$

where every E_j is measurable, bounded, and has finite measure.

It remains to approximate one indicator. By inner and outer regularity of Lebesgue measure, for any $\eta > 0$ there are a compact set $K \subseteq E$ and an open bounded set $O \supseteq E$ such that

$$|O \setminus K| < \eta.$$

Choose $\phi \in C_c(\mathbb{R}^d)$ with

$$0 \le \phi \le 1, \qquad \phi = 1 \text{ on } K, \qquad \operatorname{supp}\phi \subseteq O.$$

For instance, one may use the distance functions to K and to $\mathbb{R}^d \setminus O$. Since $\mathbf{1}_E$ and ϕ can differ only on $O \setminus K$,

$$\|\mathbf{1}_E - \phi\|_p^p \le |O \setminus K| < \eta.$$

Choosing the errors for the finitely many E_j sufficiently small produces $v \in C_c(\mathbb{R}^d)$ with $\|s - v\|_p < \varepsilon/3$. This proves density of C_c.

Finally let ρ_δ be a mollifier. If $v \in C_c$, then $\rho_\delta * v \in C_c^\infty$, its support stays in a fixed compact set for small δ, and uniform continuity gives

$$\|\rho_\delta * v - v\|_\infty \longrightarrow 0.$$

The fixed finite support then gives convergence in L^p. Combining the three approximations proves density of C_c^∞ as well. □

3.3 Convolution and mollifiers

Theorem 3.3.1 (Young's convolution inequality). Let $1 \le p \le \infty$, $f \in L^1(\mathbb{R}^d)$, and $g \in L^p(\mathbb{R}^d)$. Then

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x-y)g(y)\, \mathrm{d}y$$

is defined for almost every x, belongs to L^p, and satisfies

$$\|f * g\|_p \le \|f\|_1 \|g\|_p.$$

Proof. For $p = 1$, Tonelli's theorem and translation invariance give

$$\begin{aligned} \|f * g\|_1 &\le \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} |f(x-y)|\,|g(y)|\, \mathrm{d}y\, \mathrm{d}x \\ &= \|f\|_1\|g\|_1. \end{aligned}$$

For $p = \infty$, for almost every x,

$$|(f*g)(x)| \le \int |f(x-y)|\,|g(y)|\, \mathrm{d}y \le \|g\|_\infty \|f\|_1.$$

For $1 < p < \infty$, Hölder's inequality with respect to the measure $|f(x-y)|\, \mathrm{d}y$ gives

$$|(f*g)(x)|^p \le \|f\|_1^{p-1} \int |f(x-y)|\,|g(y)|^p\, \mathrm{d}y.$$

Integrating in x and applying Tonelli yields the result. □

Proposition 3.3.2 (Differentiation of convolution). If $f \in C_c^k(\mathbb{R}^d)$ and $g \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, then $f * g \in C^k$ wherever the convolution is defined, and

$$D^\alpha (f * g) = (D^\alpha f) * g \qquad (|\alpha| \le k).$$

Proof. Fix a compact set $K \subset \mathbb{R}^d$. If $x \in K$ and $f(x-y) \neq 0$, then

$$y \in x - \operatorname{supp} f \subseteq K - \operatorname{supp} f,$$

which is compact. Thus all relevant values of g lie in one fixed compact set on which g is integrable. For a coordinate vector e_j,

$$\frac{(f*g)(x+he_j) - (f*g)(x)}{h} = \int_{\mathbb{R}^d} \frac{f(x+he_j-y) - f(x-y)}{h} g(y)\, \mathrm{d}y.$$

For small h, the difference quotient is supported in one fixed compact set and is bounded there by a constant multiple of $|g(y)|$, by the mean-value theorem. Dominated convergence therefore gives

$$\partial_j(f * g)(x) = ((\partial_j f) * g)(x).$$

The same domination, now applied to differences of the last integral, gives continuity in x. Iterating the argument proves the formula for every $|\alpha| \leq k$. □

Definition 3.3.3. A sequence (ρ_n) is an *approximate identity* or a sequence of *mollifiers* if

$$\rho_n \in C_c^\infty(\mathbb{R}^d), \qquad \rho_n \geq 0, \qquad \int \rho_n = 1, \qquad \operatorname{supp} \rho_n \subseteq B(0, 1/n).$$

A standard choice is $\rho_n(x) = n^d \rho(nx)$ for a fixed nonnegative $\rho \in C_c^\infty(B(0,1))$ with integral one.

Proposition 3.3.4. Let (ρ_n) be mollifiers.

(a). If $f \in C_c(\mathbb{R}^d)$, then $\rho_n * f \to f$ uniformly.

(b). If $f \in L^p(\mathbb{R}^d)$ with $1 \leq p < \infty$, then $\rho_n * f \to f$ in L^p.

Proof. For continuous compactly supported f,

$$|(\rho_n * f)(x) - f(x)| \leq \int \rho_n(y) |f(x-y) - f(x)| \, dy,$$

and uniform continuity makes the right-hand side tend uniformly to zero. The L^p statement follows by approximating f by a function in C_c, using Young's inequality for the two error terms. □

Corollary 3.3.5 (Vanishing weak gradient)**.** Let $\Omega \subseteq \mathbb{R}^d$ be open and connected, and let $u \in L^1_{\mathrm{loc}}(\Omega)$. If

$$\int_\Omega u \, \partial_j \varphi \, dx = 0 \qquad (\varphi \in C_c^\infty(\Omega),\ 1 \leq j \leq d),$$

then u is almost everywhere equal to a constant on Ω.

Proof. Fix a ball $B(x_0, 2r) \Subset \Omega$ and extend u arbitrarily outside $B(x_0, 2r)$ for the purpose of convolution. For $0 < \varepsilon < r$, put $u_\varepsilon = \rho_\varepsilon * u$ on $B(x_0, r)$. For $1 \leq j \leq d$ and $x \in B(x_0, r)$,

$$\partial_j u_\varepsilon(x) = \int u(y) \, \partial_j \rho_\varepsilon(x-y) \, dy = 0,$$

because $y \mapsto \rho_\varepsilon(x-y)$ is a test function supported in $B(x_0, 2r)$. Hence u_ε is constant on the connected ball $B(x_0, r)$; write this constant as c_ε.

Choose a smaller ball $B_0 \Subset B(x_0, r)$ of positive measure. Since $u_\varepsilon \to u$ in $L^1(B_0)$, the constants (c_ε) form a Cauchy net:

$$|B_0| \, |c_\varepsilon - c_\delta| = \|c_\varepsilon - c_\delta\|_{L^1(B_0)} \leq \|u_\varepsilon - u\|_{L^1(B_0)} + \|u - u_\delta\|_{L^1(B_0)}.$$

Thus $c_\varepsilon \to c$, and $u = c$ almost everywhere on B_0, hence on $B(x_0, r)$ by the same L^1 convergence. We have proved local constancy on every compactly contained ball. Along a finite chain of overlapping balls, the constants agree on intersections of positive measure. Since an open connected subset of $\mathbb{R}^d$ is path connected by polygonal chains of balls, one constant propagates throughout Ω. □

Chapter 4 Hilbert Spaces

Throughout the complex case, inner products are taken linear in the first variable and conjugate linear in the second. All formulas below use this convention.

4.1 Orthogonality and direct sums

Proposition 4.1.1 (Orthogonal density criterion). Let H be a Hilbert space and let $M \subseteq H$ be a linear subspace. Then

$$\overline{M} = H \quad \Longleftrightarrow \quad M^{\perp} = \{0\}.$$

Proof. If $\overline{M} = H$, continuity of the inner product makes every vector orthogonal to M orthogonal to all of H, hence zero. Conversely, if $\overline{M} \neq H$, the projection theorem gives a nonzero component of some $x \notin \overline{M}$ in $(\overline{M})^{\perp} = M^{\perp}$. □

Proposition 4.1.2. For any nonempty subset S of an inner-product space,

$$S^{\perp} = \{x : \langle x, s \rangle = 0 \text{ for every } s \in S\}$$

is a closed linear subspace. In a Hilbert space,

$$S^{\perp} = (\overline{\operatorname{span} S})^{\perp}, \qquad S^{\perp\perp} = \overline{\operatorname{span} S}.$$

Moreover, $S \cap S^{\perp} = \{0\}$ whenever S is a subspace.

Proof. For each fixed $s \in S$, the map $x \mapsto \langle x, s \rangle$ is continuous and linear (or conjugate linear, according to the chosen convention). Hence

$$S^{\perp} = \bigcap_{s \in S} \operatorname{Ker}\big(x \mapsto \langle x, s \rangle\big)$$

is a closed linear subspace. Orthogonality to S is equivalent, by linearity and continuity, to orthogonality to $\overline{\operatorname{span} S}$, proving the first identity.

Put $M = \overline{\operatorname{span} S}$. Clearly $M \subseteq M^{\perp\perp}$. If $x \notin M$, the projection theorem gives

$$x = P_M x + (x - P_M x), \qquad 0 \neq x - P_M x \in M^{\perp}.$$

Thus x is not orthogonal to every element of $M^{\perp}$, because

$$\langle x, x - P_M x \rangle = \|x - P_M x\|^2 \neq 0.$$

Hence $x \notin M^{\perp\perp}$ and $M^{\perp\perp} = M$. Finally, if $x \in S \cap S^{\perp}$ and S is a subspace, then $\|x\|^2 = \langle x, x \rangle = 0$, so $x = 0$. □

Theorem 4.1.3 (Orthogonal direct-sum theorem). Let M be a closed linear subspace of a Hilbert space H. Then

$$H = M \oplus M^{\perp}.$$

Thus every $x \in H$ has a unique decomposition

$$x = P_M x + (x - P_M x), \qquad P_M x \in M, \quad x - P_M x \in M^{\perp}.$$

Proof. Apply the projection theorem to the closed convex set M. For $x \in H$, let $y = P_M x$. Since M is a linear subspace, the variational inequality may be tested with $y \pm v$ and, over $\mathbb{C}$, with $y \pm iv$, where $v \in M$. It follows that

$$\langle x - y, v \rangle = 0 \qquad (v \in M),$$

so $x - y \in M^{\perp}$. Hence $H = M + M^{\perp}$.
The sum is direct: if $z \in M \cap M^{\perp}$, then $\|z\|^2 = \langle z, z \rangle = 0$. Thus the decomposition is unique. Equivalently, if $x = y_1 + z_1 = y_2 + z_2$ with $y_i \in M$ and $z_i \in M^{\perp}$, then $y_1 - y_2 = z_2 - z_1 \in M \cap M^{\perp} = \{0\}$. □

4.2 Inner products and projections

Definition 4.2.1. An inner product on a real vector space is a symmetric positive-definite bilinear form. On a complex vector space it is a positive-definite sesquilinear form, linear in the first variable and conjugate linear in the second. The induced norm is

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

A complete inner-product space is a *Hilbert space*.

Proposition 4.2.2. Every inner product satisfies the Cauchy–Schwarz inequality

$$|\langle x, y \rangle| \le \|x\| \, \|y\|$$

and the parallelogram identity

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

Conversely, a norm comes from an inner product if and only if it satisfies the parallelogram identity; the inner product is recovered by polarization.

Theorem 4.2.3 (Projection theorem). Let K be a nonempty closed convex subset of a Hilbert space H. For every $f \in H$ there is a unique $u \in K$ satisfying

$$\|f - u\| = \operatorname{dist}(f, K).$$

It is characterized by the variational inequality

$$\operatorname{Re}\langle f-u, v-u\rangle \leq 0 \qquad (v \in K).$$

We write $u = P_K f$.

Proof. Let $d = \operatorname{dist}(f, K)$ and choose a minimizing sequence $(u_n) \subseteq K$. Convexity gives $(u_n + u_m)/2 \in K$. The parallelogram identity yields

$$\|u_n - u_m\|^2 = 2\|f-u_n\|^2 + 2\|f-u_m\|^2 - 4\left\|f - \frac{u_n+u_m}{2}\right\|^2,$$

which tends to zero. Thus (u_n) is Cauchy and converges to some $u \in K$; continuity gives $\|f-u\| = d$. For $v \in K$ and $0 < t \leq 1$, the point $u + t(v-u)$ lies in K. Minimality implies

$$\|f-u\|^2 \leq \|f-u-t(v-u)\|^2 = \|f-u\|^2 - 2t\operatorname{Re}\langle f-u, v-u\rangle + t^2\|v-u\|^2.$$

Divide by t and let $t \downarrow 0$ to obtain the inequality. Conversely, the inequality gives, for $v \in K$,

$$\|f-v\|^2 = \|f-u\|^2 + \|u-v\|^2 + 2\operatorname{Re}\langle f-u, u-v\rangle \geq \|f-u\|^2.$$

If u_1, u_2 both satisfy the variational inequalities, use $v = u_2$ in the first and $v = u_1$ in the second; adding gives $\|u_1 - u_2\|^2 \leq 0$. □

Proposition 4.2.4 (Nonexpansiveness of metric projection)**.** For every nonempty closed convex $K \subseteq H$,

$$\|P_K f - P_K g\| \leq \|f-g\| \qquad (f, g \in H).$$

Proof. Apply the variational inequality to $P_K f$ with competitor $P_K g$ and to $P_K g$ with competitor $P_K f$. Adding gives

$$\|P_K f - P_K g\|^2 \leq \operatorname{Re}\langle P_K f - P_K g, f-g\rangle \leq \|P_K f - P_K g\|\,\|f-g\|.$$

□

Corollary 4.2.5 (Orthogonal projections)**.** If M is a closed linear subspace, then P_M is linear, bounded, and $\|P_M\| \leq 1$. It is characterized by

$$P_M f \in M, \qquad f - P_M f \in M^\perp.$$

If $M \neq \{0\}$, then $\|P_M\| = 1$.

Proof. The variational inequality may be tested with $u \pm v$ and, in the complex case, $u \pm iv$ for $v \in M$, so it reduces to orthogonality. Uniqueness of the orthogonal decomposition gives linearity. Nonexpansiveness gives the norm bound. □

4.3 Riesz representation

Theorem 4.3.1 (Riesz representation theorem)**.** For every $\ell \in H^*$ there is a unique $y \in H$ such that

$$\ell(x) = \langle x, y \rangle \qquad (x \in H)$$

(with the variables adjusted to the chosen complex convention), and $\|\ell\| = \|y\|$.

Proof. If $\ell = 0$, take $y = 0$. Otherwise $M = \operatorname{Ker} \ell$ is a closed proper subspace. Choose $0 \neq z \in M^{\perp}$. For each $x \in H$, the vector

$$x - \frac{\ell(x)}{\ell(z)} z$$

belongs to M, hence is orthogonal to z. Therefore

$$\ell(x) = \frac{\ell(z)}{\|z\|^2} \langle x, z \rangle ,$$

which has the required form. Cauchy–Schwarz gives $\|\ell\| \leq \|y\|$, while testing at $x = y/\|y\|$ gives equality.

□

4.4 Orthogonal sums and orthonormal bases

Definition 4.4.1. Let $(E_n)_{n\geq 1}$ be closed subspaces of a Hilbert space H. Their Hilbert direct sum is

$$\bigoplus_{n=1}^{\infty} E_n = \left\{ (u_n) : u_n \in E_n, \sum_{n=1}^{\infty} \|u_n\|^2 < \infty \right\}$$

with inner product $\langle (u_n), (v_n) \rangle = \sum_n \langle u_n, v_n \rangle$.

Lemma 4.4.2 (Orthogonal series)**.** If (v_n) is an orthogonal sequence in H and $\sum_n \|v_n\|^2 < \infty$, then $\sum_n v_n$ converges in H and

$$\left\| \sum_{n=1}^{\infty} v_n \right\|^2 = \sum_{n=1}^{\infty} \|v_n\|^2.$$

Proof. For $m > n$, orthogonality gives

$$\left\| \sum_{k=n+1}^{m} v_k \right\|^2 = \sum_{k=n+1}^{m} \|v_k\|^2,$$

so the partial sums are Cauchy. The norm identity follows by passage to the limit. □

Theorem 4.4.3 (Projections onto mutually orthogonal subspaces)**.** Let (E_n) be mutually orthogonal closed subspaces and let $P_n = P_{E_n}$. For all $u \in H$,

$$\sum_{n=1}^{\infty} \|P_n u\|^2 \leq \|u\|^2,$$

and the series $\sum_n P_n u$ converges. If $E = \overline{\operatorname{span} \bigcup_n E_n}$, then

$$u = P_E u = \sum_{n=1}^{\infty} P_n u, \qquad \|P_E u\|^2 = \sum_{n=1}^{\infty} \|P_n u\|^2.$$

Proof. Finite orthogonality gives

$$\left\| \sum_{n=1}^{N} P_n u \right\|^2 = \sum_{n=1}^{N} \|P_n u\|^2 = \left\langle u, \sum_{n=1}^{N} P_n u \right\rangle \leq \|u\| \left\| \sum_{n=1}^{N} P_n u \right\|.$$

This yields Bessel's bound and convergence. The limit lies in E, and the difference from u is orthogonal to every E_n, hence to E; it is therefore $P_E u$. □

Definition 4.4.4. A sequence (e_n) is an *orthonormal basis* of H if

$$\langle e_n, e_m \rangle = \delta_{nm} \qquad \text{and} \qquad \overline{\operatorname{span}\{e_n : n \geq 1\}} = H.$$

Theorem 4.4.5 (Fourier expansion and Parseval identity)**.** If (e_n) is an orthonormal basis, then for every $u \in H$,

$$u = \sum_{n=1}^{\infty} \langle u, e_n \rangle \, e_n, \qquad \|u\|^2 = \sum_{n=1}^{\infty} |\langle u, e_n \rangle|^2.$$

Conversely, for every $(a_n) \in \ell^2$ the series $\sum_n a_n e_n$ converges to a unique $u \in H$ satisfying

$$\langle u, e_n \rangle = a_n, \qquad \|u\|^2 = \sum_n |a_n|^2.$$

Proof. Apply the theorem on mutually orthogonal subspaces to $E_n = \operatorname{span}\{e_n\}$. The orthogonal projection onto E_n is

$$P_n u = \langle u, e_n \rangle \, e_n$$

for the convention in which the first variable is linear. Since the closed linear span of the e_n is all of H, the preceding projection theorem gives

$$u = \sum_{n=1}^{\infty} P_n u = \sum_{n=1}^{\infty} \langle u, e_n \rangle \, e_n$$

in norm, and its norm identity is precisely Parseval's formula.

Conversely, let $(a_n) \in \ell^2$ and put $v_n = a_n e_n$. The vectors v_n are orthogonal and

$$\sum_n \|v_n\|^2 = \sum_n |a_n|^2 < \infty.$$

The orthogonal-series lemma therefore gives a vector $u = \sum_n a_n e_n \in H$ and

$$\|u\|^2 = \sum_n |a_n|^2.$$

Continuity of the inner product and orthonormality yield $\langle u, e_k \rangle = a_k$. If another vector has the same coefficients, its difference from u is orthogonal to every e_k and hence to their dense linear span; the orthogonal density criterion makes that difference zero. □

Theorem 4.4.6. Every separable Hilbert space has a countable orthonormal basis.

Proof. Let (x_n) be a countable dense subset of H. Starting with the first nonzero x_n, discard each later vector that already belongs to the span of the retained predecessors. This leaves a finite or countable linearly independent sequence (y_n) whose linear span contains every x_n and is therefore dense in H. Apply Gram–Schmidt:

$$z_n = y_n - \sum_{k<n} \langle y_n, e_k \rangle \, e_k, \qquad e_n = \frac{z_n}{\|z_n\|}.$$

Linear independence ensures $z_n \neq 0$. Inductively, the e_n are orthonormal and

$$\operatorname{span}\{e_1, \dots, e_n\} = \operatorname{span}\{y_1, \dots, y_n\}.$$

Thus their total linear span is dense in H, so they form an orthonormal basis. $\square$

Chapter 5 Spectral Theory

Throughout this chapter, scalar fields are complex whenever the spectrum of an operator is discussed.

5.1 Closed operators, resolvents, and spectra

Definition 5.1.1. Let X be a complex Banach space and let $T\colon \mathrm{D}(T) \subseteq X \to X$ be linear. A number $\lambda \in \mathbb{C}$ is a *regular value* if

$$T - \lambda I\colon \mathrm{D}(T) \to X$$

is bijective and its inverse

$$R_\lambda(T) = (T - \lambda I)^{-1}\colon X \to X$$

is bounded. The *resolvent set* $\rho(T)$ is the set of regular values, and the *spectrum* is

$$\sigma(T) = \mathbb{C} \setminus \rho(T).$$

The spectrum is partitioned into:

- the point spectrum, where $T - \lambda I$ is not injective;
- the continuous spectrum, where it is injective and has dense but non-surjective range;
- the residual spectrum, where it is injective but its range is not dense.

Proposition 5.1.2. If T is closed and $T - \lambda I$ is bijective, then $R_\lambda(T)$ is automatically bounded. In particular, for a closed operator the definition of $\rho(T)$ may omit boundedness of the inverse.

Proof. The inverse has graph obtained from $\mathrm{Graph}(T - \lambda I)$ by interchanging the two coordinates, so it is closed. Its domain is all of the Banach space X; the closed graph theorem makes it bounded. □

Theorem 5.1.3 (Neumann series)**.** Let $A \in \mathcal{B}(X, X)$ with $\|A\| < 1$. Then $I - A$ is invertible and

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n$$

in operator norm. More generally, if $\lambda_0 \in \rho(T)$ and $|\lambda - \lambda_0|\,\|R_{\lambda_0}(T)\| < 1$, then $\lambda \in \rho(T)$ and

$$R_\lambda(T) = R_{\lambda_0}(T) \sum_{n=0}^{\infty} \big((\lambda - \lambda_0) R_{\lambda_0}(T)\big)^n.$$

Proof. The partial sums telescope: $(I-A)\sum_{n=0}^{N} A^n = I - A^{N+1}$, and $A^{N+1} \to 0$. For the resolvent, factor

$$T - \lambda I = \big(I - (\lambda - \lambda_0)R_{\lambda_0}(T)\big)(T - \lambda_0 I).$$

□

Corollary 5.1.4. The resolvent set is open and the spectrum is closed. If $T \in \mathcal{B}(X, X)$, then

$$\sigma(T) \subseteq \{\lambda : |\lambda| \le \|T\|\}.$$

Proof. Openness follows from the local Neumann-series formula. If $|\lambda| > \|T\|$, then

$$T - \lambda I = -\lambda\left(I - \frac{T}{\lambda}\right)$$

is invertible by the Neumann series. □

Theorem 5.1.5 (Resolvent identity). For $\lambda, \mu \in \rho(T)$,

$$R_\lambda(T) - R_\mu(T) = (\lambda - \mu)R_\lambda(T)R_\mu(T).$$

In particular, the resolvents commute.

Proof. Use $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$ with $A = T - \lambda I$ and $B = T - \mu I$. □

Theorem 5.1.6. For every bounded operator on a nonzero complex Banach space, the spectrum is nonempty and compact.

Proof. Closedness and boundedness were proved above. If $\sigma(T) = \varnothing$, then for each $x \in X$ and $f \in X^*$ the scalar function

$$\lambda \longmapsto f(R_\lambda(T)x)$$

is entire. For $|\lambda| > \|T\|$, the Neumann expansion gives $\|R_\lambda(T)\| \le (|\lambda| - \|T\|)^{-1}$, so this entire function tends to zero at infinity. Liouville's theorem makes it identically zero. Since functionals separate points, $R_\lambda(T) = 0$, impossible for an inverse. □

Definition 5.1.7. For $T \in \mathcal{B}(X, X)$, the *spectral radius* is

$$r(T) = \sup\{|\lambda| : \lambda \in \sigma(T)\}.$$

Proposition 5.1.8. Let $T \in \mathcal{B}(X, X)$. If $\lambda \in \rho(T)$, then $R_\lambda(T)$ commutes with T. More generally, $R_\lambda(T)$ commutes with every bounded operator that commutes with T. Consequently, all resolvents commute with one another.

Proof. From

$$(T - \lambda I)R_\lambda(T) = R_\lambda(T)(T - \lambda I) = I$$

we obtain

$$TR_\lambda(T) = I + \lambda R_\lambda(T) = R_\lambda(T)T.$$

Now let $S \in \mathcal{B}(X, X)$ commute with T. Then S commutes with $T - \lambda I$, and hence

$$(T - \lambda I)SR_\lambda(T) = S(T - \lambda I)R_\lambda(T) = S.$$

Applying $R_\lambda(T)$ on the left gives $SR_\lambda(T) = R_\lambda(T)S$. Taking $S = R_\mu(T)$ proves that any two resolvents commute. □

5.2 Spectral mapping and the spectral radius formula

Theorem 5.2.1 (Polynomial spectral mapping theorem)**.** Let $T \in \mathcal{B}(X, X)$ and let p be a complex polynomial. Then

$$\sigma(p(T)) = p(\sigma(T)).$$

Proof. First let $\mu \notin p(\sigma(T))$. Factor

$$p(z) - \mu = c \prod_{j=1}^{m} (z - \lambda_j).$$

Every λ_j lies in $\rho(T)$, so $p(T) - \mu I = c \prod_j (T - \lambda_j I)$ is invertible.

Conversely, let $\mu = p(\lambda)$ with $\lambda \in \sigma(T)$. Polynomial division gives

$$p(T) - p(\lambda)I = (T - \lambda I)q(T),$$

and the two factors commute. If the left side were invertible, then $(T - \lambda I)$ would have both a bounded left inverse and a bounded right inverse obtained by combining $q(T)$ with that inverse, hence would itself be invertible. This contradicts $\lambda \in \sigma(T)$. □

Proposition 5.2.2 (Linear independence of eigenvectors)**.** Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Proof. If $\sum_{j=1}^{n} a_j x_j = 0$ is a shortest nontrivial relation and $Tx_j = \lambda_j x_j$, apply $T - \lambda_n I$. The resulting relation among $x_1, \ldots, x_{n-1}$ has coefficients $a_j(\lambda_j - \lambda_n)$ and is nontrivial, a contradiction. □

Theorem 5.2.3 (Spectral radius formula)**.** For $T \in \mathcal{B}(X, X)$,

$$r(T) = \lim_{n\to\infty} \|T^n\|^{1/n} = \inf_{n\geq 1} \|T^n\|^{1/n}.$$

Proof. By spectral mapping, $r(T)^n = r(T^n) \leq \|T^n\|$, so $r(T) \leq \inf_n \|T^n\|^{1/n}$.

Let $R > r(T)$. The resolvent is analytic on $|\lambda| > r(T)$ and has the Laurent expansion at infinity

$$R_\lambda(T) = -\sum_{n=0}^{\infty} \frac{T^n}{\lambda^{n+1}}$$

for $|\lambda| > \|T\|$. Cauchy's formula, applied on the circle $|\lambda| = R$, gives

$$T^n = -\frac{1}{2\pi i} \int_{|\lambda|=R} \lambda^n R_\lambda(T) \, d\lambda.$$

Hence $\|T^n\| \leq C_R R^{n+1}$, and $\limsup_n \|T^n\|^{1/n} \leq R$. Letting $R \downarrow r(T)$ gives the reverse inequality. The equality with the infimum follows from submultiplicativity of $\|T^n\|$ and Fekete's lemma applied to $\log \|T^n\|$. □

5.3 Compact operators

Definition 5.3.1. A linear map $T\colon X \to Y$ is compact if it maps bounded sets to relatively compact sets. Equivalently, for every bounded sequence (x_n) in X, the sequence (Tx_n) has a convergent subsequence in Y.

Lemma 5.3.2. Every compact linear operator is bounded. The identity operator on a normed space is compact if and only if the space is finite dimensional.

Proof. If T were unbounded, choose x_n with $\|x_n\| \le 1$ and $\|Tx_n\| \ge n$; then (Tx_n) has no convergent subsequence. The statement about the identity is the Riesz lemma characterization of finite-dimensional normed spaces. □

Theorem 5.3.3 (Sequential criterion)**.** A linear map $T\colon X \to Y$ is compact if and only if every bounded sequence (x_n) has a subsequence (x_{n_k}) for which (Tx_{n_k}) converges.

Proof. If T maps bounded sets to relatively compact sets and (x_n) is bounded, then all Tx_n lie in a set with compact closure, so a subsequence converges.

Conversely, let $B \subseteq X$ be bounded. Every sequence in $T(B)$ has the form (Tx_n) with $x_n \in B$, and therefore has a convergent subsequence. Thus $T(B)$ is relatively sequentially compact. To pass to the closure, let $(y_n) \subseteq \overline{T(B)}$ and choose $z_n \in T(B)$ with $\|y_n - z_n\| < 1/n$. A subsequence of (z_n) converges, and the corresponding subsequence of (y_n) has the same limit. Hence $\overline{T(B)}$ is sequentially compact. In a metric space, sequential compactness and compactness are equivalent, so $\overline{T(B)}$ is compact and T is compact. □

Theorem 5.3.4. Every bounded finite-rank operator is compact. If Y is Banach, then $\mathcal{K}(X,Y)$ is closed in $\mathcal{B}(X,Y)$ for the operator norm.

Proof. A bounded set in a finite-dimensional range is relatively compact. For the closure statement, let $T_n \to T$ in operator norm with each T_n compact. Let (x_j) be bounded, say $\|x_j\| \le M$. Compactness of T_1 gives a subsequence on which T_1x_j converges. From that subsequence choose one on which T_2x_j converges, and continue. If (y_j) is the diagonal subsequence, then for every fixed n the sequence (T_ny_j) converges.

Fix $\varepsilon > 0$ and choose n so large that
$$2M\|T - T_n\| < \frac{2\varepsilon}{3}.$$
For sufficiently large j,k, $\|T_ny_j - T_ny_k\| < \varepsilon/3$, and hence
$$\begin{aligned}\|Ty_j - Ty_k\| &\le \|(T-T_n)y_j\| + \|T_ny_j - T_ny_k\| + \|(T_n - T)y_k\| \\ &< \varepsilon.\end{aligned}$$
Thus (Ty_j) is Cauchy and converges because Y is Banach. The sequential criterion proves that T is compact.

□

Definition 5.3.5. A metric-space subset B is *totally bounded* if, for every $\varepsilon > 0$, it is covered by finitely many balls of radius ε.

Lemma 5.3.6. A metric space is compact if and only if it is complete and totally bounded. Every totally bounded set is separable, and every subset of a totally bounded set is totally bounded.

Proof. Suppose first that M is compact. Compact metric spaces are complete. For each $\varepsilon > 0$, the open cover $\{B(x, \varepsilon) : x \in M\}$ has a finite subcover, so M is totally bounded.

Conversely, assume that M is complete and totally bounded, and let (x_n) be a sequence in M. Cover M by finitely many balls of radius 2^{-1}. One of them contains infinitely many terms; retain that subsequence. Cover M by finitely many balls of radius 2^{-2} and retain an infinite subsequence lying in one of them. Continuing in this way and taking the diagonal subsequence (x_{n_k}), we have

$$d(x_{n_j}, x_{n_k}) \le 2^{1-m} \qquad (j, k \ge m),$$

because both points lie in the same ball of radius 2^{-m}. Thus the diagonal subsequence is Cauchy and converges in M by completeness. Hence M is sequentially compact, and therefore compact.

If B is totally bounded, choose for each n a finite $1/n$-net F_n. The countable set $\bigcup_n F_n$ is dense in B, so B is separable. If $A \subseteq B$, a finite $\varepsilon/2$-net for B gives a finite ε-net for A after discarding balls that miss A and choosing one point of A from each remaining ball. Thus every subset of a totally bounded set is totally bounded. $\square$

Theorem 5.3.7. If $T\colon X \to Y$ is compact, then $\operatorname{Ran}(T)$ is separable.

Proof. Write

$$\operatorname{Ran}(T) = \bigcup_{n=1}^{\infty} T(nB_X).$$

Each $T(nB_X)$ is totally bounded and therefore separable. A countable union of countable dense sets is dense in the range. $\square$

Theorem 5.3.8 (Compact extension from a dense subspace)**.** Let X and Y be Banach spaces, let $D \subseteq X$ be dense, and let $T\colon D \to Y$ be bounded. Its unique bounded extension $\widetilde{T}\colon X \to Y$ is compact if T maps the unit ball of D to a totally bounded subset of Y.

Proof. First construct the extension. If $x \in X$, choose $d_k \in D$ with $d_k \to x$. Since T is bounded,

$$\|Td_k - Td_\ell\| \le \|T\| \, \|d_k - d_\ell\|,$$

so (Td_k) is Cauchy and converges in the Banach space Y. Define $\widetilde{T}x = \lim_k Td_k$. The estimate above shows that this value is independent of the approximating sequence, and it also gives linearity, $\|\widetilde{T}\| = \|T\|$, and uniqueness.

Now let (x_n) be bounded, say $\|x_n\| \le M$. Choose $d_n \in D$ with

$$\|x_n - d_n\| < \frac{1}{n}.$$

Then $\|d_n\| \le M + 1$ for all large n. The set $T((M+1)B_D)$ is totally bounded because it is a scalar multiple of $T(B_D)$. Hence (Td_n) has a Cauchy subsequence (Td_{n_k}). Moreover,

$$\|\widetilde{T}x_{n_k} - Td_{n_k}\| \le \|T\| \, \|x_{n_k} - d_{n_k}\| \longrightarrow 0.$$

Thus $(\widetilde{T}x_{n_k})$ is Cauchy and converges in Y. By the sequential criterion, $\widetilde{T}$ is compact. $\square$

Theorem 5.3.9 (Schauder theorem)**.** A bounded linear operator $T\colon X \to Y$ is compact if and only if $T^*\colon Y^* \to X^*$ is compact.

Proof. Assume first that T is compact. Fix $\varepsilon > 0$. Since $T(B_X)$ is totally bounded, choose $x_1, \ldots, x_m \in B_X$ such that for every $x \in B_X$ there is j with

$$\|Tx - Tx_j\| < \varepsilon.$$

Consider the finite-dimensional coordinate map

$$A\colon Y^* \to \mathbb{K}^m, \qquad A(y^*) = (y^*(Tx_1), \ldots, y^*(Tx_m)).$$

The set $A(B_{Y^*})$ is bounded in $\mathbb{K}^m$, hence totally bounded. Choose $y_1^*, \ldots, y_N^* \in B_{Y^*}$ so that their coordinate vectors form an ε-net for $A(B_{Y^*})$ in the supremum norm.
Given $y^* \in B_{Y^*}$, choose k with

$$|(y^* - y_k^*)(Tx_j)| < \varepsilon \qquad (1 \le j \le m).$$

For $x \in B_X$, choose j as above. Since $\|y^* - y_k^*\| \le 2$,

$$\begin{aligned} |(T^*y^* - T^*y_k^*)(x)| &= |(y^* - y_k^*)(Tx)| \\ &\le |(y^* - y_k^*)(Tx_j)| + 2\|Tx - Tx_j\| \\ &< 3\varepsilon. \end{aligned}$$

Taking the supremum over $x \in B_X$ gives $\|T^*y^* - T^*y_k^*\| < 3\varepsilon$. Thus $T^*(B_{Y^*})$ is totally bounded; its closure is complete in the Banach space X^*, hence compact. Therefore T^* is compact.
Conversely, if T^* is compact, the implication just proved shows that T^{**} is compact. The canonical embeddings satisfy

$$T^{**}J_X = J_Y T.$$

If (x_n) is bounded in X, compactness of T^{**} gives a subsequence for which $J_Y T x_{n_k}$ converges in Y^{**}. Because $J_Y(Y)$ is closed and J_Y is an isometry, (Tx_{n_k}) converges in Y. Hence T is compact. $\square$

Lemma 5.3.10 (Ideal property)**.** For normed spaces X, Y, compact operators form a two-sided operator ideal: if $T \in \mathcal{K}(X, Y)$ and A, B are bounded with compatible domains, then ATB is compact.

Proof. Let (x_n) be bounded. Since B is bounded, (Bx_n) is bounded; compactness of T gives a subsequence for which (TBx_{n_k}) converges. Continuity of A then makes $(ATBx_{n_k})$ converge. The sequential criterion proves that ATB is compact. $\square$

5.4 The spectrum of a compact operator

Theorem 5.4.1 (Riesz spectral theorem for compact operators)**.** Let T be a compact operator on an infinite-dimensional complex Banach space. Then:

(a). every nonzero point of $\sigma(T)$ is an eigenvalue;

(b). for each $\lambda \neq 0$, the eigenspace $\mathrm{Ker}(T - \lambda I)$ is finite dimensional;

(c). $\sigma(T)$ is at most countable, and 0 is its only possible accumulation point.

Proof. For $\lambda \neq 0$, put $N_\lambda = \mathrm{Ker}(T - \lambda I)$. On this subspace, $T = \lambda I$, so

$$I_{N_\lambda} = \lambda^{-1} T|_{N_\lambda}$$

is compact. The identity on a normed space is compact only in finite dimension; hence N_λ is finite dimensional.

Fix $\varepsilon > 0$ and suppose that there are infinitely many distinct eigenvalues λ_n with $|\lambda_n| \geq \varepsilon$. Let

$$M_n = \mathrm{span}\{v_1, \ldots, v_n\}, \qquad Tv_j = \lambda_j v_j,$$

where $v_j \neq 0$. Eigenvectors for distinct eigenvalues are linearly independent, so $M_{n-1} \subsetneq M_n$. Riesz's lemma gives $x_n \in M_n$ with

$$\|x_n\| = 1, \qquad \mathrm{dist}(x_n, M_{n-1}) > \frac{1}{2}.$$

The subspaces M_n are T-invariant, and

$$(T - \lambda_n I)x_n \in M_{n-1}.$$

For $m < n$, also $Tx_m \in M_m \subseteq M_{n-1}$. Therefore

$$Tx_n - Tx_m = \lambda_n x_n - z_{m,n} \qquad \text{for some } z_{m,n} \in M_{n-1},$$

and hence

$$\|Tx_n - Tx_m\| = |\lambda_n| \left\| x_n - \frac{z_{m,n}}{\lambda_n} \right\| > \frac{\varepsilon}{2}.$$

Thus (Tx_n) has no Cauchy subsequence, contradicting compactness of T. Only finitely many eigenvalues can therefore satisfy $|\lambda| \geq \varepsilon$. Taking $\varepsilon = 1/k$ proves countability of the nonzero eigenvalues and shows that only 0 can be an accumulation point.

The remaining assertion—that every nonzero spectral value is an eigenvalue—follows from the Fredholm alternative proved below. □

Proposition 5.4.2. Let T be compact and $\lambda \neq 0$. Then $\mathrm{Ran}(T - \lambda I)$ is closed.

Proof. Let $N = \mathrm{Ker}(T - \lambda I)$, which is finite dimensional, and choose a closed complement $X = N \oplus M$. The restriction $S = (T - \lambda I)|_M$ is bounded below. Otherwise there are $x_n \in M$ with $\|x_n\| = 1$ and $Sx_n \to 0$.

Compactness gives a subsequence for which Tx_n converges, and

$$x_n = \lambda^{-1}(Tx_n - Sx_n)$$

then converges to some $x \in M$ with $\|x\| = 1$ and $Sx = 0$, contradicting $M \cap N = \{0\}$. A bounded-below operator has closed range, and $\operatorname{Ran} S = \operatorname{Ran}(T - \lambda I)$. $\square$

Theorem 5.4.3 (Stabilization and Riesz decomposition)**.** Let T be compact and $\lambda \neq 0$. There is $m \geq 1$ such that

$$\operatorname{Ker}(T - \lambda I)^m = \operatorname{Ker}(T - \lambda I)^{m+1} = \cdots,$$

$$\operatorname{Ran}(T - \lambda I)^m = \operatorname{Ran}(T - \lambda I)^{m+1} = \cdots,$$

and

$$X = \operatorname{Ker}(T - \lambda I)^m \oplus \operatorname{Ran}(T - \lambda I)^m.$$

Proof. Write

$$S = T - \lambda I, \qquad N_n = \operatorname{Ker} S^n, \qquad R_n = \operatorname{Ran} S^n,$$

with $N_0 = \{0\}$ and $R_0 = X$. The spaces N_n are closed. The ranges are closed as well: by the binomial formula,

$$S^n = (-\lambda)^n I + K_n,$$

where K_n is a finite sum of positive powers of the compact operator T and is therefore compact. The earlier closed-range proposition applies to S^n because $(-\lambda)^n \neq 0$.

Stabilization of the kernels. Suppose, contrary to the claim, that $N_{n-1} \subsetneq N_n$ for every $n \geq 1$. Riesz's lemma gives $x_n \in N_n$ such that

$$\|x_n\| = 1, \qquad \operatorname{dist}(x_n, N_{n-1}) > \frac{1}{2}.$$

If $n > m$, then $x_m \in N_m \subseteq N_{n-1}$, $Sx_n \in N_{n-1}$, and $Tx_m = (S + \lambda I)x_m \in N_m \subseteq N_{n-1}$. Consequently

$$Tx_n - Tx_m = \lambda x_n + Sx_n - Tx_m = \lambda x_n - z_{m,n}$$

for some $z_{m,n} \in N_{n-1}$. Hence

$$\|Tx_n - Tx_m\| = |\lambda| \left\| x_n - \frac{z_{m,n}}{\lambda} \right\| > \frac{|\lambda|}{2}.$$

This contradicts compactness of T. Thus $N_r = N_{r+1}$ for some r. Once equality occurs it persists: if $N_r = N_{r+1}$ and $x \in N_{r+2}$, then $Sx \in N_{r+1} = N_r$, so $x \in N_{r+1} = N_r$; induction gives $N_r = N_{r+k}$ for all $k \geq 1$.

Stabilization of the ranges. Suppose that $R_n \supsetneq R_{n+1}$ for every $n \geq 0$. Since the R_n are closed, Riesz's lemma gives $x_n \in R_n$ such that

$$\|x_n\| = 1, \qquad \operatorname{dist}(x_n, R_{n+1}) > \frac{1}{2}.$$

For $n > m$, the descending-chain relation gives $x_n \in R_n \subseteq R_{m+1}$. Also $Sx_m \in R_{m+1}$ and $Tx_n = (S + \lambda I)x_n \in R_n \subseteq R_{m+1}$. Therefore

$$Tx_m - Tx_n = \lambda x_m - z_{m,n} \qquad (z_{m,n} \in R_{m+1}),$$

so

$$\|Tx_m - Tx_n\| > \frac{|\lambda|}{2},$$

again contradicting compactness. Hence $R_s = R_{s+1}$ for some s, and then $R_s = R_{s+k}$ for every $k \geq 1$. This is the null-space/range argument appearing in the handwritten proof. To make its final step explicit, choose

$$m \geq \max\{r, s, 1\}.$$

Then $N_m = N_{2m}$ and $R_m = R_{2m}$. Given $x \in X$, the vector $S^m x$ lies in $R_m = R_{2m}$, so there is $z \in X$ with

$$S^m x = S^{2m} z.$$

Thus

$$x = (x - S^m z) + S^m z,$$

where $x - S^m z \in N_m$ and $S^m z \in R_m$. Therefore $X = N_m + R_m$. If $x \in N_m \cap R_m$, write $x = S^m y$. Then $0 = S^m x = S^{2m} y$, so $y \in N_{2m} = N_m$ and hence $x = S^m y = 0$. The sum is direct, proving

$$X = N_m \oplus R_m.$$

□

5.5 Fredholm alternative

Lemma 5.5.1 (Biorthogonal vectors)**.** If $f_1, \dots, f_n \in X^*$ are linearly independent, then there exist $x_1, \dots, x_n \in X$ such that

$$f_i(x_j) = \delta_{ij}.$$

Proof. The map $Q\colon X \to \mathbb{K}^n$, $Qx = (f_1(x), \dots, f_n(x))$, is surjective; otherwise a nonzero functional on $\mathbb{K}^n$ annihilating $Q(X)$ would give a nontrivial linear relation among the f_i. Choose x_j with $Qx_j = e_j$. □

Theorem 5.5.2 (Fredholm alternative)**.** Let $T \in \mathcal{K}(X, X)$ and let $\lambda \neq 0$. Put $S = \lambda I - T$. Then:

(a). S is injective if and only if it is surjective; in this case S^{-1} is bounded;

(b). $\dim \operatorname{Ker} S = \dim \operatorname{Ker} S^* < \infty$;

(c). the equation $Sx = y$ is solvable if and only if

$$f(y) = 0 \qquad (f \in \operatorname{Ker} S^*);$$

(d). the equation $S^* f = g$ is solvable if and only if

$$g(x) = 0 \qquad (x \in \operatorname{Ker} S).$$

Here $S^* = \lambda I - T^*$ for the Banach-space adjoint defined on linear functionals. Under the Hilbert-space adjoint identification, the corresponding spectral parameter is conjugated according to the inner-product convention.

Proof. Since $S = \lambda I - T$ differs only by a sign from $T - \lambda I$, the closed-range and stabilization results above apply to S and all its powers.

We first follow the null-space/range argument of the notes. If S is injective and $\operatorname{Ran} S \neq X$, then every inclusion

$$X \supsetneq \operatorname{Ran} S \supsetneq \operatorname{Ran} S^2 \supsetneq \cdots$$

is strict. Indeed, if $\operatorname{Ran} S^n = \operatorname{Ran} S^{n+1}$ for some $n \geq 1$ and $y \in \operatorname{Ran} S^{n-1}$, then $Sy \in \operatorname{Ran} S^n = \operatorname{Ran} S^{n+1}$, so $Sy = S(S^n z)$ for some z; injectivity gives $y = S^n z \in \operatorname{Ran} S^n$. Repeating this backward would yield $X = \operatorname{Ran} S$, a contradiction. The strict chain contradicts range stabilization. Hence injectivity implies surjectivity.

Conversely, suppose S is surjective but has $0 \neq x_1 \in \operatorname{Ker} S$. Choose recursively x_{n+1} with $Sx_{n+1} = x_n$. Then

$$x_n \in \operatorname{Ker} S^n \setminus \operatorname{Ker} S^{n-1},$$

so the kernels increase strictly forever, contradicting kernel stabilization. Thus surjectivity implies injectivity. If either condition holds, S is a bounded bijection of Banach spaces and the bounded inverse theorem gives $S^{-1} \in \mathcal{B}(X, X)$.

We next compute the dimensions. Choose m from the Riesz decomposition and write

$$X = N_m \oplus R_m, \qquad N_m = \operatorname{Ker} S^m, \qquad R_m = \operatorname{Ran} S^m.$$

The space N_m is finite dimensional, because $S^m = cI + K$ with $c \neq 0$ and K compact. Since the ranges have stabilized, $S(R_m) = R_m$; the direct-sum relation also shows that $S|_{R_m}$ is injective. Moreover,

$$\operatorname{Ran} S = S(N_m) \oplus R_m.$$

Consequently,

$$\begin{aligned} \operatorname{codim} \operatorname{Ran} S &= \dim N_m - \dim S(N_m) \\ &= \dim \operatorname{Ker}(S|_{N_m}) = \dim \operatorname{Ker} S. \end{aligned}$$

Because $\operatorname{Ran} S$ is closed,

$$\operatorname{Ker} S^* = (\operatorname{Ran} S)^{\perp}$$

is naturally the dual of the finite-dimensional quotient $X / \operatorname{Ran} S$. Therefore

$$\dim \operatorname{Ker} S^* = \operatorname{codim} \operatorname{Ran} S = \dim \operatorname{Ker} S < \infty.$$

Finally, the standard annihilator identities give

$$\operatorname{Ran} S = {}^{\perp}\operatorname{Ker} S^*$$

because $\operatorname{Ran} S$ is closed. Hence $Sx = y$ is solvable exactly when $f(y) = 0$ for every $f \in \operatorname{Ker} S^*$. By Schauder's theorem, T^* is compact, so $S^* = \lambda I - T^*$ also has closed range. The dual annihilator identity then gives

$$\operatorname{Ran} S^* = (\operatorname{Ker} S)^{\perp},$$

which is precisely the stated solvability criterion for $S^* f = g$. $\square$

Corollary 5.5.3. Every nonzero spectral value of a compact operator is an eigenvalue.

Proof. If $\lambda \neq 0$ and $\lambda I - T$ were injective, the Fredholm alternative would make it surjective and hence invertible. Thus a nonzero point of the spectrum must have nontrivial kernel. $\square$

5.6 Arzelà–Ascoli and integral operators

Theorem 5.6.1 (Arzelà–Ascoli theorem)**.** Let K be a compact metric space. A bounded equicontinuous family $\mathcal{F} \subseteq C(K)$ is relatively compact in the supremum norm. Equivalently, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence.

Theorem 5.6.2 (Continuous-kernel integral operators)**.** Let $J = [a, b]$ and let $k \in C(J \times J)$. Define

$$(Tf)(s) = \int_a^b k(s,t)f(t)\,\mathrm{d}t, \qquad f \in C(J).$$

Then $T \colon C(J) \to C(J)$ is compact.

Proof. For $\|f\|_\infty \leq 1$,

$$\|Tf\|_\infty \leq (b-a)\|k\|_\infty,$$

so the image of the unit ball is uniformly bounded. Uniform continuity of k gives

$$|Tf(s_1) - Tf(s_2)| \leq \int_a^b |k(s_1,t) - k(s_2,t)|\,\mathrm{d}t,$$

which tends to zero uniformly in f as $s_1 \to s_2$. Thus the image is equicontinuous, and Arzelà–Ascoli gives relative compactness. $\square$

Chapter 6 Operator Semigroups and Hille–Yosida

6.1 Strongly continuous semigroups and generators

Definition 6.1.1. Let X be a Banach space. A family $(T(t))_{t\geq 0} \subseteq \mathcal{B}(X, X)$ is a *strongly continuous semigroup*, or C_0-semigroup, if

$$T(0) = I, \qquad T(t+s) = T(t)T(s),$$

and

$$\lim_{t\downarrow 0} T(t)x = x \qquad (x \in X).$$

Its *infinitesimal generator* is

$$Ax = \lim_{t\downarrow 0} \frac{T(t)x - x}{t},$$

with domain

$$\mathrm{D}(A) = \left\{ x \in X : \lim_{t\downarrow 0} \frac{T(t)x - x}{t} \text{ exists in } X \right\}.$$

Proposition 6.1.2. Let A be the generator of a C_0-semigroup $(T(t))$. Then:

(a). A is linear and $\mathrm{D}(A)$ is dense in X;

(b). $T(t)\,\mathrm{D}(A) \subseteq \mathrm{D}(A)$ and $AT(t)x = T(t)Ax$ for $x \in \mathrm{D}(A)$;

(c). for $x \in \mathrm{D}(A)$, the orbit is continuously differentiable and

$$\frac{\mathrm{d}}{\mathrm{d}t} T(t)x = AT(t)x = T(t)Ax;$$

(d). A is closed.

Proof. Linearity follows from the limit. For density, set

$$x_h = \frac{1}{h} \int_0^h T(s)x \, \mathrm{d}s.$$

Strong continuity gives $x_h \to x$, while

$$\frac{T(t)x_h - x_h}{t} = \frac{1}{h}\left(\frac{1}{t}\int_h^{h+t} T(s)x \, \mathrm{d}s - \frac{1}{t}\int_0^t T(s)x \, \mathrm{d}s \right) \longrightarrow \frac{T(h)x - x}{h}.$$

Thus $x_h \in \mathrm{D}(A)$.

For $x \in \mathrm{D}(A)$,

$$\frac{T(h)T(t)x - T(t)x}{h} = T(t)\frac{T(h)x - x}{h} \longrightarrow T(t)Ax,$$

which proves invariance and commutation. The derivative at arbitrary t follows from the semigroup law; continuity of $s \mapsto T(s)Ax$ makes it a continuous derivative.

Finally, suppose $x_n \to x$ and $Ax_n \to y$. The fundamental theorem of calculus gives

$$T(t)x_n - x_n = \int_0^t T(s)Ax_n \, ds.$$

Passing to the limit yields

$$T(t)x - x = \int_0^t T(s)y \, ds.$$

Divide by t and let $t \downarrow 0$ to obtain $x \in D(A)$ and $Ax = y$. □

Definition 6.1.3. A C_0-semigroup is a *contraction semigroup* if $\|T(t)\| \leq 1$ for every $t \geq 0$.

6.2 Resolvents of generators

Lemma 6.2.1 (Laplace-transform resolvent)**.** Let $(T(t))$ be a contraction semigroup with generator A. For $\operatorname{Re}\lambda > 0$, define

$$R(\lambda, A)x = \int_0^\infty e^{-\lambda t} T(t)x \, dt.$$

Then the Bochner integral is well defined,

$$R(\lambda, A) = (\lambda I - A)^{-1}, \qquad \|R(\lambda, A)\| \leq \frac{1}{\operatorname{Re}\lambda}.$$

For real $\lambda > 0$,

$$\|\lambda R(\lambda, A)\| \leq 1.$$

Proof. The norm estimate follows from

$$\int_0^\infty e^{-\operatorname{Re}\lambda t} \|T(t)x\| \, dt \leq \frac{\|x\|}{\operatorname{Re}\lambda}.$$

For $s > 0$, the semigroup law and a change of variables give

$$\frac{T(s) - I}{s} R(\lambda, A)x = \frac{1}{s} \int_0^\infty e^{-\lambda t} \big(T(t+s) - T(t)\big)x \, dt \longrightarrow \lambda R(\lambda, A)x - x.$$

Hence $R(\lambda, A)x \in D(A)$ and $(\lambda I - A)R(\lambda, A)x = x$. If $x \in D(A)$, integration by parts in $e^{-\lambda t}T(t)x$ gives $R(\lambda, A)(\lambda I - A)x = x$. Thus it is the inverse. □

Corollary 6.2.2. The generator of a contraction semigroup is densely defined and closed, $(0, \infty) \subseteq \rho(A)$, and

$$\|\lambda(\lambda I - A)^{-1}\| \leq 1 \qquad (\lambda > 0).$$

6.3 The contraction Hille–Yosida theorem

Theorem 6.3.1 (Hille–Yosida theorem, contraction form)**.** Let $A\colon \mathrm{D}(A) \subseteq X \to X$ be linear. Then A generates a contraction C_0-semigroup if and only if

(a). A is densely defined and closed;

(b). $(0, \infty) \subseteq \rho(A)$;

(c). $\|\lambda(\lambda I - A)^{-1}\| \leq 1$ for every $\lambda > 0$.

Proof. Necessity is exactly the Laplace-transform resolvent lemma, together with the general facts that a generator is densely defined and closed. We prove sufficiency.

Write $R_\lambda = (\lambda I - A)^{-1}$ and, for $\lambda > 0$, define the bounded Yosida approximation

$$A_\lambda = \lambda A R_\lambda = \lambda^2 R_\lambda - \lambda I.$$

The bounded operator A_λ generates the uniformly continuous semigroup

$$T_\lambda(t) = e^{tA_\lambda}.$$

Since $\|\lambda R_\lambda\| \leq 1$,

$$\begin{aligned}\|T_\lambda(t)\| &= \left\| e^{-\lambda t} \sum_{n=0}^{\infty} \frac{(\lambda^2 t)^n}{n!} R_\lambda^n \right\| \\ &\leq e^{-\lambda t} \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} = 1.\end{aligned}$$

Thus all approximating semigroups are contractions.

We next record the approximation property. If $y \in \mathrm{D}(A)$, then

$$\lambda R_\lambda y - y = R_\lambda A y,$$

so $\lambda R_\lambda y \to y$. Since $\|\lambda R_\lambda\| \leq 1$ and $\mathrm{D}(A)$ is dense, the same convergence holds for every $y \in X$. Therefore, for $x \in \mathrm{D}(A)$,

$$A_\lambda x = \lambda R_\lambda A x \longrightarrow Ax.$$

All resolvents commute, and hence the operators A_λ commute. For $x \in \mathrm{D}(A)$, the bounded-operator variation formula gives

$$T_\lambda(t)x - T_\mu(t)x = \int_0^t T_\lambda(t-s) T_\mu(s)(A_\lambda - A_\mu)x \, \mathrm{d}s,$$

whence

$$\sup_{0 \leq t \leq t_0} \|T_\lambda(t)x - T_\mu(t)x\| \leq t_0 \|(A_\lambda - A_\mu)x\| \longrightarrow 0.$$

For arbitrary $x \in X$, approximate x by an element of $\mathrm{D}(A)$ and use the contraction bounds. Thus, for each x, $(T_\lambda(t)x)$ is Cauchy uniformly on compact time intervals. Define

$$T(t)x = \lim_{\lambda \to \infty} T_\lambda(t)x.$$

Passing to the limit in $T_\lambda(t+s) = T_\lambda(t)T_\lambda(s)$ proves the semigroup law, and the norm estimate passes to the limit. Strong continuity follows first on $\mathrm{D}(A)$ from

$$T_\lambda(t)x - x = \int_0^t T_\lambda(s)A_\lambda x \, \mathrm{d}s$$

and the uniform boundedness of $A_\lambda x$ for large λ, and then on all of X by density and the contraction bound.

Let G be the generator of $(T(t))$. For $x \in \mathrm{D}(A)$, uniform convergence on compact time intervals, $A_\lambda x \to Ax$, and the contraction estimate allow passage to the limit in the last integral:

$$T(t)x - x = \int_0^t T(s)Ax \, \mathrm{d}s.$$

Dividing by t and letting $t \downarrow 0$ gives $x \in \mathrm{D}(G)$ and $Gx = Ax$; hence $A \subseteq G$.

To prove equality, fix $\mu > 0$. Both $\mu I - A$ and $\mu I - G$ are bijective. For $y \in X$, put $x = R(\mu, A)y$. Then $x \in \mathrm{D}(A) \subseteq \mathrm{D}(G)$ and

$$(\mu I - G)x = (\mu I - A)x = y.$$

Thus $R(\mu, G)y = R(\mu, A)y$ for every y. Equal resolvents have the same range, so $\mathrm{D}(G) = \mathrm{D}(A)$, and then $G = A$. □

Corollary 6.3.2 (Uniqueness and the abstract Cauchy problem)**.** A generator determines its C_0-semigroup uniquely. If $x_0 \in \mathrm{D}(A)$, then

$$u(t) = T(t)x_0$$

is the unique classical solution of

$$u'(t) = Au(t), \qquad u(0) = x_0.$$

Moreover, $\mathrm{D}(A^2)$ is a core for A and hence is dense in $\mathrm{D}(A)$ for the graph norm.

Proof. Existence follows from the orbit differentiation formula. If u is another classical solution, fix $t > 0$ and define $F(s) = T(t-s)u(s)$ for $0 \le s \le t$. Since $u(s) \in \mathrm{D}(A)$ and $u'(s) = Au(s)$,

$$F'(s) = -AT(t-s)u(s) + T(t-s)Au(s) = 0.$$

Hence $u(t) = F(t) = F(0) = T(t)u(0)$, proving uniqueness and, consequently, uniqueness of the semigroup generated by A.

For the core assertion, let $x \in \mathrm{D}(A)$ and put

$$x_\lambda = \lambda R(\lambda, A)x.$$

The resolvent maps X into $\mathrm{D}(A)$ and commutes with A on $\mathrm{D}(A)$, so

$$x_\lambda \in \mathrm{D}(A^2), \qquad Ax_\lambda = \lambda R(\lambda, A)Ax.$$

The resolvent approximation gives

$$x_\lambda \to x, \qquad Ax_\lambda \to Ax.$$

Thus $x_\lambda \to x$ in the graph norm, proving that $\mathrm{D}(A^2)$ is a core. □

Theorem 6.3.3 (Euler approximation)**.** If A generates a contraction semigroup, then for every $x \in X$ and $t \geq 0$,

$$T(t)x = \lim_{n\to\infty} \left(I - \frac{t}{n}A\right)^{-n} x.$$

Proof. The case $t = 0$ is immediate. Fix $t > 0$ and put $\lambda_n = n/t$. Repeated use of the Laplace resolvent formula and the semigroup law gives

$$R(\lambda_n, A)^n x = \frac{1}{(n-1)!} \int_0^\infty s^{n-1} e^{-\lambda_n s} T(s)x \, \mathrm{d}s.$$

Therefore

$$\begin{aligned} \left(I - \frac{t}{n}A\right)^{-n} x &= (\lambda_n R(\lambda_n, A))^n x \\ &= \int_0^\infty k_{n,t}(s) T(s)x \, \mathrm{d}s, \end{aligned}$$

where

$$k_{n,t}(s) = \frac{\lambda_n^n}{(n-1)!} s^{n-1} e^{-\lambda_n s}$$

is the density of a Gamma random variable S_n with

$$\mathbb{E}S_n = t, \qquad \mathrm{Var}(S_n) = \frac{t^2}{n}.$$

The density has integral one. Hence

$$\left\| \left(I - \frac{t}{n}A\right)^{-n} x - T(t)x \right\| \leq \int_0^\infty k_{n,t}(s) \|T(s)x - T(t)x\| \, \mathrm{d}s.$$

Given $\varepsilon > 0$, strong continuity gives $\delta > 0$ such that the integrand is $< \varepsilon$ whenever $|s - t| < \delta$. On the complement it is at most $2\|x\|$, because T is contractive. Chebyshev's inequality gives

$$\int_{|s-t|\geq\delta} k_{n,t}(s) \, \mathrm{d}s \leq \frac{t^2}{n\delta^2} \longrightarrow 0.$$

Thus the displayed norm tends to zero, proving the Euler formula. □

6.4 Exponentially bounded semigroups

Lemma 6.4.1. Every C_0-semigroup is exponentially bounded: there are $M \geq 1$ and $\omega \in \mathbb{R}$ such that

$$\|T(t)\| \leq M e^{\omega t} \qquad (t \geq 0).$$

Proof. Strong continuity and uniform boundedness give $M_0 = \sup_{0\leq t\leq 1} \|T(t)\| < \infty$. Writing $t = n + s$ with $n \in \mathbb{N}_0$ and $0 \leq s < 1$ gives, by the semigroup law,

$$\|T(t)\| = \|T(1)^n T(s)\| \leq M_0^{n+1}.$$

Because $T(0) = I$, one has $M_0 \geq 1$. Taking

$$M = M_0, \qquad \omega = \log M_0$$

and using $n \le t$ yields

$$\|T(t)\| \le M_0^{n+1} \le M_0 e^{t \log M_0} = M e^{\omega t}.$$

□

Lemma 6.4.2 (Higher resolvent powers). If $\|T(t)\| \le M e^{\omega t}$ and A is its generator, then for $\operatorname{Re}\lambda > \omega$ and $n \ge 1$,

$$R(\lambda, A)^n x = \frac{1}{(n-1)!} \int_0^\infty t^{n-1} e^{-\lambda t} T(t)x \, \mathrm{d}t,$$

and therefore

$$\|R(\lambda, A)^n\| \le \frac{M}{(\operatorname{Re}\lambda - \omega)^n}.$$

Proof. The formula for $n = 1$ is the Laplace representation. Suppose it holds for n. Since

$$\int_0^\infty e^{-\operatorname{Re}\lambda t} \|T(t)x\| \, \mathrm{d}t \le M\|x\| \int_0^\infty e^{-(\operatorname{Re}\lambda - \omega)t} \, \mathrm{d}t < \infty,$$

the relevant Bochner integrals are absolutely integrable and Fubini's theorem applies. Using $T(s)T(t) = T(s+t)$,

$$\begin{aligned}
R(\lambda, A)^{n+1} x &= \frac{1}{(n-1)!} \int_0^\infty \int_0^\infty t^{n-1} e^{-\lambda(s+t)} T(s+t)x \, \mathrm{d}s \, \mathrm{d}t \\
&= \frac{1}{(n-1)!} \int_0^\infty e^{-\lambda r} T(r)x \left(\int_0^r t^{n-1} \, \mathrm{d}t \right) \mathrm{d}r \\
&= \frac{1}{n!} \int_0^\infty r^n e^{-\lambda r} T(r)x \, \mathrm{d}r.
\end{aligned}$$

This proves the representation by induction. Taking norms and putting $a = \operatorname{Re}\lambda - \omega > 0$ gives

$$\begin{aligned}
\|R(\lambda, A)^n x\| &\le \frac{M\|x\|}{(n-1)!} \int_0^\infty t^{n-1} e^{-at} \, \mathrm{d}t \\
&= \frac{M}{a^n} \|x\|,
\end{aligned}$$

because the last integral equals $(n-1)!/a^n$. □

Theorem 6.4.3 (Hille–Yosida–Phillips theorem). Let A be densely defined and closed. It generates a C_0-semigroup satisfying

$$\|T(t)\| \le M e^{\omega t}$$

if and only if $(\omega, \infty) \subseteq \rho(A)$ and

$$\|R(\lambda, A)^n\| \le \frac{M}{(\lambda - \omega)^n} \qquad (\lambda > \omega, \; n \ge 1).$$

Proof. Necessity is the higher-resolvent-power lemma. We prove sufficiency by the Yosida construction used in the handwritten proof.

For $\lambda > \max\{\omega, 0\}$, write $R_\lambda = (\lambda I - A)^{-1}$ and define

$$B_\lambda = \lambda A R_\lambda = \lambda^2 R_\lambda - \lambda I, \qquad T_\lambda(t) = e^{t B_\lambda}.$$

The assumed power estimates give

$$\begin{aligned}
\|T_\lambda(t)\| &= \left\| e^{-\lambda t} \sum_{n=0}^{\infty} \frac{(\lambda^2 t)^n}{n!} R_\lambda^n \right\| \\
&\le M e^{-\lambda t} \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{\lambda^2 t}{\lambda - \omega} \right)^n \\
&= M \exp\left(\frac{\lambda \omega}{\lambda - \omega} t \right).
\end{aligned}$$

Here $M \ge 1$ may be assumed, so the $n = 0$ term is covered by the same bound. For every fixed $t_0 > 0$, the right-hand side is bounded uniformly for $0 \le t \le t_0$ and all sufficiently large λ.

We now show convergence of the generators on the original domain. If $y \in \mathrm{D}(A)$, then

$$\lambda R_\lambda y - y = R_\lambda A y, \qquad \|R_\lambda A y\| \le \frac{M}{\lambda - \omega} \|Ay\| \longrightarrow 0.$$

Since $\mathrm{D}(A)$ is dense and $\|\lambda R_\lambda\| \le M\lambda/(\lambda - \omega)$ is uniformly bounded for large λ, it follows that $\lambda R_\lambda y \to y$ for every $y \in X$. Hence, for $x \in \mathrm{D}(A)$,

$$B_\lambda x = \lambda R_\lambda A x \longrightarrow Ax.$$

The resolvent identity makes all B_λ commute. Thus, for $x \in \mathrm{D}(A)$,

$$T_\lambda(t)x - T_\mu(t)x = \int_0^t T_\lambda(s) T_\mu(t-s)(B_\lambda - B_\mu)x \, \mathrm{d}s.$$

The uniform semigroup bounds on compact time intervals and $B_\lambda x \to Ax$ show that $(T_\lambda(t)x)$ is Cauchy uniformly for t in compact intervals. Density of $\mathrm{D}(A)$ and the same uniform bounds extend the conclusion to every $x \in X$. Define

$$T(t)x = \lim_{\lambda \to \infty} T_\lambda(t)x.$$

As in the contraction proof, uniform convergence on compact time intervals allows passage to the limit in the semigroup law. It also gives strong continuity: first for $x \in \mathrm{D}(A)$ by integrating $T_\lambda(s)B_\lambda x$, and then for general x by density. Finally,

$$\|T(t)\| \le \liminf_{\lambda \to \infty} M \exp\left(\frac{\lambda \omega}{\lambda - \omega} t \right) = M e^{\omega t}.$$

Let G be the generator of T. For $x \in \mathrm{D}(A)$, dominated convergence in

$$T_\lambda(t)x - x = \int_0^t T_\lambda(s) B_\lambda x \, \mathrm{d}s$$

gives

$$T(t)x - x = \int_0^t T(s) A x \, \mathrm{d}s.$$

Thus $A \subseteq G$. Choose any $\mu > \omega$. The growth estimate implies $\mu \in \rho(G)$ by the Laplace-transform formula, while the hypothesis gives $\mu \in \rho(A)$. If $y \in X$ and $x = R(\mu, A)y$, then $x \in \mathrm{D}(A) \subseteq \mathrm{D}(G)$ and

$$(\mu I - G)x = (\mu I - A)x = y.$$

Hence $R(\mu, G) = R(\mu, A)$, so $G = A$. This completes the sufficiency proof. □

6.5 Examples

Theorem 6.5.1 (Translation semigroup)**.** Let

$$X = C_0([0,\infty)) = \{f \in C([0,\infty)) : f(t) \to 0 \text{ as } t \to \infty\}$$

with the supremum norm, and define

$$(T(t)f)(x) = f(x+t).$$

Then $(T(t))$ is a contraction C_0-semigroup. Its generator is

$$Af = f',$$

with domain

$$D(A) = \{f \in C_0([0,\infty)) : f \text{ is } C^1 \text{ and } f' \in C_0([0,\infty))\}.$$

Proof. Uniform continuity of every $f \in C_0([0,\infty))$ gives strong continuity. For $f \in D(A)$, the difference quotient converges uniformly to f'. For the converse, suppose

$$\frac{T(h)f - f}{h} \longrightarrow g \quad \text{uniformly as } h \downarrow 0.$$

Fix $x \geq 0$ and $t > 0$, and divide $[0,t]$ into n equal pieces of length $h = t/n$. Telescoping gives

$$\begin{aligned} f(x+t) - f(x) &= \sum_{k=0}^{n-1} \big(f(x+(k+1)h) - f(x+kh)\big) \\ &= h \sum_{k=0}^{n-1} \frac{f(x+kh+h) - f(x+kh)}{h}. \end{aligned}$$

The difference quotients converge to g uniformly in their spatial variable, so the last Riemann sum converges, as $n \to \infty$, to

$$f(x+t) - f(x) = \int_0^t g(x+s)\,\mathrm{d}s,$$

because g is continuous as a uniform limit of continuous functions. Hence f is right differentiable at every $x \geq 0$ and $f'(x) = g(x)$; at interior points the same identity with nearby endpoints gives the ordinary two-sided derivative. Thus $f \in C^1$, with $f' = g \in C_0$. □

Proposition 6.5.2 (Positive self-adjoint operators)**.** Let H be a Hilbert space and let A be densely defined, self-adjoint, and positive:

$$\langle Ax, x\rangle \geq 0 \qquad (x \in D(A)).$$

Then $-A$ generates a contraction C_0-semigroup.

Proof. For $x \in D(A)$ and $\lambda > 0$,

$$\begin{aligned} \|(\lambda I + A)x\|\,\|x\| &\geq \operatorname{Re}\langle(\lambda I + A)x, x\rangle \\ &= \lambda\|x\|^2 + \langle Ax, x\rangle \geq \lambda\|x\|^2. \end{aligned}$$

Thus $\lambda I + A$ is injective, bounded below by λ, and has closed range. If y is orthogonal to this range, self-adjointness gives

$$y \in \mathrm{D}((\lambda I + A)^*) = \mathrm{D}(A) \quad \text{and} \quad 0 = (\lambda I + A)^* y = (\lambda I + A) y,$$

so the same lower bound forces $y = 0$. The range is therefore dense and closed, hence all of H. It follows that

$$\|\lambda(\lambda I + A)^{-1}\| \leq 1.$$

The contraction Hille–Yosida theorem, applied to $-A$, now yields a contraction semigroup. The sign is essential: a positive operator itself generally produces growth rather than contraction. □

6.6 Dissipative operators and Lumer–Phillips

Definition 6.6.1. For $x \in X$, the normalized duality set is

$$\mathcal{J}(x) = \{x^* \in X^* : x^*(x) = \|x\|^2,\ \|x^*\| = \|x\|\}.$$

It is nonempty by Hahn–Banach. A densely defined operator A is *accretive* if, for every $x \in \mathrm{D}(A)$, there is $x^* \in \mathcal{J}(x)$ with

$$\operatorname{Re} x^*(Ax) \geq 0.$$

It is *dissipative* if $-A$ is accretive.

Proposition 6.6.2. An operator A is dissipative if and only if

$$\|(\lambda I - A)x\| \geq \lambda \|x\| \qquad (x \in \mathrm{D}(A),\ \lambda > 0).$$

Proof. Suppose first that A is dissipative. Choose $x^* \in \mathcal{J}(x)$ with $\operatorname{Re} x^*(Ax) \leq 0$. Then

$$\begin{aligned}\|(\lambda I - A)x\|\,\|x\| &\geq \operatorname{Re} x^*((\lambda I - A)x)\\ &= \lambda\|x\|^2 - \operatorname{Re} x^*(Ax) \geq \lambda \|x\|^2,\end{aligned}$$

which proves the norm inequality.

Conversely, assume the inequality. For $t > 0$, take $\lambda = 1/t$ to obtain

$$\|x - tAx\| \geq \|x\|.$$

Assume $x \neq 0$. For each $t > 0$, Hahn–Banach gives $f_t \in X^*$ such that

$$\|f_t\| = 1, \qquad f_t(x - tAx) = \|x - tAx\|;$$

in the complex case the functional is multiplied by a unimodular scalar so that the displayed value is real and nonnegative. Then

$$\begin{aligned}\|x\| &\leq \|x - tAx\|\\ &= \operatorname{Re} f_t(x) - t \operatorname{Re} f_t(Ax)\\ &\leq \|x\| - t \operatorname{Re} f_t(Ax),\end{aligned}$$

and hence $\operatorname{Re} f_t(Ax) \le 0$.

Let t tend to zero through any sequence. Banach–Alaoglu gives a weak-$*$ convergent subnet, still denoted f_t, with limit $f \in B_{X^*}$. Moreover,

$$|f_t(x) - \|x\|| \le |f_t(x) - f_t(x - tAx)| + \big|\|x - tAx\| - \|x\|\big| \le 2t\|Ax\|,$$

so $f(x) = \|x\|$. It follows that $\|f\| = 1$, and weak-$*$ convergence also gives

$$\operatorname{Re} f(Ax) \le 0.$$

Thus $x^* = \|x\| f$ belongs to $\mathcal{J}(x)$ and verifies dissipativity. The case $x = 0$ is immediate. □

Theorem 6.6.3 (Lumer–Phillips theorem). Let A be densely defined on a Banach space. Then A generates a contraction C_0-semigroup if and only if

(a). A is dissipative;

(b). $\operatorname{Ran}(\lambda_0 I - A) = X$ for some $\lambda_0 > 0$.

In this case the range condition holds for every $\lambda > 0$.

Proof. If A generates a contraction semigroup, the positive resolvent is onto by the Laplace formula. For $x \in \mathrm{D}(A)$ and $x^* \in \mathcal{J}(x)$,

$$\operatorname{Re} x^*(T(t)x - x) \le \|x\| \, \|T(t)x\| - \|x\|^2 \le 0.$$

Divide by t and let $t \downarrow 0$ to obtain $\operatorname{Re} x^*(Ax) \le 0$, so A is dissipative.

Conversely, assume dissipativity and $\operatorname{Ran}(\lambda_0 I - A) = X$. The norm inequality gives injectivity of $\lambda I - A$ for every $\lambda > 0$ and

$$\|(\lambda I - A)^{-1}\| \le \frac{1}{\lambda}$$

on its range. In particular, $R(\lambda_0, A)$ is a bounded operator on all of X. This also proves that A is closed: if $x_n \to x$ and $Ax_n \to y$, then

$$x_n = R(\lambda_0, A)(\lambda_0 x_n - Ax_n) \longrightarrow R(\lambda_0, A)(\lambda_0 x - y),$$

so x belongs to $\mathrm{D}(A)$ and $Ax = y$.

Surjectivity propagates along the positive axis. If $\mu I - A$ is onto, then for $|\lambda - \mu| < \mu$,

$$\lambda I - A = \big(I + (\lambda - \mu)R(\mu, A)\big)(\mu I - A),$$

and $\|(\lambda - \mu)R(\mu, A)\| < 1$. The Neumann series therefore makes $\lambda I - A$ onto. Starting from λ_0, these overlapping intervals cover $(0, \infty)$ (for example, repeatedly move from μ to $3\mu/2$). Thus

$$(0, \infty) \subseteq \rho(A), \qquad \|\lambda R(\lambda, A)\| \le 1.$$

The contraction Hille–Yosida theorem now shows that A generates a contraction semigroup. □

Corollary 6.6.4. Let A be densely defined and closed on a Banach space. If both A and its Banach adjoint A^* are dissipative, then A generates a contraction C_0-semigroup.

Proof. Dissipativity makes $\mathrm{Ran}(I - A)$ closed. If it were not dense, Hahn–Banach would give $0 \neq f \in X^*$ annihilating it. Then $f \in \mathrm{D}(A^*)$ and $(I - A^*)f = 0$, contradicting the dissipative inequality for A^*. Hence the range is dense and closed, so it equals X. Apply Lumer–Phillips. □

Definition 6.6.5. Let A be closed. A subspace $D \subseteq \mathrm{D}(A)$ is a *core* for A if the graph of $A|_D$ is dense in $\mathrm{Graph}(A)$, equivalently, if D is dense in $\mathrm{D}(A)$ for the graph norm $\|x\| + \|Ax\|$.

Theorem 6.6.6 (Invariant cores)**.** Let A generate a C_0-semigroup $(T(t))$. Suppose $D \subseteq \mathrm{D}(A)$ is dense in X and $T(t)D \subseteq D$ for every $t \geq 0$. Then D is a core for A.

Proof. Choose λ larger than a growth bound of the semigroup, so that

$$R(\lambda, A)x = \int_0^\infty e^{-\lambda t} T(t)x \, \mathrm{d}t.$$

Let $\overline{A|_D}$ denote the graph closure of the restriction. We first show that $R(\lambda, A)d$ belongs to its domain for every $d \in D$. Approximate the Laplace integral by finite Riemann sums

$$z_n = \sum_j c_{n,j} T(t_{n,j})d.$$

Invariance gives $z_n \in D$. Since $d \in \mathrm{D}(A)$ and $AT(t)d = T(t)Ad$, the same Riemann sums after applying A converge to

$$\int_0^\infty e^{-\lambda t} T(t)Ad \, \mathrm{d}t = AR(\lambda, A)d.$$

Thus $z_n \to R(\lambda, A)d$ in the graph norm, as required.

Now let $x \in X$ and choose $d_n \in D$ with $d_n \to x$. The resolvent is bounded from X into $\mathrm{D}(A)$ equipped with the graph norm, because

$$AR(\lambda, A) = \lambda R(\lambda, A) - I.$$

Hence

$$R(\lambda, A)d_n \longrightarrow R(\lambda, A)x \quad \text{in the graph norm.}$$

The graph closure of $A|_D$ therefore contains $R(\lambda, A)X = \mathrm{D}(A)$. The reverse inclusion is automatic because A is closed. Thus $\overline{A|_D} = A$, and D is a core. □

Theorem 6.6.7 (Gaussian heat semigroup)**.** On $X = C_0(\mathbb{R}^d)$, define for $t > 0$

$$(T(t)f)(x) = \frac{1}{(4\pi t)^{d/2}} \int_{\mathbb{R}^d} e^{-|x-y|^2/(4t)} f(y) \, \mathrm{d}y, \qquad T(0) = I.$$

Then $(T(t))$ is a contraction C_0-semigroup. Its generator is the closure of

$$\Delta \colon C_c^\infty(\mathbb{R}^d) \to C_0(\mathbb{R}^d),$$

and $C_c^\infty(\mathbb{R}^d)$ is a core.

Proof. Let
$$G_t(x) = (4\pi t)^{-d/2} e^{-|x|^2/(4t)}.$$
The kernels are nonnegative, have integral one, and satisfy $G_t * G_s = G_{t+s}$; the last identity follows by completing the square (or by Fourier transform). Therefore $T(t+s) = T(t)T(s)$ and
$$\|T(t)f\|_\infty \leq \|f\|_\infty.$$
Each $T(t)f$ belongs to C_0: continuity follows by translating the kernel, and vanishing at infinity follows first for compactly supported f and then for general $f \in C_0$ by uniform approximation. Since (G_t) is an approximate identity, $T(t)f \to f$ uniformly for $f \in C_0$. Thus T is a contraction C_0-semigroup.

Let G denote its generator. For $u \in \mathcal{S}(\mathbb{R}^d)$, differentiation under the integral (equivalently, Fourier transform) gives
$$\frac{\mathrm{d}}{\mathrm{d}t} T(t)u = \Delta T(t)u = T(t)\Delta u.$$
Hence $\mathcal{S} \subseteq \mathrm{D}(G)$ and $Gu = \Delta u$ there. The Schwartz space is dense in C_0 and invariant under $T(t)$, so the invariant-core theorem shows that $\mathcal{S}$ is a core for G.

Finally, C_c^∞ is graph dense in $\mathcal{S}$ for the operator Δ. Indeed, if $\chi_R(x) = \chi(x/R)$ with $\chi = 1$ near the origin, then $\chi_R u \in C_c^\infty$ and
$$\chi_R u \to u, \qquad \Delta(\chi_R u) \to \Delta u$$
uniformly, because
$$\Delta(\chi_R u) - \Delta u = (\chi_R - 1)\Delta u + 2\nabla\chi_R \cdot \nabla u + (\Delta\chi_R)u$$
and u is Schwartz. Consequently C_c^∞ is also a core, and G is the closure of Δ on that domain. □

Remark 6.6.8. The same formula is not strongly continuous on all of $C_b(\mathbb{R}^d)$: functions in C_b need not be uniformly continuous. The correct standard spaces are $C_0(\mathbb{R}^d)$ or the bounded uniformly continuous functions.

Theorem 6.6.9 (Poisson semigroup). Let
$$P_t(x) = c_d \frac{t}{(t^2 + |x|^2)^{(d+1)/2}}, \qquad t > 0,$$
where c_d is chosen so that $\int_{\mathbb{R}^d} P_t = 1$, and define
$$T(t)f = P_t * f, \qquad T(0) = I.$$
Then $(T(t))_{t\geq 0}$ is a contraction C_0-semigroup on $C_0(\mathbb{R}^d)$. If A denotes its generator, then
$$\mathcal{S}(\mathbb{R}^d) \subseteq \mathrm{D}(A^2)$$
and, for every $u \in \mathcal{S}(\mathbb{R}^d)$,
$$Au = -(-\Delta)^{1/2}u, \qquad A^2 u = -\Delta u,$$
where $(-\Delta)^{1/2}$ is the Fourier multiplier with symbol $|\xi|$.

Proof. The kernel P_t is nonnegative and has integral one. Hence, for $f \in C_0(\mathbb{R}^d)$,

$$\|T(t)f\|_\infty \le \|P_t\|_1 \|f\|_\infty = \|f\|_\infty.$$

Convolution with an L^1 kernel maps C_0 into C_0. Indeed, continuity follows from

$$|(P_t * f)(x+h) - (P_t * f)(x)| \le \|P_t\|_1 \|f(\cdot + h) - f\|_\infty,$$

and functions in C_0 are uniformly continuous. To see vanishing at infinity, first take f compactly supported and split the integral into a large compact set and its complement; then pass to arbitrary $f \in C_0$ by uniform approximation with compactly supported continuous functions.

With the usual Fourier-transform normalization,

$$\widehat{P}_t(\xi) = e^{-t|\xi|}.$$

Consequently

$$\widehat{P_t * P_s} = e^{-(t+s)|\xi|} = \widehat{P_{t+s}}.$$

Both sides belong to $L^1(\mathbb{R}^d)$, so uniqueness of the Fourier transform gives $P_t * P_s = P_{t+s}$ and therefore $T(t)T(s) = T(t+s)$.

The family (P_t) is an approximate identity. In fact, after the change of variables $y = tz$,

$$\int_{|y|\ge\delta} P_t(y)\,\mathrm{d}y = \int_{|z|\ge\delta/t} P_1(z)\,\mathrm{d}z \longrightarrow 0 \qquad (t \downarrow 0)$$

for every $\delta > 0$. Given $f \in C_0$ and $\varepsilon > 0$, choose $\delta > 0$ so that $|f(x-y) - f(x)| < \varepsilon$ whenever $|y| < \delta$, uniformly in x. Then

$$\begin{aligned} |T(t)f(x) - f(x)| &\le \int_{|y|<\delta} P_t(y)|f(x-y) - f(x)|\,\mathrm{d}y \\ &\quad + \int_{|y|\ge\delta} P_t(y)|f(x-y) - f(x)|\,\mathrm{d}y \\ &\le \varepsilon + 2\|f\|_\infty \int_{|y|\ge\delta} P_t(y)\,\mathrm{d}y. \end{aligned}$$

The right-hand side tends uniformly to ε, and hence $\|T(t)f - f\|_\infty \to 0$. Thus $(T(t))$ is a contraction C_0-semigroup.

Now let $u \in \mathcal{S}(\mathbb{R}^d)$. Fourier inversion gives

$$\frac{T(t)u - u}{t} = \mathcal{F}^{-1}\left(\frac{e^{-t|\xi|} - 1}{t}\,\widehat{u}(\xi)\right).$$

For every $t > 0$,

$$\left|\frac{e^{-t|\xi|} - 1}{t}\right| \le |\xi|,$$

and $|\xi|\,|\widehat{u}(\xi)|$ is integrable. Dominated convergence in L^1_ξ, followed by the L^1 Fourier-inversion estimate, yields uniform convergence

$$\frac{T(t)u - u}{t} \longrightarrow \mathcal{F}^{-1}(-|\xi|\widehat{u}) = -(-\Delta)^{1/2}u.$$

Therefore $u \in \mathrm{D}(A)$ and $Au = -(-\Delta)^{1/2}u$. Put

$$g(\xi) = -|\xi|\widehat{u}(\xi), \qquad Au = \mathcal{F}^{-1}g.$$

Here $g \in L^1$, so $Au \in C_0$. Moreover, Fubini's theorem is applicable because $P_t \in L^1$ and $g \in L^1$; it gives

$$T(t)Au = \mathcal{F}^{-1}(e^{-t|\xi|}g).$$

Consequently

$$\frac{T(t)Au - Au}{t} = \mathcal{F}^{-1}\left(\frac{e^{-t|\xi|} - 1}{t}(-|\xi|\widehat{u})\right).$$

The multiplier is dominated by $|\xi|^2|\widehat{u}(\xi)| \in L^1$. Hence

$$\frac{T(t)Au - Au}{t} \longrightarrow \mathcal{F}^{-1}(|\xi|^2\widehat{u}) = -\Delta u$$

uniformly. Thus $u \in \mathrm{D}(A^2)$ and $A^2u = -\Delta u$. □

Chapter 7 Sobolev Spaces

7.1 One-dimensional Sobolev spaces

Definition 7.1.1. Let $I \subseteq \mathbb{R}$ be an open interval and let $1 \le p \le \infty$. A function $u \in L^p(I)$ belongs to $W^{1,p}(I)$ if there is $v \in L^p(I)$ such that

$$\int_I u\varphi' \,\mathrm{d}x = -\int_I v\varphi \,\mathrm{d}x \qquad (\varphi \in C_c^\infty(I)).$$

The function v is unique almost everywhere and is denoted by u'. The norm is

$$\|u\|_{W^{1,p}(I)} = \|u\|_{L^p(I)} + \|u'\|_{L^p(I)}.$$

We write $H^1(I) = W^{1,2}(I)$.

Theorem 7.1.2. The space $W^{1,p}(I)$ is Banach for every $1 \le p \le \infty$, separable for $1 \le p < \infty$, and reflexive for $1 < p < \infty$.

Proof. The map

$$u \longmapsto (u, u')$$

is an isometry from $W^{1,p}(I)$ into $L^p(I) \times L^p(I)$. Its range is closed: if $u_n \to u$ and $u_n' \to v$ in L^p, then for every test function

$$\int u\varphi' = \lim_n \int u_n\varphi' = -\lim_n \int u_n'\varphi = -\int v\varphi.$$

Completeness, separability, and reflexivity therefore follow from the corresponding product-space properties of L^p. □

Theorem 7.1.3 (One-dimensional fundamental theorem)**.** Let $u \in W^{1,p}(I)$. There is an absolutely continuous representative $\tilde{u}$ such that

$$\tilde{u}(x) - \tilde{u}(y) = \int_y^x u'(t) \,\mathrm{d}t \qquad (x, y \in I).$$

In particular, every one-dimensional Sobolev function has a continuous representative.

Proof. Fix $y_0 \in I$ and define

$$v(x) = \int_{y_0}^x u'(t) \,\mathrm{d}t.$$

Then v is absolutely continuous and has weak derivative u'. Hence $u - v$ has weak derivative zero. The lemma below shows that $u - v$ is almost everywhere constant, so u agrees almost everywhere with $v +$

c. ☐

Lemma 7.1.4. If $f \in L^1_{\mathrm{loc}}(I)$ and

$$\int_I f\varphi' \, \mathrm{d}x = 0 \qquad (\varphi \in C_c^\infty(I)),$$

then f is almost everywhere constant on I.

Proof. Choose $\psi \in C_c^\infty(I)$ with $\int_I \psi = 1$. For any $\varphi \in C_c^\infty(I)$, the function $\varphi - (\int \varphi)\psi$ has integral zero and hence is the derivative of a compactly supported smooth function. Therefore

$$\int f\varphi = \left(\int f\psi\right)\left(\int \varphi\right),$$

which says that the distribution defined by f is the constant $c = \int f\psi$. ☐

Proposition 7.1.5 (Dual characterization of a weak derivative)**.** Let $u \in L^p(I)$ and $1 < p \leq \infty$. Then $u \in W^{1,p}(I)$ if and only if there is $C > 0$ such that

$$\left|\int_I u\varphi' \, \mathrm{d}x\right| \leq C\|\varphi\|_{L^{p'}(I)} \qquad (\varphi \in C_c^\infty(I)),$$

with $p' = 1$ when $p = \infty$. The smallest such C is $\|u'\|_p$.

Proof. One direction is Hölder's inequality. Conversely, the map $\varphi \mapsto -\int u\varphi'$ is bounded on the dense subspace $C_c^\infty(I) \subset L^{p'}(I)$. Extend it to $L^{p'}$ and apply the L^p duality theorem to obtain $v \in L^p$ with $-\int u\varphi' = \int v\varphi$. ☐

Remark 7.1.6. At $p = 1$, the displayed estimate with the L^∞ norm controls the distributional derivative only as a finite measure and characterizes bounded variation. It does not by itself force that measure to have an L^1 density. This is why the theorem is stated for $p > 1$ (and for $p = \infty$ via $(L^1)^* = L^\infty$).

Remark 7.1.7. For $p = 1$, the absolutely continuous representative is exactly the classical one: a function belongs to $W^{1,1}(I)$ precisely when it is almost everywhere equal to an absolutely continuous function whose classical derivative lies in L^1. Merely bounded variation is weaker; BV functions may have jump or singular derivatives.

Theorem 7.1.8 (Lipschitz characterization)**.** A function $u \in L^\infty(I)$ belongs to $W^{1,\infty}(I)$ if and only if it has a Lipschitz representative. Moreover,

$$\mathrm{Lip}(u) = \|u'\|_\infty.$$

Proof. The fundamental theorem gives $|u(x) - u(y)| \leq \|u'\|_\infty |x - y|$. Conversely, a Lipschitz function is absolutely continuous, differentiable almost everywhere, and its derivative is bounded by its Lipschitz constant. Integration by parts then identifies that derivative as the weak derivative. ☐

Theorem 7.1.9 (Difference-quotient criterion on $\mathbb{R}$)**.** Let $1 < p < \infty$ and $u \in L^p(\mathbb{R})$. Then $u \in W^{1,p}(\mathbb{R})$ if and only if there is $C > 0$ such that

$$\|u(\cdot + h) - u\|_p \leq C|h| \qquad (h \in \mathbb{R}).$$

The least admissible C equals $\|u'\|_p$.

Proof. If $u \in W^{1,p}$, then for the absolutely continuous representative,

$$u(x+h) - u(x) = \int_0^h u'(x+s)\, ds.$$

Minkowski's integral inequality gives the estimate with $C = \|u'\|_p$. Conversely, for $\varphi \in C_c^\infty(\mathbb{R})$,

$$\int u(x) \frac{\varphi(x-h) - \varphi(x)}{h}\, dx = \int \frac{u(x+h) - u(x)}{h} \varphi(x)\, dx.$$

The right-hand side is bounded by $C\|\varphi\|_{p'}$. Letting $h \to 0$ and using the preceding dual characterization yields a weak derivative in L^p. For $p = 1$, the analogous uniform translation bound characterizes bounded variation rather than $W^{1,1}$, so the endpoint cannot be inserted without an additional absolute-continuity condition. □

7.2 Extension and smooth approximation

Theorem 7.2.1 (Extension from an interval). Let I be an interval and $1 \le p \le \infty$. There is a bounded linear operator

$$E\colon W^{1,p}(I) \to W^{1,p}(\mathbb{R})$$

such that $Eu = u$ almost everywhere on I and

$$\|Eu\|_{W^{1,p}(\mathbb{R})} \le C_I \|u\|_{W^{1,p}(I)}.$$

For a bounded interval, Eu may be chosen with support in a fixed bounded neighbourhood of I.

Proof. The construction in the notes is reflection. We give it explicitly. For $I = (0, \infty)$, use the absolutely continuous representative and set

$$(Eu)(x) = u(|x|), \qquad x \in \mathbb{R}.$$

Then, almost everywhere,

$$(Eu)'(x) = \begin{cases} u'(x), & x > 0, \\ -u'(-x), & x < 0, \end{cases}$$

and hence, for $p < \infty$,

$$\|Eu\|_p^p = 2\|u\|_p^p, \qquad \|(Eu)'\|_p^p = 2\|u'\|_p^p.$$

The corresponding supremum-norm estimates hold when $p = \infty$. Translation handles every half-line, while $I = \mathbb{R}$ needs no extension.

For a bounded interval, an affine change of variables reduces the problem to $I = (0,1)$. Define on $(-1,2)$

$$v(x) = \begin{cases} u(-x), & -1 < x < 0, \\ u(x), & 0 \le x \le 1, \\ u(2-x), & 1 < x < 2. \end{cases}$$

The one-dimensional fundamental theorem shows that $v \in W^{1,p}(-1,2)$, with derivative obtained by changing signs on the reflected pieces, and

$$\|v\|_{W^{1,p}(-1,2)} \le C\|u\|_{W^{1,p}(0,1)}.$$

Choose $\eta \in C_c^\infty(-1,2)$ with $\eta = 1$ on $[0,1]$, and set $Eu = \eta v$ on $(-1,2)$ and $Eu = 0$ elsewhere. The product rule gives

$$(Eu)' = \eta' v + \eta v'$$

and hence the required bound. Every operation is linear, and the support is contained in the fixed compact set $\operatorname{supp}\eta$. Undoing the affine change of variables completes the construction for an arbitrary bounded interval. □

Lemma 7.2.2 (Mollification of Sobolev functions)**.** Let $u \in W^{1,p}(\mathbb{R})$ and let $\rho \in C_c^\infty(\mathbb{R})$. Then

$$\rho * u \in W^{1,p}(\mathbb{R}), \qquad (\rho * u)' = \rho * u'.$$

Proof. For a test function φ, Fubini's theorem and the definition of the weak derivative give

$$\int (\rho * u)\varphi' = \int u\,(\tilde{\rho} * \varphi') = -\int u'\,(\tilde{\rho} * \varphi) = -\int (\rho * u')\varphi,$$

where $\tilde{\rho}(x) = \rho(-x)$. Young's inequality gives the L^p bounds. □

Theorem 7.2.3 (Smooth approximation)**.** For $1 \le p < \infty$, $C_c^\infty(\mathbb{R})$ is dense in $W^{1,p}(\mathbb{R})$. More generally, smooth functions are dense in $W^{1,p}(I)$, and on a bounded interval they may be obtained by extending, mollifying, and restricting.

Proof. We first work on $\mathbb{R}$. Choose $\chi \in C_c^\infty(\mathbb{R})$ with $0 \le \chi \le 1$, $\chi = 1$ on $[-1,1]$, and $\operatorname{supp}\chi \subset [-2,2]$, and put $\chi_R(x) = \chi(x/R)$. Then

$$\chi_R u \longrightarrow u \quad \text{in } L^p, \qquad \chi_R u' \longrightarrow u' \quad \text{in } L^p$$

by dominated convergence. Moreover,

$$(\chi_R u)' = \chi_R u' + \chi_R' u, \qquad \|\chi_R' u\|_p \le \frac{\|\chi'\|_\infty}{R}\|u\|_p \longrightarrow 0.$$

Thus $\chi_R u \to u$ in $W^{1,p}$, and each $\chi_R u$ has compact support.
Now fix a compactly supported $v \in W^{1,p}(\mathbb{R})$. For mollifiers ρ_ε, the convolution lemma gives

$$(\rho_\varepsilon * v)' = \rho_\varepsilon * v',$$

and the L^p approximate-identity theorem gives

$$\rho_\varepsilon * v \to v, \qquad \rho_\varepsilon * v' \to v'$$

in L^p. Hence $\rho_\varepsilon * v \to v$ in $W^{1,p}$; for small ε the convolutions are still compactly supported. A diagonal choice of R and ε proves density of $C_c^\infty(\mathbb{R})$.
For a general interval, extend u by the preceding extension theorem, approximate the extension by smooth functions on $\mathbb{R}$, and restrict back to I. □

7.3 One-dimensional Sobolev embeddings

Theorem 7.3.1 (Sobolev embedding on an interval)**.** Let I be an interval of finite length and $1 \leq p \leq \infty$. Every $u \in W^{1,p}(I)$ has a continuous representative and

$$\|u\|_{L^\infty(I)} \leq C_I \|u\|_{W^{1,p}(I)}.$$

Thus $W^{1,p}(I) \hookrightarrow L^\infty(I)$ continuously.

If I is bounded, then

(a). $W^{1,p}(I) \hookrightarrow C(\overline{I})$ compactly for $1 < p \leq \infty$;

(b). $W^{1,1}(I) \hookrightarrow L^2(I)$ compactly.

Proof. Let $L = |I|$. We use the absolutely continuous representative of u. First suppose $1 \leq p < \infty$. Since $|u|^p$ is integrable, there is a point $y \in \overline{I}$ such that

$$|u(y)|^p \leq \frac{1}{L}\|u\|_p^p.$$

For $p = 1$, the fundamental theorem immediately gives, for every $x \in \overline{I}$,

$$|u(x)| \leq |u(y)| + \int_I |u'(t)|\,\mathrm{d}t \leq \frac{1}{L}\|u\|_1 + \|u'\|_1.$$

Now let $1 < p < \infty$. The function $|u|^p$ is absolutely continuous and, almost everywhere,

$$\left|\frac{\mathrm{d}}{\mathrm{d}t}|u(t)|^p\right| \leq p|u(t)|^{p-1}|u'(t)|;$$

in the complex case this follows from $\frac{\mathrm{d}}{\mathrm{d}t}|u|^p = p|u|^{p-2}\operatorname{Re}(\overline{u}\,u')$ at points where $u \neq 0$, with the usual continuous interpretation at $u = 0$. Therefore

$$\begin{aligned}|u(x)|^p &\leq |u(y)|^p + p\int_I |u|^{p-1}|u'| \\ &\leq \frac{1}{L}\|u\|_p^p + p\|u\|_p^{p-1}\|u'\|_p.\end{aligned}$$

Young's inequality, $pa^{p-1}b \leq (p-1)a^p + b^p$, now yields

$$\|u\|_\infty^p \leq \left(\frac{1}{L} + p - 1\right)\|u\|_p^p + \|u'\|_p^p,$$

which is the desired continuous embedding. For $p = \infty$ the conclusion is immediate from the definition of the $W^{1,\infty}$ norm.

Let next I be bounded. If $1 < p < \infty$, Hölder's inequality and the fundamental theorem give

$$|u(x) - u(y)| \leq \int_x^y |u'(t)|\,\mathrm{d}t \leq \|u'\|_p|x-y|^{1/p'}.$$

For $p = \infty$ the same estimate holds with exponent one. Thus a bounded set in $W^{1,p}(I)$ is uniformly bounded and equicontinuous on $\overline{I}$; Arzelà–Ascoli gives relative compactness in $C(\overline{I})$.

For $p = 1$, extend each function by the bounded extension operator to one fixed bounded interval containing $\bar{I}$. A bounded family is uniformly bounded in both L^1 and L^∞, and the fundamental theorem gives the uniform translation estimate

$$\|u(\cdot + h) - u\|_1 \le |h| \, \|u'\|_1$$

(up to a harmless fixed constant caused by the extension). The Fréchet–Kolmogorov criterion therefore gives relative compactness in L^1. If $u_n \to u$ in L^1 along a subsequence and $\sup_n \|u_n\|_\infty \le M$, pass once more to a subsequence converging almost everywhere. Then $|u| \le M$ almost everywhere, so

$$\|u_n - u\|_2^2 \le \|u_n - u\|_\infty \|u_n - u\|_1 \le 2M\|u_n - u\|_1 \longrightarrow 0.$$

Hence the embedding into $L^2(I)$ is compact. □

Corollary 7.3.2 (Decay on an unbounded interval)**.** Let I be unbounded and let $1 \le p < \infty$. If $u \in W^{1,p}(I)$, then its continuous representative satisfies

$$u(x) \longrightarrow 0$$

at every infinite endpoint of I.

Proof. We prove the assertion at a right-hand infinite endpoint; the left-hand case is identical after reflection.

Suppose first that $p = 1$. For $x > y$ in I, the fundamental theorem gives

$$|u(x) - u(y)| \le \int_y^x |u'(t)| \, dt.$$

Because $u' \in L^1(I)$, the right-hand side tends to zero uniformly when $x, y \to +\infty$. Hence $u(x)$ has a finite limit L at the endpoint. If $L \ne 0$, then $|u(x)| \ge |L|/2$ on a tail of infinite length, contradicting $u \in L^1(I)$. Thus $L = 0$.

Now let $1 < p < \infty$. Hölder's inequality gives the global modulus of continuity

$$|u(x) - u(y)| \le \|u'\|_{L^p(I)} |x - y|^{1/p'}.$$

Thus u is uniformly continuous. If $u(x)$ did not tend to zero, there would be an $\varepsilon > 0$ and a sequence $x_n \to +\infty$ with $|u(x_n)| \ge \varepsilon$. Uniform continuity supplies $\delta > 0$ such that $|u(x) - u(x_n)| < \varepsilon/2$ whenever $|x - x_n| < \delta$. Passing to a subsequence, the intervals $(x_n - \delta/2, x_n + \delta/2)$ are pairwise disjoint and contained in I. On each of them $|u| \ge \varepsilon/2$, so

$$\int_I |u|^p \ge \sum_n \delta \left(\frac{\varepsilon}{2}\right)^p = \infty,$$

a contradiction. Therefore $u(x) \to 0$. □

Remark 7.3.3. On an unbounded interval, the embedding $W^{1,p}(I) \hookrightarrow L^\infty(I)$ is not compact. Translates of one fixed nonzero compactly supported smooth function are bounded in $W^{1,p}$ but have no uniformly convergent subsequence.

7.4 Product and chain rules

Proposition 7.4.1 (One-dimensional differentiation rules)**.** Let $1 \le p \le \infty$.

(a). If $u, v \in W^{1,p}(I) \cap L^\infty(I)$, then $uv \in W^{1,p}(I)$ and

$$(uv)' = u'v + uv'.$$

The integration-by-parts identity

$$\int_a^b u'v = u(b)v(b) - u(a)v(a) - \int_a^b uv'$$

holds for bounded $I = (a, b)$.

(b). If u is real valued, $G \in C^1(\mathbb{R})$, $G(0) = 0$, and G' is bounded, then $G \circ u \in W^{1,p}(I)$ and

$$(G \circ u)' = (G' \circ u)u'.$$

Proof. Take the absolutely continuous representatives furnished by the one-dimensional fundamental theorem. On every compact subinterval, the classical product rule for absolutely continuous functions gives

$$(uv)' = u'v + uv' \quad \text{almost everywhere.}$$

The assumption $u, v \in L^\infty$ makes both terms on the right belong to $L^p(I)$, while $uv \in L^p(I)$. Thus the local identity is a global weak identity, and $uv \in W^{1,p}(I)$. If $I = (a, b)$ is bounded, the product uv is absolutely continuous on $[a, b]$, so the ordinary fundamental theorem gives

$$\int_a^b (u'v + uv')\, \mathrm{d}x = u(b)v(b) - u(a)v(a),$$

which is the stated integration-by-parts formula.

For the chain rule, assume u is real valued, as in the statement. Since G' is bounded and $G(0) = 0$, the mean-value theorem gives $|G(s)| \le \|G'\|_\infty |s|$, so $G \circ u \in L^p(I)$. The composition of an absolutely continuous function with a C^1 Lipschitz function is absolutely continuous, and the classical chain rule holds almost everywhere:

$$(G \circ u)' = (G' \circ u)u'.$$

The right-hand side belongs to L^p, proving the Sobolev chain rule. □

7.5 Higher-order spaces, traces, and negative norms

Definition 7.5.1. For an integer $m \ge 1$,

$$W^{m,p}(I) = \{u \in L^p(I) : u^{(j)} \in L^p(I),\ 1 \le j \le m\},$$

with norm

$$\|u\|_{W^{m,p}(I)} = \sum_{j=0}^{m} \|u^{(j)}\|_p.$$

Equivalent ℓ^p combinations of these terms give the same topology. When $p = 2$ we write $H^m(I) = W^{m,2}(I)$.

Definition 7.5.2. For $1 \le p < \infty$,

$$W_0^{1,p}(I) = \overline{C_c^\infty(I)}^{W^{1,p}(I)}.$$

Theorem 7.5.3 (Trace characterization)**.** Let $I = (a, b)$ be bounded and let $1 \le p < \infty$. Then

$$W_0^{1,p}(I) = \{u \in W^{1,p}(I) : u(a) = u(b) = 0\},$$

where the boundary values are taken from the continuous representative.

Proof. The Sobolev estimate makes the trace map

$$u \longmapsto (u(a), u(b))$$

continuous on $W^{1,p}(I)$. Every function in $C_c^\infty(I)$ has zero trace, so the closure of that space is contained in the right-hand side.

Conversely, let $u \in W^{1,p}(I)$ with $u(a) = u(b) = 0$. Choose smooth cutoffs η_ε such that

$$0 \le \eta_\varepsilon \le 1, \qquad \eta_\varepsilon = 0 \text{ when } \operatorname{dist}(x, \partial I) \le \varepsilon, \qquad \eta_\varepsilon = 1 \text{ when } \operatorname{dist}(x, \partial I) \ge 2\varepsilon,$$

and $|\eta_\varepsilon'| \le C/\varepsilon$. Put $u_\varepsilon = \eta_\varepsilon u$. The L^p tails of u and u' near the endpoints tend to zero, so

$$(1 - \eta_\varepsilon)u \to 0, \qquad (1 - \eta_\varepsilon)u' \to 0 \quad \text{in } L^p.$$

It remains to control $\eta_\varepsilon' u$. Near a, the zero trace and the fundamental theorem give

$$|u(x)| \le \int_a^x |u'(t)| \, dt.$$

If $p > 1$, Hölder's inequality yields

$$|u(x)|^p \le (x - a)^{p-1} \int_a^{a+2\varepsilon} |u'(t)|^p \, dt,$$

after which integration over $(a, a + 2\varepsilon)$ gives

$$\int_a^{a+2\varepsilon} |\eta_\varepsilon'(x) u(x)|^p \, dx \le C \int_a^{a+2\varepsilon} |u'(t)|^p \, dt \longrightarrow 0.$$

For $p = 1$, Fubini gives the endpoint estimate explicitly:

$$\begin{aligned}
\int_a^{a+2\varepsilon} |\eta_\varepsilon'(x) u(x)| \, dx &\le \frac{C}{\varepsilon} \int_a^{a+2\varepsilon} \int_a^x |u'(t)| \, dt \, dx \\
&= \frac{C}{\varepsilon} \int_a^{a+2\varepsilon} (a + 2\varepsilon - t)|u'(t)| \, dt \\
&\le 2C \int_a^{a+2\varepsilon} |u'(t)| \, dt \longrightarrow 0.
\end{aligned}$$

The endpoint b is identical. Hence $u_\varepsilon \to u$ in $W^{1,p}$. Each u_ε has compact support in I; mollifying at a scale smaller than its distance from the boundary gives elements of $C_c^\infty(I)$ converging to it in $W^{1,p}$. A diagonal sequence proves the reverse inclusion. □

Theorem 7.5.4 (Poincaré inequality)**.** For a bounded interval I and $1 \le p < \infty$, there is $C_I > 0$ such that

$$\|u\|_p \le C_I \|u'\|_p \qquad (u \in W_0^{1,p}(I)).$$

The same estimate holds for zero-trace functions in $W^{1,\infty}(I)$. Consequently, $\|u'\|_p$ is an equivalent norm on $W_0^{1,p}(I)$.

Proof. For the continuous representative and $u(a) = 0$,

$$|u(x)| \le \int_a^x |u'(t)| \, \mathrm{d}t.$$

Hölder's inequality and integration in x give the result. □

Definition 7.5.5. For $m \ge 1$, $W_0^{m,p}(I)$ is the closure of $C_c^\infty(I)$ in $W^{m,p}(I)$. For $1 < p < \infty$, the dual of $W_0^{1,p}(I)$ is denoted

$$W^{-1,p'}(I),$$

and the dual of $H_0^1(I)$ is denoted $H^{-1}(I)$.

Theorem 7.5.6 (Representation of $W^{-1,p'}$)**.** Let $1 < p < \infty$. A functional F belongs to $W^{-1,p'}(I)$ if and only if there are $f_0, f_1 \in L^{p'}(I)$ such that

$$\langle F, u \rangle = \int_I f_0 u \, \mathrm{d}x + \int_I f_1 u' \, \mathrm{d}x \qquad (u \in W_0^{1,p}(I)).$$

Moreover,

$$\|F\|_{W^{-1,p'}} = \inf\{\|f_0\|_{p'} + \|f_1\|_{p'} : (f_0, f_1) \text{ represents } F\}$$

up to the harmless choice of an equivalent product norm. If I is bounded, Poincaré's inequality allows a representation with $f_0 = 0$.

Proof. Embed $W_0^{1,p}(I)$ isometrically as the closed subspace

$$G = \{(u, u') : u \in W_0^{1,p}(I)\} \subset L^p(I) \times L^p(I).$$

The functional F induces a bounded functional on G. Hahn–Banach extends it to the product, and L^p duality represents that extension by $(f_0, f_1) \in L^{p'} \times L^{p'}$. Conversely, Hölder's inequality makes every such formula continuous. On a bounded interval, use the derivative embedding $u \mapsto u'$ and Poincaré's inequality instead. □

7.6 Sobolev spaces in several variables

Definition 7.6.1. Let $\Omega \subseteq \mathbb{R}^N$ be open. A function $u \in L^p(\Omega)$ belongs to $W^{1,p}(\Omega)$ if, for each $1 \le i \le N$, there is $g_i \in L^p(\Omega)$ satisfying

$$\int_\Omega u\,\partial_i\varphi \,\mathrm{d}x = -\int_\Omega g_i\varphi \,\mathrm{d}x \qquad (\varphi \in C_c^\infty(\Omega)).$$

We write $\partial_i u = g_i$ and $\nabla u = (\partial_1 u, \dots, \partial_N u)$. The norm

$$\|u\|_{W^{1,p}(\Omega)} = \|u\|_p + \sum_{i=1}^N \|\partial_i u\|_p$$

may be replaced by any equivalent finite-dimensional combination of these terms.

Proposition 7.6.2. The space $W^{1,p}(\Omega)$ is Banach for every $1 \le p \le \infty$, separable for $1 \le p < \infty$, and reflexive for $1 < p < \infty$. The space $H^1(\Omega)$ is Hilbert under

$$\langle u, v\rangle_{H^1} = \int_\Omega u\bar{v}\,\mathrm{d}x + \sum_{i=1}^N \int_\Omega \partial_i u\,\overline{\partial_i v}\,\mathrm{d}x.$$

Proof. Equip

$$E_p = L^p(\Omega)^{N+1}$$

with the product norm corresponding to the displayed Sobolev norm, and define

$$J\colon W^{1,p}(\Omega) \longrightarrow E_p, \qquad Ju = (u, \partial_1 u, \dots, \partial_N u).$$

The map J is an isometry onto its range. We show that this range is closed. Suppose

$$Ju_n \longrightarrow (u, g_1, \dots, g_N) \quad \text{in } E_p.$$

For every $\varphi \in C_c^\infty(\Omega)$ and every i,

$$\begin{aligned}\int_\Omega u\,\partial_i\varphi\,\mathrm{d}x &= \lim_{n\to\infty}\int_\Omega u_n\,\partial_i\varphi\,\mathrm{d}x\\ &= -\lim_{n\to\infty}\int_\Omega (\partial_i u_n)\varphi\,\mathrm{d}x = -\int_\Omega g_i\varphi\,\mathrm{d}x.\end{aligned}$$

Thus $u \in W^{1,p}(\Omega)$ and $\partial_i u = g_i$, so the limit lies in $J(W^{1,p}(\Omega))$. Since E_p is Banach, its closed subspace $J(W^{1,p}(\Omega))$ is Banach, and therefore so is $W^{1,p}(\Omega)$.
For $1 \le p < \infty$, the space $L^p(\Omega)$ is separable; hence the finite product E_p and its closed subspace $J(W^{1,p})$ are separable. For $1 < p < \infty$, $L^p(\Omega)$ is reflexive, so the finite product E_p and its closed subspace are reflexive. Finally, for $p = 2$ the displayed inner product induces a norm equivalent to the chosen $W^{1,2}$ norm. Completeness just proved therefore makes $H^1(\Omega)$ a Hilbert space. □

Proposition 7.6.3 (Closedness of weak differentiation)**.** Suppose $u_n \to u$ in $L^p(\Omega)$ and $\partial_i u_n \to g_i$ in $L^p(\Omega)$ for every i. Then $u \in W^{1,p}(\Omega)$ and $\partial_i u = g_i$. The same conclusion holds when all convergences are weak in the corresponding L^p spaces.

Proof. For $\varphi \in C_c^\infty(\Omega)$, strong convergence gives

$$\int_\Omega u\,\partial_i\varphi = \lim_n \int_\Omega u_n\,\partial_i\varphi = -\lim_n \int_\Omega (\partial_i u_n)\varphi = -\int_\Omega g_i\varphi.$$

The same computation is valid under weak convergence, because $\partial_i\varphi$ and φ belong to the appropriate dual $L^{p'}$ space (with the usual endpoint interpretation). This is exactly the definition of the weak derivative. □

Definition 7.6.4. For open sets $\omega, \Omega \subseteq \mathbb{R}^N$, the notation $\omega \Subset \Omega$ means that $\overline{\omega}$ is compact and contained in Ω.

Lemma 7.6.5 (Local mollification)**.** If $u \in W^{1,p}(\Omega)$ and $\omega \Subset \Omega$, then for sufficiently small ε,

$$u_\varepsilon = \rho_\varepsilon * u$$

is smooth on ω and

$$\partial_i u_\varepsilon = \rho_\varepsilon * \partial_i u \quad \text{on } \omega.$$

Proof. Choose $\varepsilon < \operatorname{dist}(\omega, \partial\Omega)$. For $x \in \omega$, the integral defining $u_\varepsilon(x)$ uses only values of u inside Ω. Differentiation of convolution makes u_ε smooth. To identify the derivative, take $\varphi \in C_c^\infty(\omega)$ and use Fubini:

$$\begin{aligned}
\int_\omega u_\varepsilon\,\partial_i\varphi &= \int_\Omega u(y)\left(\int_\omega \rho_\varepsilon(x-y)\partial_i\varphi(x)\;\mathrm{d}x\right)\mathrm{d}y \\
&= -\int_\Omega \partial_i u(y)\left(\int_\omega \rho_\varepsilon(x-y)\varphi(x)\;\mathrm{d}x\right)\mathrm{d}y \\
&= -\int_\omega (\rho_\varepsilon * \partial_i u)(x)\varphi(x)\;\mathrm{d}x.
\end{aligned}$$

Thus the two smooth functions have the same distributional derivative and are equal pointwise. □

Theorem 7.6.6 (Meyers–Serrin density theorem)**.** Let $1 \le p < \infty$. Then

$$C^\infty(\Omega) \cap W^{1,p}(\Omega)$$

is dense in $W^{1,p}(\Omega)$. If $\Omega = \mathbb{R}^N$, the approximants may be chosen in $C_c^\infty(\mathbb{R}^N)$. On a general open set, $C_c^\infty(\Omega)$ is dense only in the smaller space $W_0^{1,p}(\Omega)$, not in all of $W^{1,p}(\Omega)$.

Proof. Fix $u \in W^{1,p}(\Omega)$ and $\varepsilon > 0$. Choose an exhaustion by open sets

$$\Omega_1 \Subset \Omega_2 \Subset \cdots \Subset \Omega, \qquad \bigcup_{j=1}^\infty \Omega_j = \Omega.$$

Set

$$U_1 = \Omega_2, \qquad U_j = \Omega_{j+1} \setminus \overline{\Omega_{j-1}} \quad (j \ge 2).$$

This is a locally finite open cover of Ω. Let (ψ_j) be a smooth partition of unity subordinate to it, chosen so that $\operatorname{supp}\psi_j \Subset U_j$. By the product rule,

$$u_j := \psi_j u \in W^{1,p}(\Omega)$$

and u_j has compact support in Ω. Moreover, $u = \sum_j u_j$, locally finitely, together with the corresponding identity for weak derivatives.

Extend each u_j by zero to $\mathbb{R}^N$. Choose $\delta_j > 0$ smaller than the distance from $\operatorname{supp} u_j$ to $\partial\Omega$ and so small that

$$\|\rho_{\delta_j} * u_j - u_j\|_{W^{1,p}(\mathbb{R}^N)} < \varepsilon 2^{-j}.$$

The scales may also be chosen so that the enlarged supports remain a locally finite family. Define

$$v = \sum_{j=1}^{\infty} \rho_{\delta_j} * u_j.$$

The sum is locally finite, so $v \in C^\infty(\Omega)$. Since the series of errors is absolutely summable in $W^{1,p}$,

$$\|v - u\|_{W^{1,p}(\Omega)} \leq \sum_{j=1}^{\infty} \|\rho_{\delta_j} * u_j - u_j\|_{W^{1,p}(\mathbb{R}^N)} < \varepsilon.$$

This proves density on an arbitrary open set.

When $\Omega = \mathbb{R}^N$, choose cutoffs $\chi_R(x) = \chi(x/R)$. Then

$$\chi_R u \to u \quad \text{in } W^{1,p}(\mathbb{R}^N),$$

because the tails of u and ∇u tend to zero and $\|u \nabla \chi_R\|_p \leq C R^{-1} \|u\|_p$. Mollifying the compactly supported functions $\chi_R u$ gives approximants in $C_c^\infty(\mathbb{R}^N)$. □

Theorem 7.6.7 (Distribution and translation criteria)**.** Let $1 < p < \infty$ and $u \in L^p(\mathbb{R}^N)$. The following are equivalent:

(a). $u \in W^{1,p}(\mathbb{R}^N)$;

(b). for each i there is C_i such that

$$\left| \int_{\mathbb{R}^N} u \, \partial_i \varphi \, \mathrm{d}x \right| \leq C_i \|\varphi\|_{p'} \qquad (\varphi \in C_c^\infty);$$

(c). there is $C > 0$ such that

$$\|u(\cdot + h) - u\|_p \leq C|h| \qquad (h \in \mathbb{R}^N).$$

For $p = 1$, the analogous bounded distributional-derivative or translation condition characterizes BV, not necessarily $W^{1,1}$.

Proof. If $u \in W^{1,p}$, Hölder's inequality gives (ii) with $C_i = \|\partial_i u\|_p$. Conversely, under (ii) the functional

$$\varphi \longmapsto - \int u \, \partial_i \varphi$$

is bounded on the dense subspace $C_c^\infty \subset L^{p'}$. Hahn–Banach and L^p duality give $g_i \in L^p$ representing it, so $g_i = \partial_i u$ weakly. Thus (i) and (ii) are equivalent.

For smooth u, the fundamental theorem along the segment from x to $x + h$ gives

$$u(x + h) - u(x) = \int_0^1 h \cdot \nabla u(x + th) \, \mathrm{d}t.$$

Minkowski's inequality and translation invariance yield

$$\|u(\cdot + h) - u\|_p \le |h| \sum_{i=1}^{N} \|\partial_i u\|_p.$$

Meyers–Serrin approximation passes this estimate to every $u \in W^{1,p}$, proving (i)⇒(iii).

Finally, assume (iii), fix a coordinate vector e_i, and let $\varphi \in C_c^\infty$. A change of variables gives

$$\int u(x) \frac{\varphi(x - te_i) - \varphi(x)}{t} \, dx = \int \frac{u(x + te_i) - u(x)}{t} \varphi(x) \, dx.$$

The right-hand side has absolute value at most $C\|\varphi\|_{p'}$. Letting $t \to 0$ gives

$$\left| \int u \, \partial_i \varphi \right| \le C\|\varphi\|_{p'},$$

which is (ii). □

Proposition 7.6.8 (Differentiation rules in $\mathbb{R}^N$). Let $1 \le p \le \infty$.

(a). If $u, v \in W^{1,p}(\Omega) \cap L^\infty(\Omega)$, then $uv \in W^{1,p}(\Omega)$ and

$$\partial_i(uv) = u \, \partial_i v + v \, \partial_i u.$$

(b). If u is real valued, $G \in C^1(\mathbb{R})$, $G(0) = 0$, and G' is bounded, then $G \circ u \in W^{1,p}(\Omega)$ and

$$\partial_i(G \circ u) = (G' \circ u)\partial_i u.$$

(c). Let $\Phi \colon \Omega' \to \Omega$ be a C^1 bi-Lipschitz diffeomorphism whose derivative and inverse derivative are bounded. If $u \in W^{1,p}(\Omega)$, then $u \circ \Phi \in W^{1,p}(\Omega')$ and

$$\nabla(u \circ \Phi) = D\Phi^{\mathsf{T}} (\nabla u) \circ \Phi$$

almost everywhere.

(d). If $\eta \in C_c^\infty(\Omega)$ and $u \in W^{1,p}_{\mathrm{loc}}(\Omega)$, then $\eta u \in W^{1,p}(\Omega)$ and the product rule holds.

Proof. For $p = \infty$, Sobolev functions have locally Lipschitz representatives; the product, chain, and coordinate-change formulas then follow from the classical almost-everywhere rules for Lipschitz functions. We therefore give the approximation argument for $1 \le p < \infty$.

All statements are local, so fix $\omega \Subset \Omega$ and mollify inside a slightly larger open set. For the product rule, choose smooth u_n, v_n with

$$u_n \to u, \quad v_n \to v \quad \text{in } W^{1,p}(\omega)$$

and almost everywhere, while $\|u_n\|_\infty \le \|u\|_\infty$ and $\|v_n\|_\infty \le \|v\|_\infty$. The classical identity

$$\partial_i(u_n v_n) = u_n \partial_i v_n + v_n \partial_i u_n$$

passes to L^p. For example,

$$\|u_n \partial_i v_n - u \partial_i v\|_p \le \|u_n\|_\infty \|\partial_i v_n - \partial_i v\|_p + \|(u_n - u)\partial_i v\|_p,$$

and the second term tends to zero by dominated convergence. This proves (i) on ω, hence on Ω.

For (ii), G is Lipschitz because G' is bounded, and $G(0) = 0$ gives $G(u) \in L^p$. For smooth approximants u_n,

$$\partial_i G(u_n) = G'(u_n)\partial_i u_n.$$

Now $G(u_n) \to G(u)$ in L^p, while

$$\begin{aligned} \|G'(u_n)\partial_i u_n - G'(u)\partial_i u\|_p &\le \|G'\|_\infty \|\partial_i u_n - \partial_i u\|_p \\ &\quad + \|(G'(u_n) - G'(u))\partial_i u\|_p, \end{aligned}$$

and the last term tends to zero by dominated convergence. Closedness of weak differentiation gives the chain rule.

For (iii), first take smooth u. The classical chain rule gives the stated formula, and the change-of-variables theorem plus boundedness of $D\Phi$ and $D\Phi^{-1}$ gives

$$\|u \circ \Phi\|_{W^{1,p}(\Omega')} \le C_\Phi \|u\|_{W^{1,p}(\Omega)}.$$

Approximate a general u by smooth Sobolev functions and pass to the limit in this estimate. Assertion (iv) follows from (i) on a neighbourhood of $\operatorname{supp} \eta$; compact support then makes all terms globally L^p. □

Definition 7.6.9. For an integer $m \ge 1$, a multi-index α, and $1 \le p \le \infty$,

$$W^{m,p}(\Omega) = \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \text{ for all } |\alpha| \le m\}.$$

It is equipped with

$$\|u\|_{W^{m,p}} = \sum_{|\alpha| \le m} \|D^\alpha u\|_p.$$

Equivalent finite-dimensional combinations of these terms define the same norm topology; for $p = 2$ this is the Hilbert space $H^m(\Omega) = W^{m,2}(\Omega)$.

Chapter 8 Calculus of Variations and Distributions

8.1 Quadratic minimization

Theorem 8.1.1 (Coercive quadratic minimization)**.** Let V be a real Banach space and let $a\colon V \times V \to \mathbb{R}$ be continuous, symmetric, and coercive: there are $M, \alpha > 0$ such that

$$|a(u,v)| \le M\|u\|\,\|v\|, \qquad a(v,v) \ge \alpha\|v\|^2.$$

For $\ell \in V^*$ define

$$J_\ell(v) = \frac{1}{2}a(v,v) - \ell(v).$$

If $K \subseteq V$ is nonempty, closed, and convex, then J_ℓ has a unique minimizer $u \in K$. The solution map $\ell \mapsto u$ is Lipschitz continuous. It is linear when K is a linear subspace.

Proof. The form a is an inner product, and $\|v\|_a = \sqrt{a(v,v)}$ is equivalent to the original norm. Since V is complete in the original norm, it is complete in $\|\cdot\|_a$ and is therefore a Hilbert space. Riesz representation gives a unique $c \in V$ such that

$$\ell(v) = a(c,v).$$

Then

$$J_\ell(v) = \frac{1}{2}\|v-c\|_a^2 - \frac{1}{2}\|c\|_a^2.$$

Thus the minimizer is the metric projection $P_K^a c$ in the Hilbert norm $\|\cdot\|_a$. Existence, uniqueness, and nonexpansiveness follow from the projection theorem. To make the dependence on the right-hand side explicit, let c_i be the Riesz representative of ℓ_i with respect to a. Then

$$\|c_1 - c_2\|_a^2 = (\ell_1 - \ell_2)(c_1 - c_2) \le \|\ell_1 - \ell_2\|_{V^*}\,\|c_1 - c_2\|.$$

Coercivity gives $\|w\| \le \alpha^{-1/2}\|w\|_a$, and hence

$$\|c_1 - c_2\|_a \le \alpha^{-1/2}\|\ell_1 - \ell_2\|_{V^*}.$$

Writing $u_i = P_K^a c_i$ and using nonexpansiveness of the projection once more,

$$\|u_1 - u_2\| \le \alpha^{-1/2}\|u_1 - u_2\|_a \le \alpha^{-1}\|\ell_1 - \ell_2\|_{V^*}.$$

Thus the solution map is Lipschitz. If K is a subspace, the orthogonal projection is linear, so $\ell \mapsto u$ is linear as well. $\square$

Corollary 8.1.2 (Variational inequality)**.** The minimizer $u \in K$ is characterized by

$$a(u, v-u) \geq \ell(v-u) \qquad (v \in K).$$

If K is a linear subspace, this is equivalent to the variational equation

$$a(u,v) = \ell(v) \qquad (v \in K).$$

Proof. For $v \in K$ and $0 < t \leq 1$, minimality of u applied to $u + t(v-u)$ gives

$$0 \leq \frac{J(u+t(v-u)) - J(u)}{t} = a(u, v-u) - \ell(v-u) + \frac{t}{2} a(v-u, v-u).$$

Let $t \downarrow 0$. Conversely, expanding $J(v) - J(u)$ and using the inequality gives nonnegativity. If K is a subspace, test with $u \pm v$ to obtain equality. □

Remark 8.1.3. The first variation of $J(v) = \frac{1}{2}a(v,v) - \ell(v)$ in the direction w is

$$J'(v)w = a(v,w) - \ell(w).$$

The additional term $\frac{1}{2}a(w,w)$ in the finite difference $J(v+w) - J(v)$ is second order. Quadratic minimization on a subspace therefore produces a linear variational equation, while nonlinear functionals generally produce nonlinear Euler–Lagrange equations.

8.2 Lax–Milgram and Stampacchia

Theorem 8.2.1 (Lax–Milgram lemma)**.** Let H be a real Hilbert space and let $a\colon H \times H \to \mathbb{R}$ be a continuous bilinear form, not necessarily symmetric. Assume coercivity:

$$a(v,v) \geq \alpha \|v\|^2 \qquad (v \in H)$$

for some $\alpha > 0$. Then, for every $\ell \in H^*$, there is a unique $u \in H$ such that

$$a(u,v) = \ell(v) \qquad (v \in H).$$

Moreover,

$$\|u\| \leq \frac{1}{\alpha}\|\ell\|,$$

and the map $\ell \mapsto u$ is bounded and linear.

Proof. Riesz representation defines a bounded operator $A \in \mathcal{B}(H,H)$ by

$$a(u,v) = \langle Au, v \rangle .$$

Coercivity gives

$$\alpha\|u\|^2 \leq \langle Au, u \rangle \leq \|Au\| \, \|u\|,$$

so A is injective, bounded below, and has closed range. If $y \perp \operatorname{Ran}(A)$, then $Ay \in \operatorname{Ran}(A)$ and therefore

$$\alpha\|y\|^2 \leq a(y,y) = \langle Ay, y \rangle = 0.$$

Thus $y = 0$. Therefore the range is dense and closed, hence all of H. Solve $Au = f$, where f is the Riesz representative of ℓ. The estimate follows from the lower bound. $\square$

Theorem 8.2.2 (Stampacchia variational inequality)**.** Let H be a real Hilbert space, let $K \subseteq H$ be nonempty, closed, and convex, and let $a\colon H \times H \to \mathbb{R}$ be continuous and coercive. For every $\ell \in H^*$ there is a unique $u \in K$ such that

$$a(u, v-u) \geq \ell(v-u) \qquad (v \in K).$$

The solution depends Lipschitz continuously on ℓ.

Proof. Let A and f be the Riesz representatives of a and ℓ. For $\rho > 0$, define

$$\Phi(u) = P_K(u - \rho(Au - f)).$$

For $u, v \in H$, nonexpansiveness of P_K gives

$$\begin{aligned}
\|\Phi(u) - \Phi(v)\|^2 &\leq \|(u-v) - \rho A(u-v)\|^2 \\
&= \|u-v\|^2 - 2\rho \langle A(u-v), u-v\rangle + \rho^2 \|A(u-v)\|^2 \\
&\leq \left(1 - 2\rho\alpha + \rho^2\|A\|^2\right)\|u-v\|^2.
\end{aligned}$$

Choose $0 < \rho < 2\alpha/\|A\|^2$. The number $q^2 = 1 - 2\rho\alpha + \rho^2\|A\|^2$ then satisfies $0 \leq q < 1$, so Φ is a contraction and has a unique fixed point $u \in K$.

It remains to identify the fixed-point equation. For $z \in H$, the characterization of the metric projection says

$$u = P_K z \quad \Longleftrightarrow \quad \langle z - u, v - u\rangle \leq 0 \quad (v \in K).$$

Taking $z = u - \rho(Au - f)$ shows that $u = \Phi(u)$ if and only if

$$\langle Au - f, v - u\rangle \geq 0 \qquad (v \in K),$$

which is precisely the stated variational inequality.

Finally, let u_i correspond to ℓ_i. Testing the inequality for u_1 with $v = u_2$ and that for u_2 with $v = u_1$, then adding, gives

$$\alpha\|u_1 - u_2\|^2 \leq (\ell_1 - \ell_2)(u_1 - u_2) \leq \|\ell_1 - \ell_2\|\,\|u_1 - u_2\|.$$

After cancelling $\|u_1 - u_2\|$ when it is nonzero, one obtains

$$\|u_1 - u_2\| \leq \frac{1}{\alpha}\|\ell_1 - \ell_2\|,$$

which proves the claimed Lipschitz dependence. $\square$

8.3 Test functions and weak derivatives

Definition 8.3.1. Let $\Omega \subseteq \mathbb{R}^N$ be open. The test-function space is

$$\mathcal{D}(\Omega) = C_c^\infty(\Omega).$$

The space $L^1_{\mathrm{loc}}(\Omega)$ consists of measurable functions that are integrable over every compact subset of Ω.

Definition 8.3.2. Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and let α be a multi-index. A function $v \in L^1_{\mathrm{loc}}(\Omega)$ is the weak derivative $D^\alpha u$ if

$$\int_\Omega v\varphi \,\mathrm{d}x = (-1)^{|\alpha|} \int_\Omega u D^\alpha \varphi \,\mathrm{d}x \qquad (\varphi \in \mathcal{D}(\Omega)).$$

8.4 Distributions

Definition 8.4.1. A *distribution* on Ω is a linear functional $T\colon \mathcal{D}(\Omega) \to \mathbb{K}$ such that, for every compact $K \subseteq \Omega$, there are $C_K > 0$ and $m_K \in \mathbb{N}_0$ with

$$|T(\varphi)| \le C_K \max_{|\alpha| \le m_K} \sup_{x \in K} |D^\alpha \varphi(x)|$$

for every $\varphi \in \mathcal{D}(\Omega)$ supported in K. The space of all distributions is denoted by $\mathcal{D}'(\Omega)$.

Remark 8.4.2 (Topology)**.** A sequence φ_n converges to φ in $\mathcal{D}(\Omega)$ if there is one compact $K \Subset \Omega$ containing all supports and

$$\sup_{x \in K} |D^\alpha(\varphi_n - \varphi)(x)| \longrightarrow 0$$

for every multi-index α. The full test-function topology is an inductive-limit topology and is generally not metrizable. A sequence $T_n \in \mathcal{D}'(\Omega)$ converges distributionally to T if

$$T_n(\varphi) \longrightarrow T(\varphi) \qquad (\varphi \in \mathcal{D}(\Omega)).$$

Proposition 8.4.3 (Regular distributions)**.** Every $u \in L^1_{\mathrm{loc}}(\Omega)$ defines a distribution T_u by

$$T_u(\varphi) = \int_\Omega u\varphi \,\mathrm{d}x.$$

The map $u \mapsto T_u$ is injective modulo equality almost everywhere.

Proof. If $\operatorname{supp} \varphi \subseteq K$, then

$$|T_u(\varphi)| \le \|u\|_{L^1(K)} \|\varphi\|_{L^\infty(K)},$$

so T_u is a distribution of order zero on each compact set.

To prove injectivity, suppose $T_u = 0$. Fix a ball $B(x_0, 2r) \Subset \Omega$ and a standard mollifier ρ_ε with $0 < \varepsilon < r$. For every $x \in B(x_0, r)$, the function

$$y \longmapsto \rho_\varepsilon(x - y)$$

belongs to $C_c^\infty(B(x_0,2r))$. Hence

$$(u*\rho_\varepsilon)(x) = \int_\Omega u(y)\rho_\varepsilon(x-y)\,\mathrm{d}y = T_u\big(\rho_\varepsilon(x-\cdot)\big) = 0.$$

Local approximation by mollifiers gives $u*\rho_\varepsilon \to u$ in $L^1(B(x_0,r))$, so $u=0$ almost everywhere on $B(x_0,r)$. Since such balls cover Ω, $u=0$ almost everywhere on Ω. Thus $u\mapsto T_u$ is injective modulo almost-everywhere equality. ☐

Example 8.4.4 (Dirac delta)**.** For $a\in\Omega$, the Dirac distribution is

$$\delta_a(\varphi) = \varphi(a).$$

It is not a regular distribution.

Proof. Suppose $\delta_a = T_u$ for some $u\in L^1_{\mathrm{loc}}$. Choose $\psi\in C_c^\infty(B(0,1))$ with $0\le\psi\le 1$ and $\psi(0)=1$, and set $\psi_k(x)=\psi(k(x-a))$. Then $\delta_a(\psi_k)=1$, whereas

$$\left|\int u\psi_k\right| \le \int_{B(a,1/k)} |u(x)|\,\mathrm{d}x \longrightarrow 0$$

by absolute continuity of the Lebesgue integral, a contradiction. ☐

Definition 8.4.5 (Distributional derivatives)**.** For $T\in\mathcal{D}'(\Omega)$ and a multi-index α, define

$$D^\alpha T(\varphi) = (-1)^{|\alpha|}T(D^\alpha\varphi).$$

Continuity of the derivative. Let $K\Subset\Omega$ and suppose that on test functions supported in K the distribution T satisfies an estimate of order m_K. Since $\operatorname{supp}(D^\alpha\varphi)\subseteq\operatorname{supp}\varphi$, we have

$$\begin{aligned}
|D^\alpha T(\varphi)| &= |T(D^\alpha\varphi)| \\
&\le C_K \max_{|\beta|\le m_K}\sup_{x\in K}|D^{\alpha+\beta}\varphi(x)| \\
&\le C_K \max_{|\gamma|\le m_K+|\alpha|}\sup_{x\in K}|D^\gamma\varphi(x)|.
\end{aligned}$$

Thus $D^\alpha T$ is a distribution, of local order at most $m_K+|\alpha|$ on K. ☐

Proposition 8.4.6. If $u\in C^{|\alpha|}(\Omega)$, then

$$D^\alpha T_u = T_{D^\alpha u}.$$

More generally, when $u\in L^1_{\mathrm{loc}}$, the regular distribution T_v is $D^\alpha T_u$ exactly when v is the weak derivative $D^\alpha u$.

Proof. Repeated integration by parts against compactly supported test functions gives

$$D^\alpha T_u(\varphi) = (-1)^{|\alpha|}\int uD^\alpha\varphi = \int D^\alpha u\,\varphi.$$

The final statement is precisely the definition of weak differentiation. ☐